PRODUCT MANUAL

ABUS wire rope hoist for double girder travelling cranes

Double-rail trolleys D, DA, DQA and DB



SERIAL NUMBER:

250-141190 / 250-141191

AT A GLANCE:

Checking the requirements: page 15

Inspecting the wire rope: page 25

Checking the load hook and bottom block: page 26

Replacing wire rope with reeving 2/1, 4/1 or 6/1: page 39

Lubricants: page 107

AN 120290EN000
2024-04-09

Original Operating Instructions

WIRE ROPE HOIST: VARIOUS TYPES, SIZES, VERSIONS AND OPTIONS

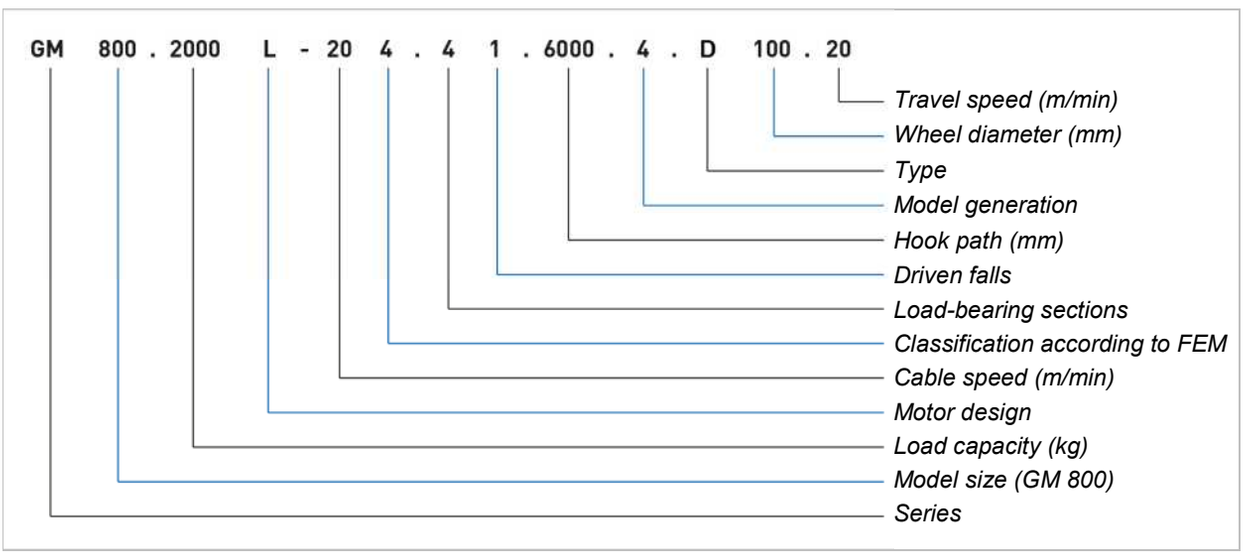
This product manual applies to wire rope hoists of various types, sizes, versions and options. The work steps described and the technical data differ depending on the wire rope hoist type, size, version and options. The areas of this product manual that do not apply to all wire rope hoists, but only under certain conditions, are framed by a dashed box. The types, sizes, versions and options to which the section applies are specified at the start of the box.

If a work step is described in a dashed box:

- At the start of the dashed box text, read the size, version or option to which this box applies.
- Note page and turn to this first page.
- Based on the images, determine which size, version or option applies to the crane in question.
- Turn back to the page with the associated dashed box for the next work steps.

- The size, version or option that applies to the crane in question can also be determined by consulting the scope of delivery or the planning documents.

TYPE DESIGNATION



WIRE ROPE HOIST (TYPE)



Double-rail trolley (D)



Double-rail trolley lowered (DA)



Double-rail trolley lowered crosswise (DQA)



Double-rail trolley with walkway (DB)

POWER SUPPLY (VERSION)

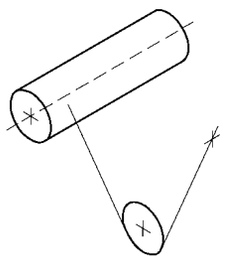


ABUPowerline energy chain

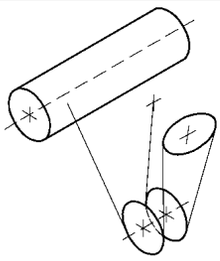


Festoon cable system

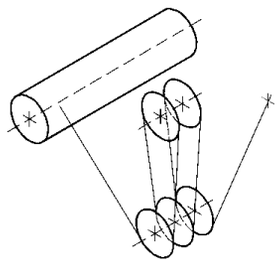
REEVING (VERSION)



Reeving 2/1

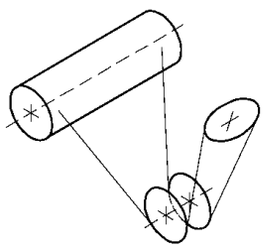


Reeving 4/1

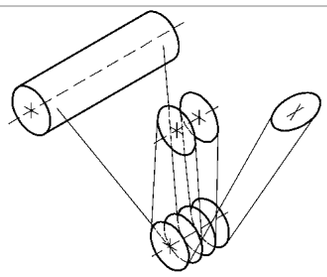


Reeving 6/1

True vertical lift hoist:



Reeving 4/2



Reeving 8/2

DRIVE (VERSION)



AZP drive



AZF drive



AZF V drive

CONTROL (VERSION)



ABUS electrics 3 with contactor control



ABUS electrics 3 with frequency converter for ABULiner travel and/or lifting



ABUControl

OVERLOAD PROTECTION FOR ABUS ELECTRICS 3 (VERSION)



LIS-SE



LIS-SV

CROSS-TYPE LIMIT SWITCH (OPTION)



Cross-type limit switch as trolley travel limit switch

CONTENTS

GENERAL INFORMATION 8

- At the start 8
- The wire rope hoist 10
- Transporting the wire rope hoist on a lorry 13
- Loading and unloading the wire rope hoist 13
- Wire rope hoist disposal 14

INSTALLING AND CONNECTING 15

- Checking the requirements 15
- Installing the wire rope hoist on the main girder 16
- Installing the trolley power supply ... 18
- Installing the second trolley 22
- Overview of tightening torques 22

INSPECTION 24

- At the start 24
- Scope of the inspection 25
- Inspecting the wire rope 25
- Checking the load hook and bottom block 26
- Inspecting the fixed point crosshead 29
- Inspecting the deflection roller crosshead 30
- Inspecting the cable guide 33
- Checking the fixed point 35
- Checking the diameter of the wheels 35
- Inspecting the wheel flange wear 36
- Checking the lift-off prevention device 37
- Inspecting the condition of the safety buffer 37

MAINTENANCE 38

- Instructions regarding safety when performing maintenance work 38
- Replacing wire rope with reeving 2/1, 4/1 or 6/1 39
- Replacing wire rope with reeving 4/2 or 8/2 57
- Exchanging the fixed point crosshead with rope socket and rope wedge 71
- Replacing deflection roller crosshead for reeving 4/1 or 6/1 81
- Replacing deflection roller crosshead for reeving 4/2 or 8/2 92
- Disassembling the powered wheel ... 96
- Installing the powered wheel 98
- Disassembling the non-powered wheel 103
- Installing the non-powered wheel .. 104
- Replacing the safety buffer 105
- Replacing the cross-type limit switch 106
- Wiring diagram of the cross-type limit switch 106
- Lubricants 107
- Overview of tightening torques 111
- ABUS Service 114

GENERAL INFORMATION

THIS SECTION APPLIES TO ANYONE WHO LIFTS LOADS USING THE CRANE, PERFORMS WORK ON THE CRANE, OR WORKS NEARBY.

AT THE START

USING THIS PRODUCT MANUAL

The following symbols are used in this product manual:



DANGER TO PERSONS!
This warning notice indicates hazards for persons.



DANGER FROM ELECTRIC SHOCK!
This warning notice indicates hazards for persons due to incorrect handling of the electrical system and electricity.



DANGER FROM FALLING SUSPENDED LOAD!
This warning notice describes hazardous situations that can lead to the suspended load falling.



NOTICE OF POSSIBLE DAMAGE!
This notice describes situations in which a component could be damaged.

➔ This is an instruction regarding an action and prompts you to perform a work step.

- This is the result of an action and describes what happens on the device.

– This is a list.

ONLY WITH...

A section that appears in a box framed by a dashed line is only applicable for certain types, versions or options. The condition to which the section applies is specified at the beginning in the heading “Only with...”.

INSTRUCTIONS FOR THE PRODUCT MANUAL

Read the product manual through carefully before beginning work. Also be sure to observe additional product manuals for accessories and components.

Afterwards, keep the product manual available in the vicinity of the crane. It must be accessible to all who work with or on the crane.

In the case of reselling or hiring out, always provide the product manual together with the crane.

PROPER USE

The wire rope hoist is intended for use on a correspondingly designed double girder travelling crane for area-wide movement of loads. The wire rope hoist travels horizontally with two end carriages on the trolley tracks of the main girders.

– The wire rope hoist is intended for a trolley power supply using the ABUPowerline energy chain or an ABUS festoon cable system.

REGULATIONS

The installation has been built and tested according to European standards, rules and regulations applicable at the time of its manufacture. The conventions applied during design and construction are specified in the Declaration of Conformity or the Declaration of Incorporation. The conventions must also be observed for installation, operation, inspection and maintenance, as must the applicable occupational health and safety regulations.

DANGER TO PERSONS!
Non-observance of the regulations can result in the death of persons or serious accidents.

To ensure that the work can be performed safely, thorough instruction in regard to this product manual and the regulations is necessary.

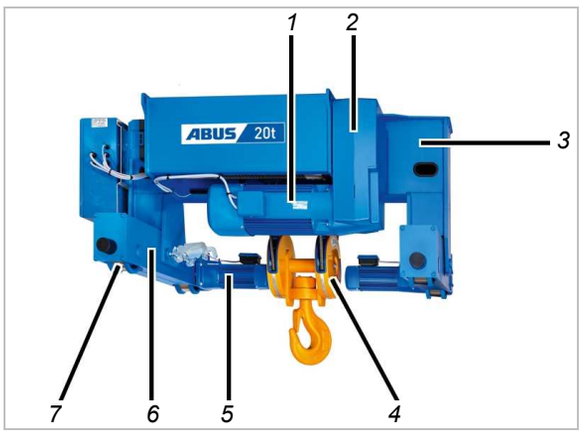
Which provision applies in individual cases will greatly depend on the crane's use and the national regulations. Review and observe all applicable and current regulations, including the occupational health and safety regulations. Refer also to the Declaration of Conformity or Declaration of Incorporation.

WARRANTY

- ABUS assumes no liability for damages that were caused by improper use, inadequately trained personnel, incorrect performance of work, nor for any changes, reconstruction or other modification of the crane or crane components which were not approved by ABUS.
- Any warranty claim becomes invalid if components were modified without consent, the crane or crane components were installed, used or serviced in any way other than that described in this product manual, or if parts other than original ABUS replacement parts were used.
- Safe operation of the crane or crane components is only guaranteed if original ABUS replacement parts are used.

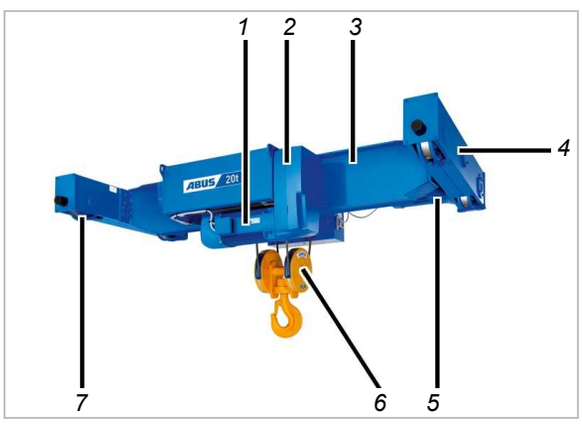
THE WIRE ROPE HOIST

DESCRIPTION OF THE DEVICE: DOUBLE-RAIL TROLLEY (D)



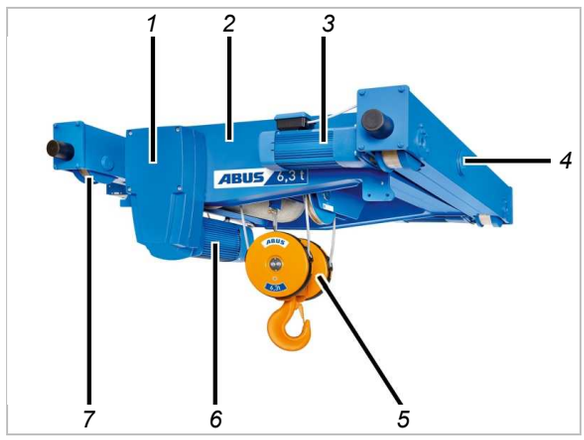
- 1: Hoist motor
- 2: Hoisting gear
- 3: Trolley frame
- 4: Bottom block and load hook
- 5: Trolley drive
- 6: End carriage
- 7: Hoist trolley

DESCRIPTION OF THE DEVICE: DOUBLE-RAIL TROLLEY (AD)



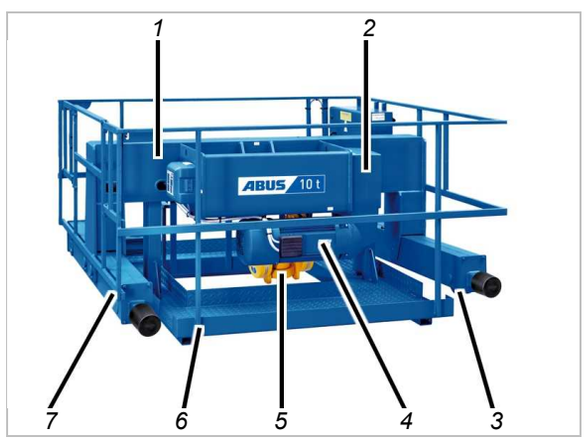
- 1: Hoist motor
- 2: Hoisting gear
- 3: Trolley frame
- 4: End carriage
- 5: Drive
- 6: Bottom block and load hook
- 7: Hoist trolley

DESCRIPTION OF THE DEVICE: DOUBLE-RAIL TROLLEY (DQA)



- 1: Hoisting gear
- 2: Trolley frame
- 3: Trolley drive
- 4: End carriage
- 5: Bottom block and load hook
- 6: Hoist motor
- 7: Hoist trolley

DESCRIPTION OF THE DEVICE: DOUBLE-RAIL TROLLEY (DB)



- 1: Trolley frame
- 2: Hoisting gear
- 3: Hoist trolley
- 4: Hoist motor
- 5: Bottom block and load hook
- 6: Walkway
- 7: End carriage

PERFORMANCE FEATURES

The double-rail trolley (D):

- The wire rope hoist has two drives for the trolley, giving it two driven wheels and two non-driven wheels.
- The trolley frame is placed onto the end carriages from above. One end carriage is permanently fixed to the trolley frame by two bolts and the other end carriage is articulated with one bolt in the trolley frame. This ensures that all four wheels always rest on the trolley track.
- Optionally, a walkway can be attached around the wire rope hoist so service and maintenance work can be performed easily and safely. (Double-rail trolley DB).
- The wire rope hoist covers a load capacity range of 1 t to 63 t.

The double-rail trolley lowered (DA):

- Unlike the double-rail trolley (D), the trolley frame for this version is placed between the end carriages, allowing an overall minimum height to be reached.
- One of the two end carriages is permanently fixed to the trolley frame with high-tensile bolts and the other end carriage is articulated with one bolt in the trolley frame. This ensures that all four wheels always rest on the trolley track.
- The wire rope hoist covers a load capacity range of 1 t to 40 t.

The double-rail trolley lowered crosswise (DQA):

- Unlike the double-rail trolley lowered (DA), on the double-rail trolley lowered crosswise (DQA), the hoist is rotated 90 leading the hook to travel in the crane travel direction.

  When the end carriages are equidistant, a longer cable drum can also be fitted, which enables a longer hook path.
- The wire rope hoist covers a load capacity range of 1 t to 40 t.

The AZP trolley drive (version):

- The drives are designed as pole-variable rotary current motors with electromagnetic disc brakes and gear units.
- The drives have a low and a high speed. The ratio of the two speeds is 1:4.
- The drive’s gear unit is a maintenance-free, two-stage or three-stage planetary gear.

The AZF trolley drive (version):

- The drives are designed as pole-variable rotary current motors with electromagnetic disc brakes and gear units.
- The drives have a low and a high speed. The ratio of the two speeds is 1:4.
- The drive's gear unit is a maintenance-free, flat gear unit with friction clutch.
- The drives of the size V420 and V500 have a visible output shaft.
- The drives are equipped with a fan blade on the motor shaft for ventilation. For very slow travel speeds, the drives can also be equipped with auxiliary fans.

The hoist drive with pole-variable hoist motor:

- The hoist drive is powered by a pole-variable 2/12-pole hoist motor with gear unit.
- The hoist motor features an electro-magnetically actuated spring-loaded brake.
- The hoist drive has a slow lifting speed and a fast lifting speed. The ratio of the two lifting speeds is 1:6.
- Using plug-in connections in the connector housing directly on the hoist drive, the hoist drive can be quickly separated from the rest of the crane’s electrics.
- The hoisting gear features a multi-stage, helical spur gear in an alloy housing. The gear wheels run in a sealed oil bath, which ensures they remain lubricated over their full service life.
- For further details on the pole-variable hoist drive, see the product manual “Pole-variable, modular ABUS hoist drive for wire rope hoists”.

The hoist:

- The wire rope of the hoist is a special wire rope with a very high breaking strength. This gives it a considerably longer service life than a normal steel rope.
- The cable guide guarantees flawless winding and unwinding of the wire rope on the cable drum. It is made of wearproof, low-friction plastic.
- The so-called “True vertical lift hoist” (optional) prevents hook travel and hook rotation by winding the wire rope from both sides of the cable drum. One half of the cable drum has a left-hand thread and the other a right-hand thread for this purpose. This allows for particularly precise movements because the load hook runs exactly vertically.

  The two cable guides are connected to the hoist drive GM 5000 made of metal and connected to the GM 7000 made of wearproof, low-friction plastic.

TECHNICAL DATA

Ambient conditions for operation:

Ambient temperature (for normal operation)	-10 °C to +40 °C
Ambient temperature (for reduced duty cycle)	+40 °C to +65 °C
Housing	IP 55
Insulation class	F

Noise emissions:

Size	Sound pressure level LP, m dB(A) at distance of 4 m	Sound pressure level LW, m dB(A)
GM 800	62	82
GM 1000	64	84
GM 2000	66	86
GM 3000	67	87
GM 5000	68	88
GM 6000	68	88
GM 7000	70	90

Table: Noise emissions based on DIN 45635, Part 61 following the substitution process with an acoustic power source

In the table, the sound pressure level LP is specified for a distance of 4 m from the wire rope hoist. To calculate the sound pressure level LW, the sound pressure level at any distance may be used.

TRANSPORTING THE WIRE ROPE HOIST ON A LORRY

ONLY WITH UNINSTALLED WIRE ROPE HOISTS

DANGER FROM FALLING COMPONENTS!
The wire rope hoist can fall from the lorry and kill or injure people if it is not correctly loaded and secured. Transport and securing of loads must be performed by trained persons.

Observe the following points, amongst others:

- ➔ Secure all loose parts and wiring.
- ➔ Secure the load according to regional regulations.
- ➔ Secure wire rope hoist to the loading area with suitable lifting tackle.
- ➔ Secure load hook so that it cannot slide back and forth during transport.
- The cable guide holds the wire rope firmly against the cable drum. This way no loops can form on the cable drum.

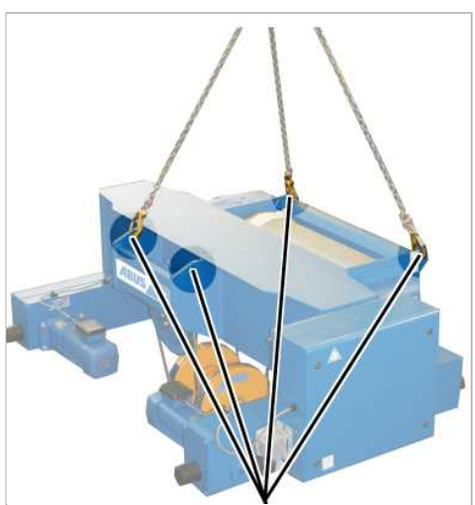
LOADING AND UNLOADING THE WIRE ROPE HOIST

ONLY WITH UNINSTALLED WIRE ROPE HOISTS

DANGER FROM FALLING COMPONENTS!
The wire rope hoist can fall and kill or injure people if it is not correctly attached and lifted. Attachment and loading should only be carried out by trained persons.

Observe the following points, amongst others:

LIFT WIRE ROPE HOIST WITH A CRANE



Suspension eyelets

- ➔ Use suitable lifting tackle with load hooks.
- ➔ Hang at least three load hooks in the suspension eyelets on the wire rope hoist.
- ➔ Observe the occupational health and safety requirements and raise the wire rope hoist.

SETTING DOWN THE WIRE ROPE HOIST

- ➔ Place wooden slats on the floor as protection.
- ➔ Place the wire rope hoist onto the wooden slats with the flange plate.

WIRE ROPE HOIST DISPOSAL

If the wire rope hoist needs to be disposed of:

- ➔ Dismantle the wire rope hoist to the extent possible.
- ➔ Observe local regulations concerning disposal and recycling.
- ➔ Dispose of the individual parts sorted by material in an environmentally sound manner.
  - – Dispose of oil and grease from the gear units as lubricants.
  - – Dispose of brake linings as multiple components (hazardous waste).
  - – Dispose of electronic components as scrap electronic parts.
  - – Dispose of cables, plug-in connections and pendant controls as scrap electronic parts.
  - – Dispose of the trolley frame, cable drum, deflection rollers, wire rope and other metal components as metal scrap.
  - – Dispose of the housing cover and housing base of the control as plastic waste.
  - – Parts of the wire rope hoist that have been repainted must be disposed of in accordance with the paint manufacturer’s specifications.



This product or electrical device may not be disposed of at the end of its service life with regular domestic waste.

INSTALLING AND CONNECTING

THIS SECTION APPLIES TO ANYONE WHO WORKS ON THE CRANE PRIOR TO ITS USE

The end user of the crane is responsible for the proper qualifications of the commissioning personnel.

DANGER TO PERSONS!
Persons can be injured if the crane is incorrectly put into operation.

If personnel other than that of the ABUS company are employed to perform the crane commissioning, it is the end user's responsibility to ensure that these persons are adequately qualified. Follow the procedures described here precisely.

Examples of qualified persons:

- Persons with comprehensive knowledge from specialist training in engineering and in the electrical systems of cranes.
- Persons with sufficient experience in the operation, installation and maintenance of cranes.
- Persons with comprehensive knowledge regarding the relevant technical rules, directives and safety regulations applicable in the respective country.
- Persons receiving regular training from ABUS.

ABUS assumes no liability for damage due to incorrectly performed commissioning work done by unqualified personnel.

ABUS recommends having the commissioning work performed by the ABUS assembly team.

CHECKING THE REQUIREMENTS

The following requirements must be met in order for the wire rope hoist to be installed:

Among other things, these points must be coordinated as part of the technical authorisation process:

- The welded box girder on which the wire rope hoist travels must meet the specifications of the enclosed dimension sheet and data sheet.
- The installation dimensions as well as the mains voltage and operating voltage must be correspondingly checked against the dimension sheet and the data sheet.
- The welded box girder that the wire rope hoist travels on must have a straight upper flange or parallel flanges.
- The trolley tracks that the wheels of the wire rope hoist travel on must be free of paint or other coatings.
- The trolley power supply must be executed as ABUPowerline energy chain or ABUS festoon cable system.

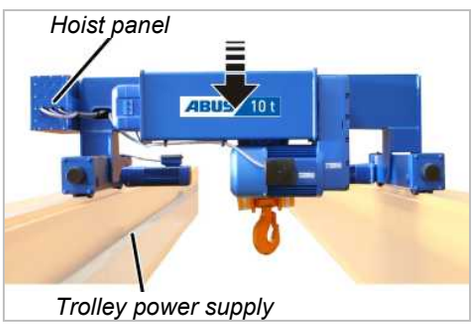
INSTALLING THE WIRE ROPE HOIST ON THE MAIN GIRDER

If the wire rope hoist has not been installed yet, continue here. Otherwise, skip this section.

LIFTING THE TROLLEY ONTO THE OVERHEAD TRAVELLING CRANE

ONLY WITH TROLLEY D, DB OR DA

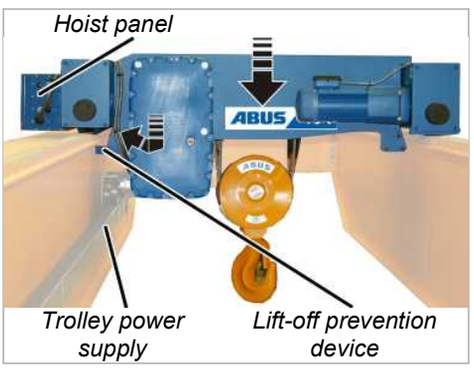
This section only applies if a trolley of construction D, DB or DA is to be placed on an overhead travelling crane.



- ➔ Rotate the trolley as in the diagram.
- ➔ Place the trolley with the wheels onto the trolley track.
- ➔ Only for guide rollers: do not fit the trolley with the guide rollers. Otherwise, they will be damaged.
- ➔ Check whether all wheels come into contact with the trolley track.

ONLY WITH TROLLEY DQA

This section only applies if a trolley of construction DQA is to be placed on an overhead travelling crane.



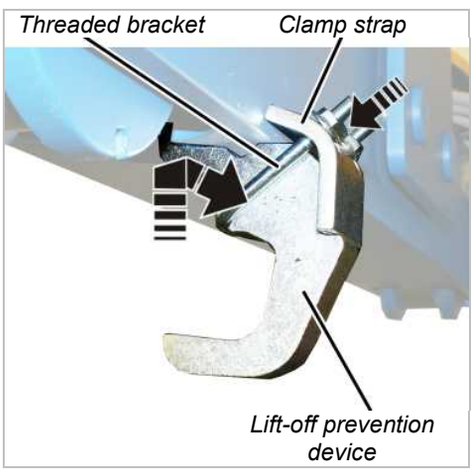
- ➔ Turn the trolley so that the hoist panel is facing the trolley power supply (energy chain or festoon cable system).
- ➔ Tilt the trolley slightly and with the lift-off prevention device move it underneath the upper flange of the main girder.
- ➔ Place the trolley with the wheels onto the trolley track.
- ➔ Only for guide rollers: do not fit the trolley with the guide rollers. Otherwise, they will be damaged.
- ➔ Check whether all wheels come into contact with the trolley track.

INSTALLING THE LIFT-OFF PREVENTION DEVICE

The lift-off prevention device is installed on the trolley. It consists of a load hook, which runs under the upper flange of the main girder. The trolley is thereby not able to fall from the main girder even in the event of accidents or similar incidents.

ONLY WITH TROLLEY D, DB OR DA

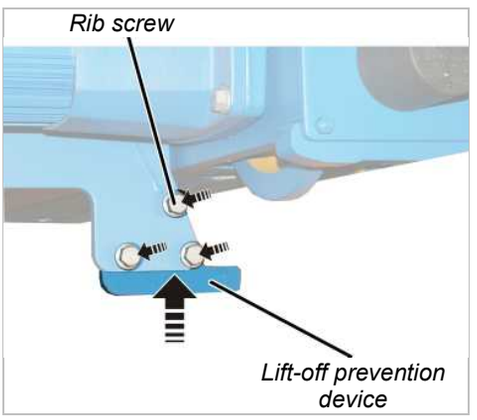
This section only applies if a trolley of construction D, DB or DA is to be placed on an overhead travelling crane.



- ➔ Connect the clamp strap and threaded bracket to the lift-off prevention device.
- ➔ Fasten the lift-off prevention device to the non-powered wheel from below in the opening.
- ➔ Screw the clamp strap tightly with M8 rib nuts (2x). 25 Nm

ONLY WITH TROLLEY DQA

This section only applies if a trolley of construction DQA is to be placed on an overhead travelling crane.



- ➔ Fasten the lift-off prevention device to the trolley frame.
- ➔ Screw on the lift-off protection device tightly with rib screws (3x).

Rib screw	Tightening torque
M12x30	115 Nm
M16x35	210 Nm

INSTALLING THE TROLLEY POWER SUPPLY

If the trolley towing arm has not been installed yet, continue here. Otherwise, skip this section.

The trolley towing arm is installed above the hoist panel. It pulls the electrical cables for the trolley power supply (festoon cable system or energy chain) and the cross-type limit switch of the trolley travel limit switch back and forth parallel to the trolley.

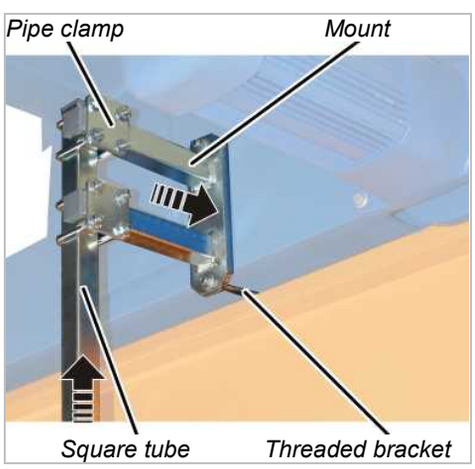
The trolley towing arm must be installed differently, depending on the main girder (I-beam or box girder) and the type of power supply (festoon cable system or energy chain).

MOUNTING THE TROLLEY TOWING ARM

ONLY WITH TROLLEY D, DA OR DB AND ENERGY CHAIN

This section only applies if a trolley of construction D, DA, or DB is connected to an overhead travelling crane with an energy chain.

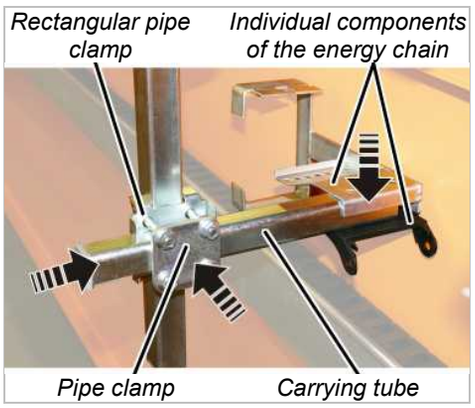
The component parts of the trolley towing arm are enclosed loose and must be assembled.



➔ Screw the mount to the end carriage using the threaded bracket and rib nuts (2x).

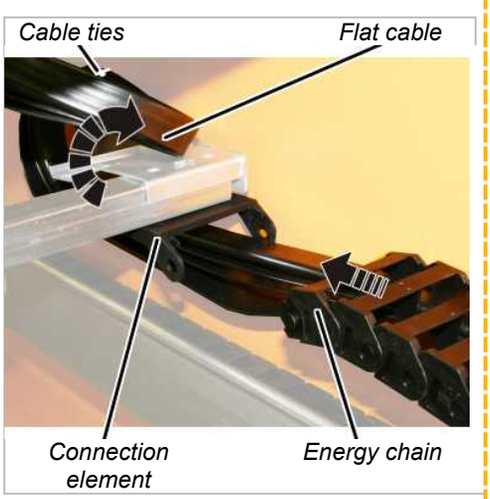
Threaded bracket	Tightening torque
M8	25 Nm
M10	50 Nm
M12	75 Nm

➔ Screw the square tube to the mount using two pipe clamps. 15 Nm



➔ Screw the carrying tube with pipe clamp and rectangular pipe clamp onto the vertical square tube (trolley towing arm). 15 Nm

➔ Fasten the individual components of the energy chain (deflection bracket and connection element) with bolts and SL safety clip to the carrying tube.



➔ Lead through the flat cable from the energy chain and guide in a curve around the deflection bracket.

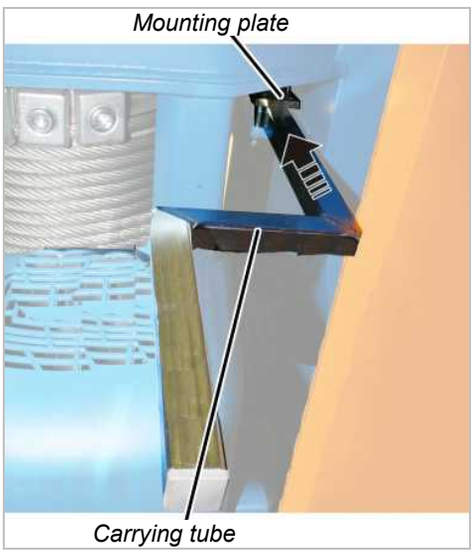
➔ Fasten the flat cable with cable ties to the upper section of the deflection bracket.

➔ Insert the energy chain into the connection element and click into place.

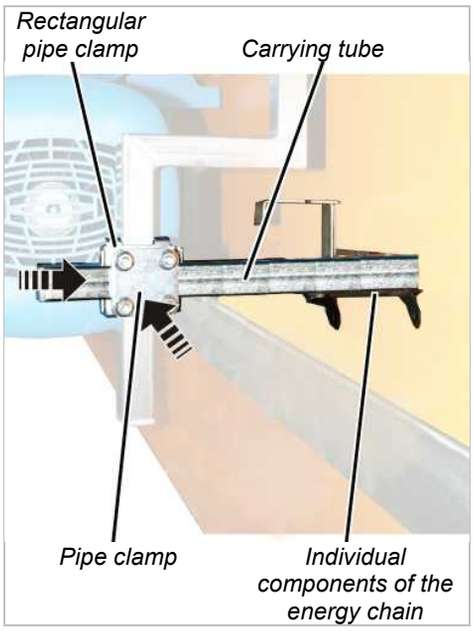
ONLY FOR TROLLEY DQA AND ENERGY CHAIN

This section only applies if a trolley of construction DQA is connected to an overhead travelling crane with an energy chain.

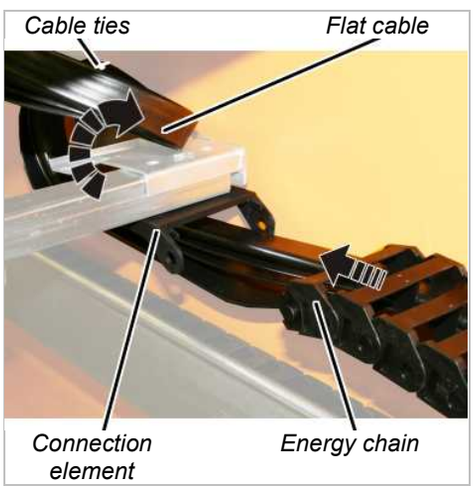
The component parts of the trolley towing arm are enclosed loose and must be assembled.



- Insert the carrying tube from below into the trolley frame.
- Screw on the mounting plate with a hexagonal head screw M8 and hexagonal nut.
- Insert the carrying tube between the mounting plate and the trolley frame.
- Tightly screw on the hexagonal nuts. 25 Nm.



- Screw the horizontal carrying tube with pipe clamp and rectangular pipe clamp onto the vertical carrying tube (trolley towing arm). 15 Nm
- Fasten the individual components of the energy chain (deflection bracket and connection element) with bolts and SL safety clip to the square tube.

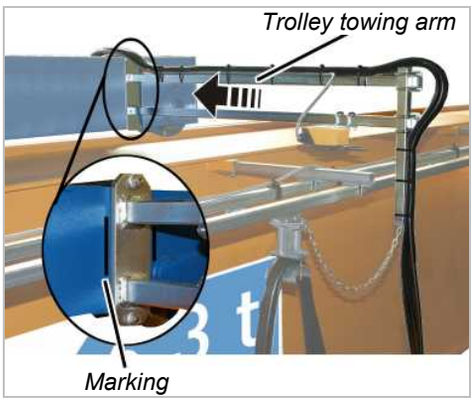


- Lead through the flat cable from the energy chain and guide in a curve around the deflection bracket.
- Fasten the flat cable with cable ties to the upper section of the deflection bracket.
- Insert the energy chain into the connection element and click into place.

ONLY FOR AN OVERHEAD TRAVELLING CRANE WITH A FESTOON CABLE SYSTEM

The trolley towing arm is fastened pre-installed to the C-rail system and must be fitted to the trolley.

For trolleys with trolley travel limit switches: the trolley travel limit switch is already completely installed.



➔ Attach the installed trolley towing arm to the end carriages.

In most cases, the end carriage has a marking at which the trolley towing arm is to be attached. The switching point of the trolley travel limit switch is already adjusted for this position and the cable lengths correspondingly adapted.

➔ Screw the trolley towing arm tightly with the threaded bracket and rib nuts (2x).

Threaded bracket	Tightening torque
M8	25 Nm
M10	50 Nm
M12	75 Nm

INSTALLING THE CROSS-TYPE LIMIT SWITCH ON THE TROLLEY TOWING ARM

ONLY WITH CROSS-TYPE LIMIT SWITCH

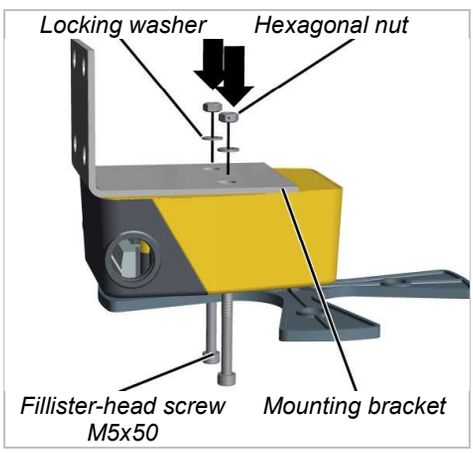
This section only applies if the wire rope hoist has a cross-type limit switch in the trolley travel direction.

Where and for which travel limit the cross-type limit switch is designed is specified on the wiring diagram.

INSTALLING THE CROSS-TYPE LIMIT SWITCH

DANGER DUE TO MALFUNCTION!
If the cross-type limit switch is screwed on too tightly, the parts on the inside can become jammed and no longer function properly.

The tightening torque of 3 Nm must be strictly observed.

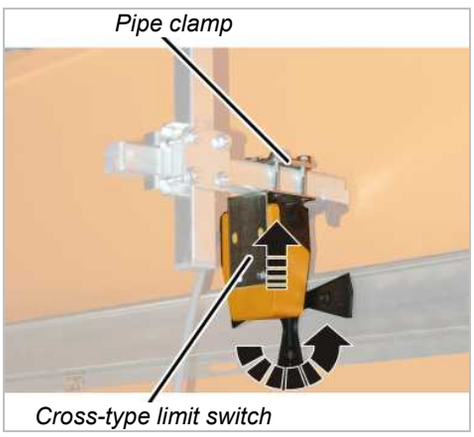


➔ Screw the mounting bracket onto the cross-type limit switch with the fillister-head screws M5x50 (2x).

➔ Screw on the mounting bracket using locking washers and M5 hexagonal nuts (2x). 3 Nm.

INSTALLING THE CROSS-TYPE LIMIT SWITCH

The cross-type limit switch is normally installed on the trolley towing arm.

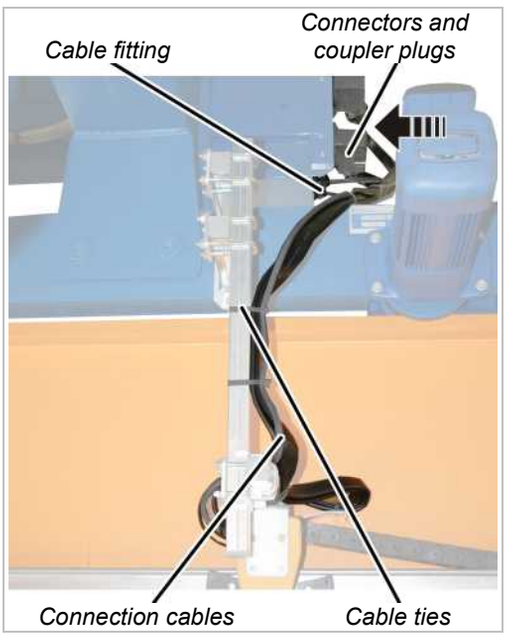


- Turn the cross-type limit switch to position 0.
- The position of the cross-type limit switch is marked with an arrow that can be rotated further, depending on the switching state.
- Move the cross-type limit switch onto the square tube in such a way that the switching lug activates the cross-type limit switch.
- Screw in the cross-type limit switch with pipe clamps (2x) tightly. 15 Nm.

CONNECTING THE CROSS-TYPE LIMIT SWITCH

- Route the connection cable (round cable or flat cable) from the cross-type limit switch to the panel.
- Route the connection cable with cable fitting into the panel.
- If necessary, connect the connection cable to a coupler plug or connector.
- Fasten connection cable with cable ties, cable grips and adhesive clips.

CONNECTING THE TROLLEY POWER SUPPLY



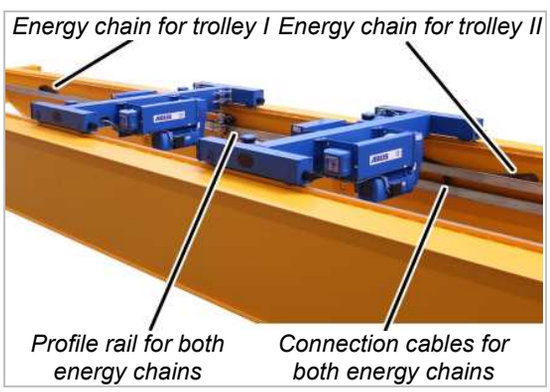
- Lay the connection cable leading from the energy chain along the square tube to the hoist panel.
- Fasten the connection cables neatly separated with cable ties.
- Plug the connector and coupler plugs into the hoist panel, or guide the connection cables through cable fittings and into the hoist panel.

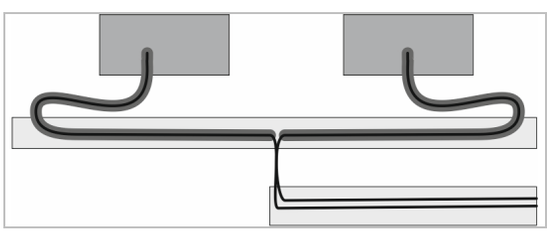
INSTALLING THE SECOND TROLLEY

ONLY FOR CRANE WITH TWO TROLLEYS

Notes on installing the second trolley:

- Where and in which direction the second trolley is installed is specified in the planning documents.
- The trolleys are labelled “Trolley I” and “Trolley II” for clear assignment.
- Installation of the second trolley and trolley power supply is identical to the first trolley. The following diagram illustrates how the trolley power supply is routed.

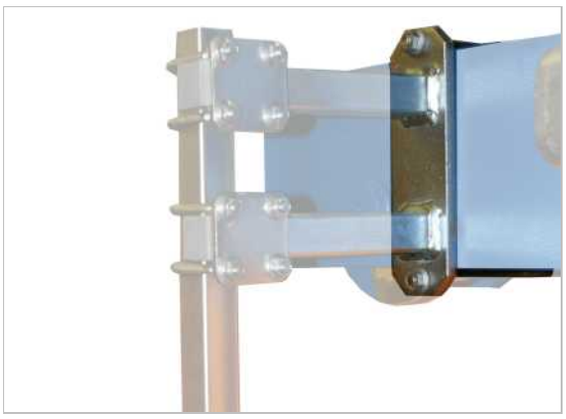




- A crane with two trolleys has two profile rails. The upper one for both energy chains, the lower one for cable guidance.
- Both trolleys have their own separate energy chain for trolley power supply. One runs from the trolley to the left, the other to the right.
- Both energy chains start in the upper profile rail exactly in the middle on the main girder. One runs clockwise from there through the profile rail to the trolley, the other anti-clockwise.
- Both energy chains merge into the connection cables in the middle of the main girder. The connection cables run in the lower profile rail to the panel.

OVERVIEW OF TIGHTENING TORQUES

MOUNTINGS ON THE END CARRIAGE AND PIPE CLAMPS



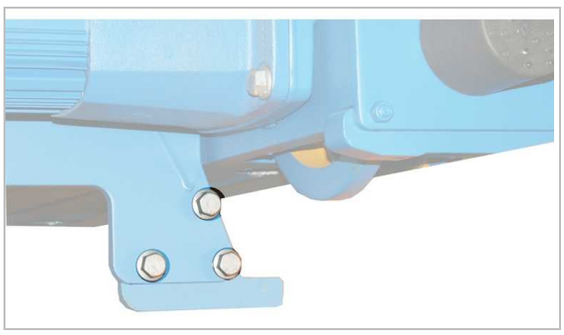
Threaded bracket	Type and size	Number	Tightening torque
M8	Rib nut M8	2x	25 Nm
M10	Rib nut M10	2x	50 Nm
M12	Rib nut M12	2x	75 Nm

LIFT-OFF PREVENTION DEVICE WITH TROLLEY D, DB OR DA



- Rib nut M8
- 2x per lift-off prevention device
- 15 Nm

LIFT-OFF PREVENTION DEVICE TROLLEY DQA



Type and size	Number	Tightening torque
Rib screw M12x30	3x	115 Nm
Rib screw M16x35	3x	210 Nm

TROLLEY TOWING ARM TROLLEY DQA



- Hexagon head screw M8 with hexagonal nut
- 6x
- 25 Nm

INSPECTION

THIS SECTION APPLIES TO ANYONE WHO INSPECTS AND ACCEPTS THE CRANE IN ACCORDANCE WITH OCCUPATIONAL HEALTH AND SAFETY REQUIREMENTS

AT THE START

TEST INTERVALS

The regular inspection is performed at least once annually.

Under certain conditions, more frequent regular inspections may be necessary. Reasons include:

- Frequent operation at the load capacity
- Working in multiple shifts
- Frequent use
- Dusty or chemically aggressive environment

The end user is responsible for checking the requirements and determining the test intervals. ABUS will gladly assist you if questions arise.

REQUIREMENTS FOR THE EXAMINER

The end user of the crane is responsible for the proper qualifications of the examiner.



DANGER TO PERSONS!
Persons can be injured if the test is performed incorrectly.

If personnel other than that of the ABUS company are employed to perform the test, it is the end user's responsibility to ensure that these persons are adequately qualified.

Examples of qualified persons:

- Persons with comprehensive knowledge from specialist training in engineering and in the electrical systems of cranes.
- Persons with sufficient experience in the operation, installation and maintenance of cranes.
- Persons with comprehensive knowledge regarding the relevant technical rules, directives and safety regulations applicable in the respective country.
- Persons receiving regular training from ABUS.

SCOPE OF THE INSPECTION

The person qualified to inspect the wire rope hoist is responsible for the scope and type of inspection.

OVERVIEW: INSPECTING THE WIRE ROPE HOIST

In addition to the points described here, all points outlined in the other supplied product manuals must also be checked.

The decision as to whether the wire rope hoist is in a flawless condition is the sole responsibility of the examiner. If any defects are found, they must be repaired. The examiner decides whether the wire rope hoist needs to undergo inspection again afterwards.

If locally applicable regulations specify further tests, these are likewise to be carried out.

Additionally, at least the following points must be checked:

- ➔ Inspect the wire rope. See page 25.
- ➔ Check the load hook and bottom block. See page 26.
- ➔ Check the edge protection on the bottom block (visual inspection).
- ➔ Inspect the fixed point crosshead. See page 29.
- ➔ Inspect the deflection roller crosshead. See page 30.
- ➔ Inspect the cable guide. See page 26.
- ➔ Check the fixed point. See page 33.
- ➔ Inspect the end clamps on the cable drum. The end clamps must be installed and must not exhibit any heavy damage.
- ➔ Check the wheel diameters. See page 35.
- ➔ Check the wheel flange wear of the wheels. See page 36.
- ➔ Check the lift-off prevention device. See page 37.
- ➔ Check the condition of the safety buffer. See page 37.
- ➔ Check the brake on the hoist drive or travel drive.

  See product manual “Pole-variable, modular ABUS hoist drive for wire rope hoists”.
- ➔ Check the braking distance of the trolley drive.

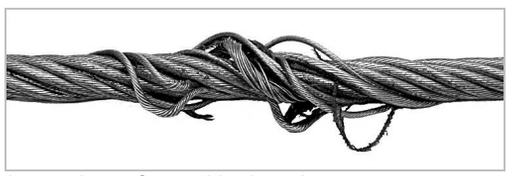
INSPECTING THE WIRE ROPE

➔ Inspect the entire wire rope for damage. The damage shown here or any similar damage should not be detectable on the wire rope.

Examples of damage:



The joining of the wire rope has opened. The inner strands of the wire rope are visible.



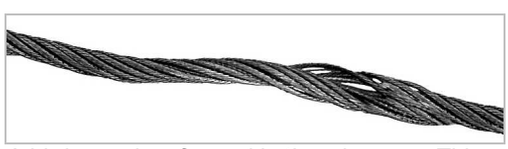
Loops have formed in the wire rope.



There is a kink in the wire rope. This can result, for example, from violent external forces being applied to the wire rope.



The wire rope is flattened at one point. This can result, for example, from the wire rope being pinched.



A bird cage has formed in the wire rope. This can result, for example, from violent untwisting of the wire rope.



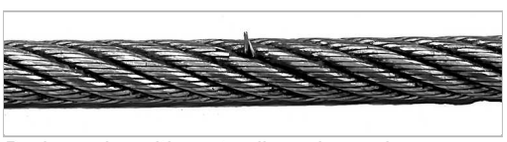
The wire rope has a corkscrew-type deformation.

➔ Check the entire wire rope for broken wires. The wire rope must not exhibit more broken wires in a length of 6 times its diameter or 30 times its diameter than is specified for these lengths on the ABUS manufacturer's certificate for wire ropes in the test log book.

Examples of broken wires:



Multiple broken wires. Broken wires are signs of normal wear of the wire rope. They occur when the wire rope is bent on the reels under load.



Broken wire with protruding wire end.

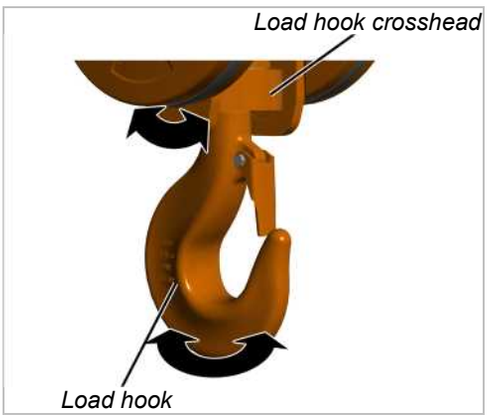
➔ If damage such as that pictured or similar damage is detected on the wire rope, replace the wire rope with a new one.

➔ If the wire rope has broken wires, but not exceeding the permissible number, shorten the test interval until the next regular inspection.

➔ If the wire rope has more broken wires than permitted, replace the wire rope with a new one.

CHECKING THE LOAD HOOK AND BOTTOM BLOCK

CHECKING FOR EASE OF MOVEMENT



➔ Rotate the load hook to the left and right.

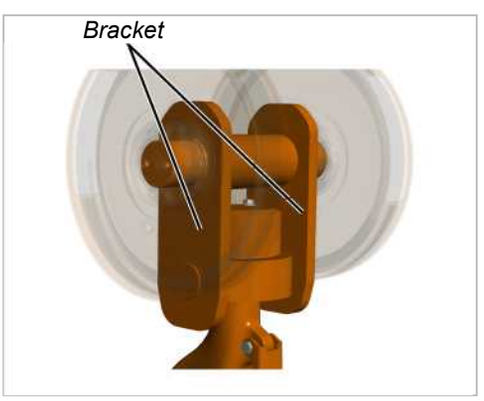
The load hook must be easy to move and freely rotatable.

➔ Slew the load hook backwards and forwards at the load hook crosshead.

The load hook crosshead must be freely slewable.

➔ If the load hook is not easy to move or freely rotatable or slewable, it must be repaired.

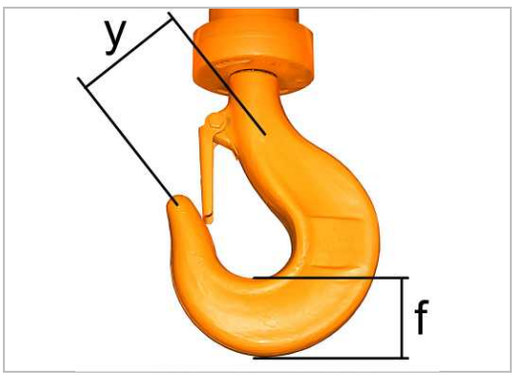
INSPECTING THE BRACKET ON THE BOTTOM BLOCK



➔ Inspect the brackets. They must not exhibit any cracking, deformation or other signs of damage.

➔ If the brackets are deformed or damaged in any other way, they must be replaced.

MEASURING DEFORMATION AND WEAR OF THE LOAD HOOK



➔ Measure the distance ‘y’ between the two stamped markings.

➔ Read the reference distance for ‘y’ from the test log book or, if applicable, from the load hook itself.

The measured distance ‘y’ must not be greater than 1.1 times the reference distance ‘y’.

With ramshorn hooks, the distances ‘y1’ and ‘y2’ from the respective engraved marking on the tip of the load hook to the marking on the hook shank must each be measured and compared separately.

➔ Measure the base height ‘f’ of the load hook.

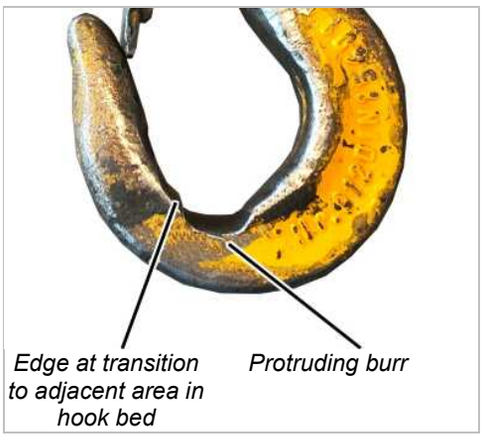
➔ Compare the measured value with the table.

The measured base height ‘f’ must not be smaller than given in the table. The table specifies the minimum base height.

Size of load hook	Type of load hook	Minimum base height ‘f’ [mm]
1	Normal hook	38
1.6	Normal hook	45.6
2.5	Ramshorn hook	47.5
2.5	Normal hook	55.1
4	Ramshorn hook	57
4	Normal hook	63.7
6	Ramshorn hook	71.3
6	Normal hook	80.8
8	Ramshorn hook	80.8
8	Normal hook	90.3
10	Ramshorn hook	90.3
10	Normal hook	100.7
12	Ramshorn hook	100.7
12	Normal hook	112.1
16	Ramshorn hook	112.1
16	Normal hook	125.4
20	Ramshorn hook	125.4
20	Normal hook	142.5
25	Ramshorn hook	142.5
25	Normal hook	161.5
32	Ramshorn hook	161.5
32	Normal hook	180.5
40	Ramshorn hook	180.5
40	Normal hook	201.4

➔ If the load hook has been widened to a greater degree than is allowed or if the base height is below the permissible limit, replace the load hook.

Do not carry out any repair welding (e.g. build-up welding) on the load hook or on the load hook nut.



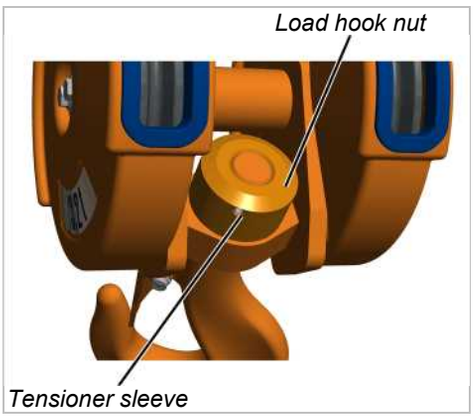
- Check the wear surfaces in the hook bed. They must merge gradually with the surrounding surfaces. They must not have any sharp score marks and edges or other surface faults.
- Inspect the edges. The side edges on the load hook should not have any exterior protruding burrs or similar surface faults.
- If the wear surfaces or side edges do have sharp score marks or edges, these may be smoothed (e.g. filed). The limit values for base height given above must be observed even after smoothing.

INSPECTING THE LOAD HOOK SURFACE

- Inspect the load hook surface. It must not have any faults, cracks or corrosion.
- If the surface of the load hook is to any extent not OK, disassemble the load hook and inspect the surface of the hook shank. It must not have any faults, cracks or corrosion.

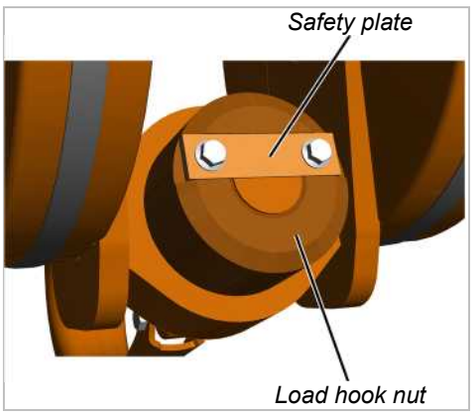
INSPECTING THE ANTI-ROTATION DEVICE OF THE LOAD HOOK NUT

With load hooks with anti-rotation device through tensioner sleeve:



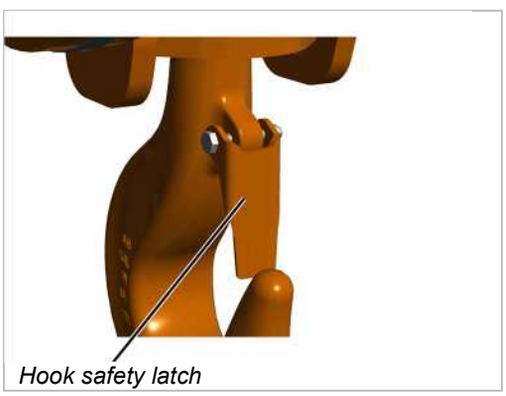
- Tilt the load hook until the anti-rotation device is clearly visible.
- Inspect the anti-rotation device. The tensioner sleeve must be present, not damaged or broken, and it must sit securely.
- If the tensioner sleeve is missing or damaged, it must be replaced.

For load hooks with anti-rotation device by means of safety plate:



- Tilt the load hook until the anti-rotation device is clearly visible.
- Inspect the anti-rotation device. The safety plate must be present, undamaged and sit securely.
- If the safety plate is missing or damaged, it must be replaced.

INSPECTING THE HOOK SAFETY LATCH

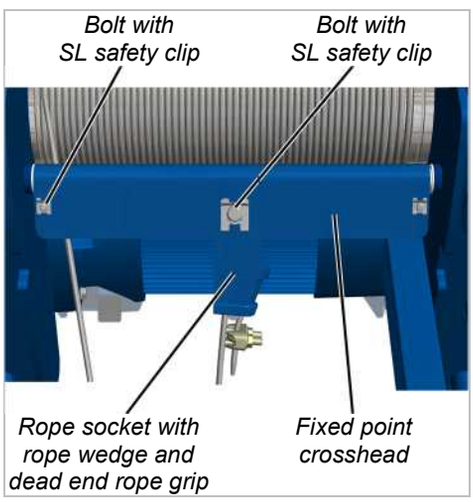


- Inspect the hook safety latch. It must be present, functional and easy to move, and must not be deformed or damaged in any other way.
- If the hook safety latch is missing, damaged or not working properly, it must be replaced.

INSPECTING THE FIXED POINT CROSSHEAD

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 800 TO GM 3000 WITH REEVING 2/1 OR 4/1

On a wire rope hoist with hoist drive GM 800 to GM 3000, the fixed point crosshead is fitted on bearing bushes using bolts and SL safety clips.



- Inspect the fastening. The fixed point crosshead must be securely fastened and all bolts must be secured with SL safety clips.
- Inspect the fixed point crosshead.

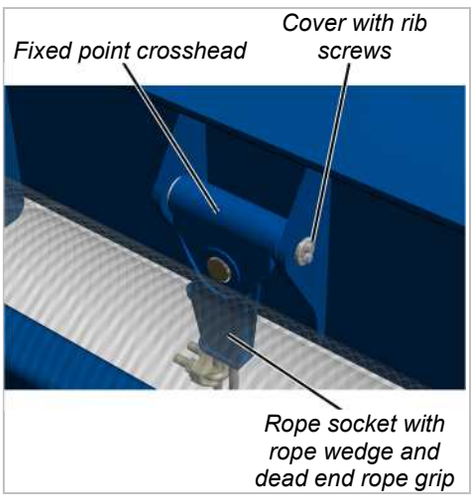
  The drilled holes and the lower opening must not exhibit any cracking or other damage.
- Check bearing bushes for damage. The bearing bushes must not exhibit any cracking or other damage.
- If components are missing or damaged: replace the components.

  See “Exchanging the fixed point crosshead with rope socket and rope wedge”, page 71.

INSPECTING THE DEFLECTION ROLLER CROSSHEAD

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 5000 TO GM 7000 WITH REEVING 2/1, 4/1 OR 6/1

On a wire rope hoist with hoist drive GM 5000 to GM 7000, the fixed point crosshead is fitted to side panels on the trolley frame on bearing bushings using a laterally inserted bolt with cover.

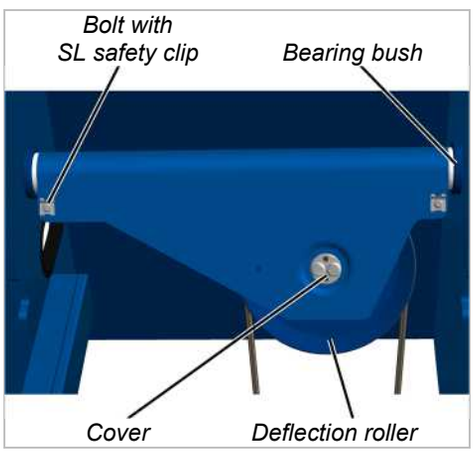


- Inspect the fastening. The fixed point crosshead must be securely fastened and the bolt must be secured with a cover.
- Inspect the fixed point crosshead.

  The drilled holes and the lower opening must not exhibit any cracking or other damage.
- Check bearing bushes for damage. The bearing bushes must not exhibit any cracking or other damage.
- If components are missing or damaged: replace the components.
- See “Exchanging the fixed point crosshead with rope socket and rope wedge”, page 71.

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 800 TO GM 6000 WITH REEVING 4/1

On a wire rope hoist with hoist drive GM 800 to GM 6000, the deflection roller crosshead is fitted on bearing bushes with bolts and SL safety clips.



- Inspect the deflection roller crosshead. The drilled holes must not exhibit any cracking or other damage.
- Inspect the deflection roller.
  It may not exhibit any deep grooves or heavy wear.

  If necessary, the deflection roller can be carefully inspected using a special template. Please contact ABUS Service for this. See page 114.
- Bolts and SL safety clips must be in place and secured.
- Check bearing bushes for damage. The bearing bushes must not exhibit any cracking or other damage.
- If components are missing or damaged: replace the components.

  See “Replacing deflection roller crosshead for reeving 4/1 or 6/1”, page 81.

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 6000 AND REEVING 6/1

On a wire rope hoist with hoist drive GM 6000 and a reeving 6/1, the two deflection rollers are fitted in the trolley frame.



- Inspect the deflection roller crosshead.
  The drilled holes must not exhibit any cracking or other damage.
- Inspect the deflection roller.
  It may not exhibit any deep grooves or heavy wear.
- Check the deflection roller crosshead suspension.

  The supporting structure must be sufficiently lubricated so the deflection roller crosshead can move back and forth.
- Bolts and SL safety clips must be in place and secured.
- If components are missing or damaged: replace the components.

  See “Replacing deflection roller crosshead for reeving 4/1 or 6/1”, page 81.

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 7000 AND REEVING 4/1

On the wire rope hoist with hoist drive GM 7000 and a reeving 4/1, the deflection roller crosshead is attached on the outside of the trolley frame.



- Inspect the deflection roller crosshead.
  The drilled holes must not exhibit any cracking or other damage.
- Inspect the deflection roller.
  It may not exhibit any deep grooves or heavy wear.
- Check the deflection roller crosshead suspension.

  The supporting structure must be sufficiently lubricated so the deflection roller crosshead can move back and forth. See page 110.
- Bolts and SL safety clips must be in place and secured.
- If components are missing or damaged: replace the components.

  See “Replacing deflection roller crosshead for reeving 4/1 or 6/1”, page 81.

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 7000 AND REEVING 6/1

On the wire rope hoist with hoist drive GM 7000 and a reeving 6/1, the deflection roller crossheads are fitted in the trolley frame.



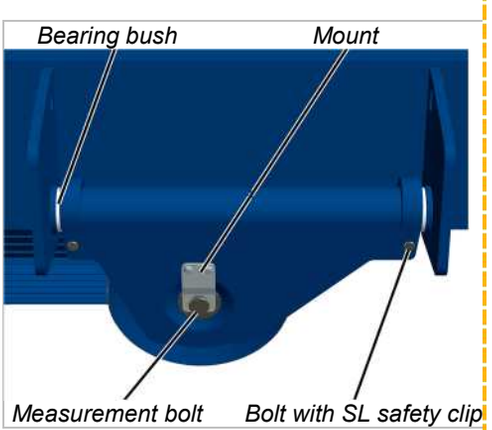
- Inspect the deflection roller crosshead. The drilled holes must not exhibit any cracking or other damage.
- Inspect the deflection roller. It may not exhibit any deep grooves or heavy wear.
- Check the deflection roller crosshead suspension.

  The supporting structure must be sufficiently lubricated so the deflection roller crosshead can move back and forth. See page 110.
- If components are missing or damaged: replace the components.

  See “Replacing deflection roller crosshead for reeving 4/1 or 6/1”, page 81.

ONLY WITH WIRE ROPE HOIST WITH REEVING 4/2 AND 8/2

This section only applies for a reeving of 4/2 or 8/2. In this case, the measurement bolt is installed in the deflection roller crosshead.



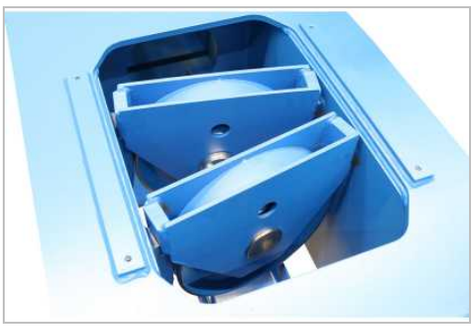
- Inspect the deflection roller crosshead. The drilled holes must not exhibit any cracking or other damage.
- Inspect the deflection roller. It may not exhibit any deep grooves or heavy wear.

  If necessary, the deflection roller can be carefully inspected using a special template. Please contact ABUS Service for this. See page 114.
- Bolts and SL safety clips must be in place and secured.
- Check bearing bushes for damage. The bearing bushes must not exhibit any cracking or other damage.
- If components are missing or damaged: replace the components.

  See “Replacing deflection roller crosshead for reeving 4/2 or 8/2”, page 92.

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 7000 AND REEVING 8/2

On a wire rope hoist with a reeving 8/2, two deflection rollers are also installed in the trolley frame.



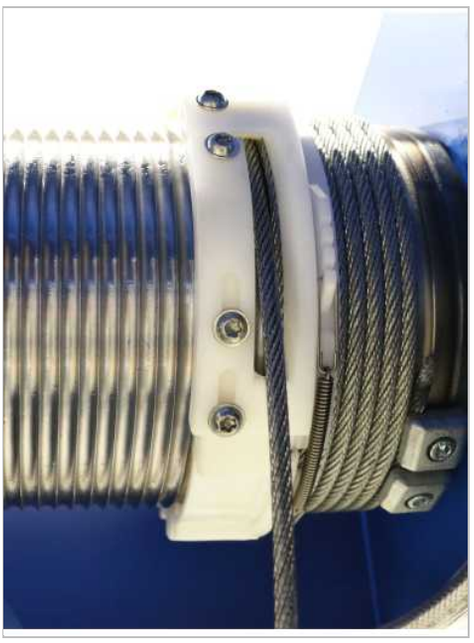
- Inspect the deflection roller crosshead. The drilled holes must not exhibit any cracking or other damage.
- Inspect the deflection roller. It may not exhibit any deep grooves or heavy wear.
- Check the deflection roller crosshead suspension.

  The supporting structure must be sufficiently lubricated so the deflection roller crosshead can move back and forth.
- If components are missing or damaged: replace the components.

  See “Replacing deflection roller crosshead for reeving 4/2 or 8/2”, page 92.

INSPECTING THE CABLE GUIDE

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 800, GM 1000, GM 2000 AND GM 3000



- Inspect the cable guide. The plastic parts must not be cracked, deformed or have abraded areas caused by the wire rope.
- Check the cable guide for ease of movement.
- Check the lubrication of the cable guide.
- If the cable guide is damaged: replace the cable guide. See “Replacing wire rope with reeving 2/1, 4/1 or 6/1”, page 39.

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 5000, GM 6000 AND GM 7000



- Inspect the cable guide. The plastic parts must not be cracked, deformed or have abraded areas caused by the wire rope.
- Check the cable guide for ease of movement.
- Check the lubrication of the cable guide.
- If the cable guide is damaged: replace the cable guide. See “Replacing wire rope with reeving 2/1, 4/1 or 6/1”, page 39.

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 5000 AND TRUE VERTICAL LIFT HOIST



- Inspect the cable guide. The plastic parts must not be cracked, deformed or have abraded areas caused by the wire rope.
- Check the cable guide for ease of movement.
- Check the lubrication of the cable guide.
- If the cable guide is damaged: replace the cable guide. See “Replacing wire rope with reeving 4/2 or 8/2”, page 57.

CHECKING THE FIXED POINT

ONLY WITH WIRE ROPE HOIST WITH REEVING 2/1, 4/1 OR 6/1

CHECKING THE ROPE WEDGE



- Measure the rope wedge protrusion (lower edge of rope wedge to lower edge of rope socket).
- Measure the rope clamp spacing (upper edge of dead end rope grip to lower edge of rope wedge).
- The measured values must not be higher than the values in the table.

Size	Rope wedge protrusion when the wire rope hoist is new	Maximum rope wedge protrusion	Minimum rope wedge protrusion
GM 800	19 mm	19 mm	18 mm
GM 1000	4 mm	4 mm	3 mm
GM 2000	2 mm	2 mm	0 mm
GM 3000	18 mm	18 mm	16 mm
GM 5000	22 mm	22 mm	19 mm
GM 6000	31 mm	31 mm	28 mm
GM 7000	28 mm	28 mm	26 mm

- If the rope wedge protrudes too far out of the rope socket, or if the dead end rope grip is too close to the rope wedge: replace rope socket and rope wedge. See page 71.

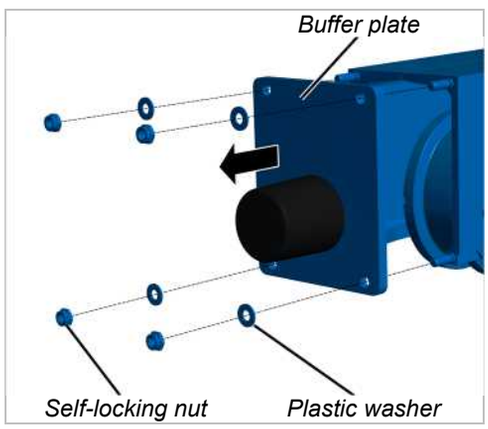
Note

The minimum rope clamp spacing specified is always exactly equal to the wire rope diameter of the wire rope hoist.

CHECKING THE DIAMETER OF THE WHEELS

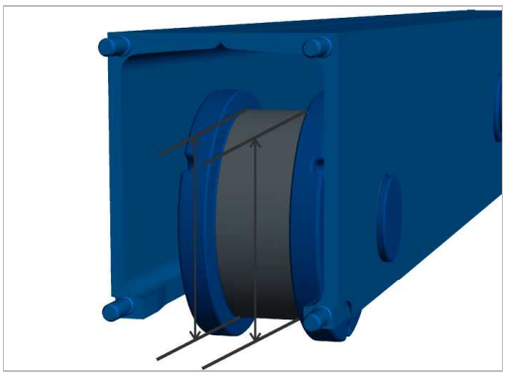
The running surfaces of the wheels become worn during operation. This reduces the diameter of the wheels.

EXPOSING THE WHEEL



- Unscrew the self-locking nuts (4x).
- From size 160: remove the plastic washers (4x).
- Remove the buffer plate.

MEASURING THE DIAMETER



➔ Measure the diameter of the wheel at multiple points over the width of the wheel.

➔ The diameter may not be smaller than is specified in the table.

If the diameter is smaller, the wheel must be replaced. See page 96.

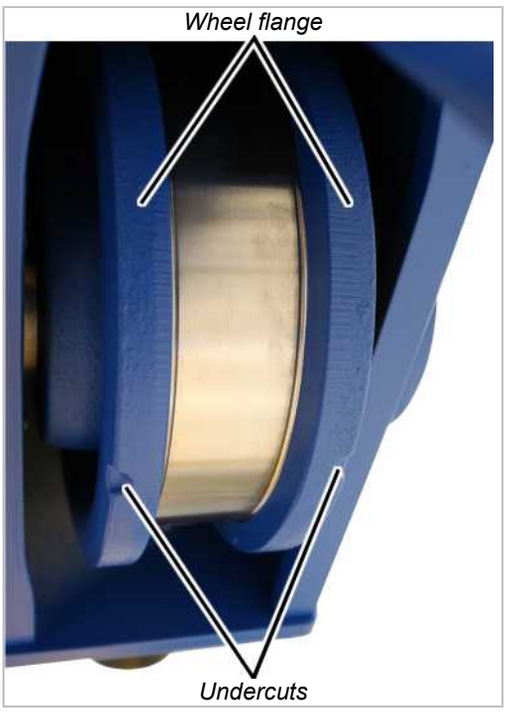
Wheel	Maximum permissible wear	Minimum diameter
130	-3 mm	97 mm
160	-3 mm	157 mm
200	-4 mm	196 mm
280	-5 mm	275 mm
350	-5 mm	345 mm
420	-5 mm	415 mm
500	-5 mm	495 mm

INSPECTING THE WHEEL FLANGE WEAR

The undercuts in the wheel flanges are a sign of wear on the wheel flange.

Note

First check the clearance between wheel flange and rail. If the clearance between wheel flange and rail is too large, the wheel flanges must be inspected.



➔ Inspect the wheel flanges of the wheels: the wheel flanges must not be worn so much that a continuous crack can be seen in the wheel flange (rather than undercuts). If they are, replace the wheels. See page 96.

CHECKING THE LIFT-OFF PREVENTION DEVICE

ONLY WITH WIRE ROPE HOIST DA, DA AND DB



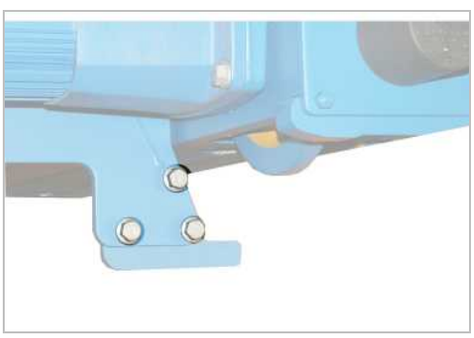
➔ Check the lift-off prevention device for wear and damage.

If the lift-off prevention device is damaged, it must be replaced. In this case, check the wire rope hoist for further damage because the wire rope hoist may have bounced on the trolley track.

➔ Test the firm seating of the lift-off prevention device with a light hammer tap.

If the rib nuts on the lift-off prevention device are loose, they must be tightened.

ONLY WITH WIRE ROPE HOIST DQA



➔ Check the lift-off prevention device for wear and damage.

If the lift-off prevention device is damaged, it must be replaced. In this case, check the wire rope hoist for further damage because the wire rope hoist may have bounced on the trolley track.

➔ Test the firm seating of the lift-off prevention device with a light hammer tap.

If the rib screws on the lift-off prevention device are loose, these must be tightened.

INSPECTING THE CONDITION OF THE SAFETY BUFFER



➔ Check all the safety buffers. The safety buffers must not be porous, pinched, bent or otherwise damaged.

➔ Check the material of the safety buffers all the way round. The plastic must not be cracked anywhere on the surface.

➔ Press the safety buffers with your finger. The safety buffers must be able to be pressed in slightly and be formable.

➔ If one of the safety buffers is damaged in any way, replace all the safety buffers on the wire rope hoist. See page 105.

➔ If one of the safety buffers cannot be pressed in, replace all the safety buffers on the wire rope hoist. See page 105.

MAINTENANCE

THIS SECTION APPLIES TO ANYONE WHO MAINTAINS, REPAIRS OR MODIFIES THE CRANE.

The end user of the crane is responsible for the selection and proper qualifications of the maintenance personnel.

DANGER TO PERSONS!
Persons can be injured if the crane is incorrectly serviced.

If personnel other than that of the ABUS company are employed to perform maintenance on the crane, it is the end user's responsibility to ensure that these persons are adequately qualified. Follow the procedures described here precisely.

Examples of qualified persons:

- Persons with comprehensive knowledge from specialist training in engineering and in the electrical systems of cranes.
- Persons with sufficient experience in the operation, installation and maintenance of cranes.
- Persons with comprehensive knowledge regarding the relevant technical rules, directives and safety regulations applicable in the respective country.
- Persons receiving regular training from ABUS.

ABUS assumes no liability for damage due to incorrectly performed maintenance work done by unqualified personnel.

ABUS recommends having the maintenance work performed by ABUS Service.

Use only original ABUS replacement parts. Otherwise, all warranty claims will be rendered invalid.

INSTRUCTIONS REGARDING SAFETY WHEN PERFORMING MAINTENANCE WORK

Observe the following safety instructions during all maintenance work on the wire rope hoist.

- Switch off the mains switch. Secure the switch to ensure that it cannot be switched back on accidentally.
- Open the fuse isolating link in the crane panel. Secure the fuse isolating link with a padlock so that it cannot be accidentally closed again.
- Use suitable lifting platforms and fall protection equipment.
- Adequately cordon off the operating range around the lifting platform.
- Switch off other cranes on the same crane track or cranes that are above or below the crane to be serviced. Secure the switches to ensure they cannot be turned back on accidentally. Other cranes could otherwise overturn the lifting platform or hit the crane to be serviced.
- Notify personnel in the area that maintenance work will be performed.
- Only trained electricians should work on the crane electrical system!
- Even after the emergency stop button has been pressed, life-threatening high voltages are still present in the crane control and in the trolley control.

These safety instructions apply specifically to the wire rope hoist:

- **Danger of crushing**!
  Fingers could be crushed on the wheels, cable drum and deflection roller should the wire rope hoist make unforeseen movements, or if work is performed inattentively.
  Secure the wire rope hoist. Work with caution!

REPLACING WIRE ROPE WITH REEVING 2/1, 4/1 OR 6/1

If the wire rope is damaged or has more broken wires than permitted (see “Inspecting the wire rope”, page 25), it must be replaced.

If the cable guide is damaged (see “Inspecting the cable guide”, page 33), it must be replaced.

The wire rope and cable guides are wearing parts that are subject to heavy loads. They subject each other to wear during operation. This is why the wire rope and cable guide should always be replaced together.

The wire rope is a special lifting rope with a very high breaking strength. This ensures it has a longer service life than a normal wire rope. This is why only genuine wire ropes from ABUS may be used.

If the rope wedge or the rope socket are damaged (see “Exchanging the fixed point crosshead with rope socket and rope wedge”, page 71), they need to be replaced. They are usually replaced when the wire rope and cable guide are replaced.

These safety instructions apply specifically to changing the wire rope:

- Cordon off a generous space for the work area under the wire rope hoist.
- Ensure the floor is swept clean.

  This reduces soiling of the new wire rope.
- Scissors lift platform with adequate surface area for setting down the bottom block.

  This is only possible with a reeving of 2/1 and 4/1.

  Observe the load capacity of the lift platform!
- If an unrolling device is available to unroll the wire rope, sufficient free space for this is required.
- Position a scrap container or similar receptacle in the immediate vicinity of the lift platform and bottom block for the old wire rope.
- For a wire rope diameter larger than 11 mm, the rope change should be performed with the aid of a second person.

PREPARING FOR WIRE ROPE REPLACEMENT

➔ Run the load hook to the highest hook position.

Only with ABUS electrics 3 with ABULiner: move the load hook under the upper limit switch range.

➔ Measure, and note, the position of the load hook.

This makes adjusting the hoist limit switch after replacing the wire rope in many cases unnecessary.

ONLY WITH ABUCONTROL

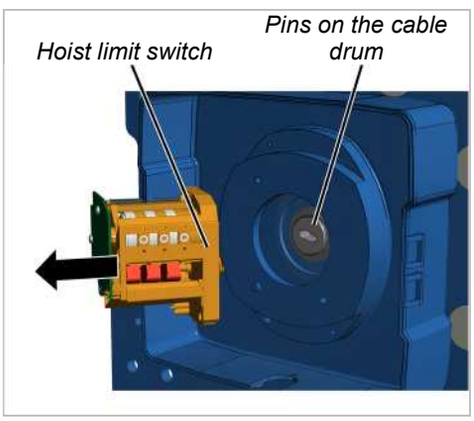
If ABUControl is installed for the wire rope hoist, refer to the product manual “ABUControl”, section “Exchanging the wire rope” to prepare for the rope change.

Further steps for the wire rope change are then described in this product manual.

ONLY FOR ABUS ELECTRICS 3 WITH CONTACTOR CONTROL OR ABULINER

DISASSEMBLING THE HOIST LIMIT SWITCH

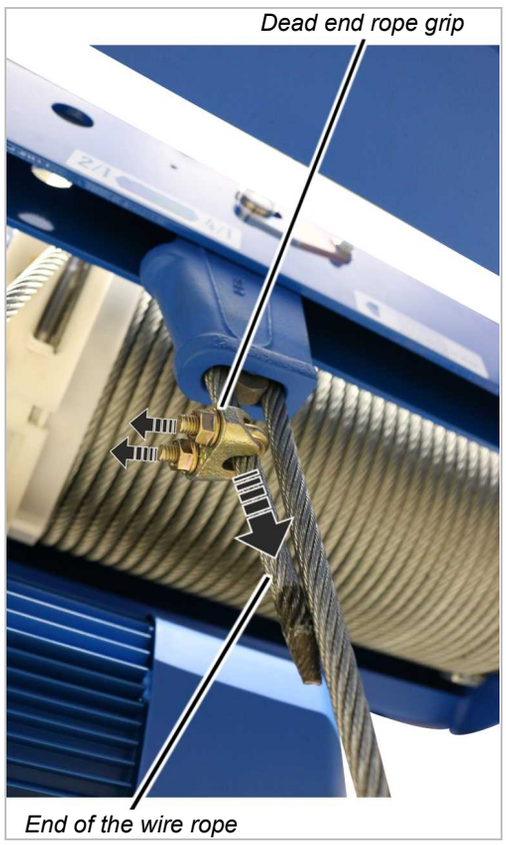
In order for the wire rope to be unwound from the cable drum using a motor, the hoist limit switch must be removed. Otherwise the cable drum will stop at the lowest hook position.



- Open the hoist limit switch housing.
- Unscrew and remove screws (2x).
- Pull off the hoist limit switch.
- Leave the shaft of the hoist limit switch in the same position, and do not turn it.

  This makes adjusting the hoist limit switch after replacing the wire rope in many cases unnecessary.
- Secure the hoist limit switch to ensure it does not fall down or hang from the connection cables.

DISASSEMBLING THE FIXED POINT



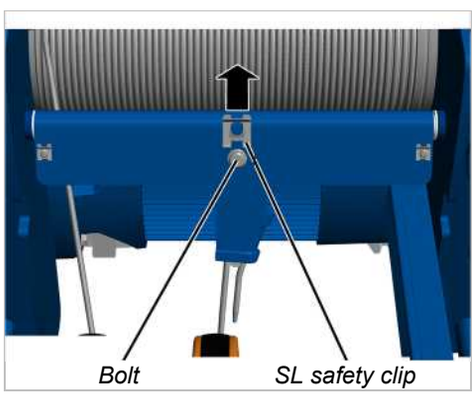
- Release the dead end rope grip.
- Pull the dead end rope grip from the end of the wire rope.

ONLY WITH WIRE ROPE HOISTS WITH LIS-SE AND EXCHANGE OF ROPE SOCKET WITH ROPE WEDGE

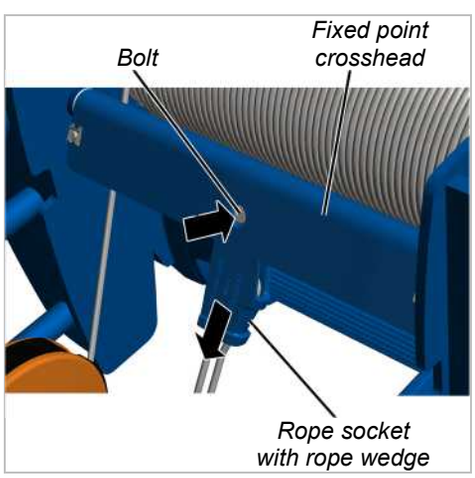
If the rope socket or the rope wedge are damaged, they must be replaced.

The removal of the rope socket on a fixed point crosshead for hoist drive GM 800 to GM 3000 is shown below. Removal from a fixed point crosshead on hoist drive GM 5000 to GM 7000 is very similar.

DISASSEMBLING THE ROPE SOCKET



➔ Pull off the SL safety clip from the bolt on the fixed point crosshead.



➔ Push the bolt out of the fixed point crosshead.

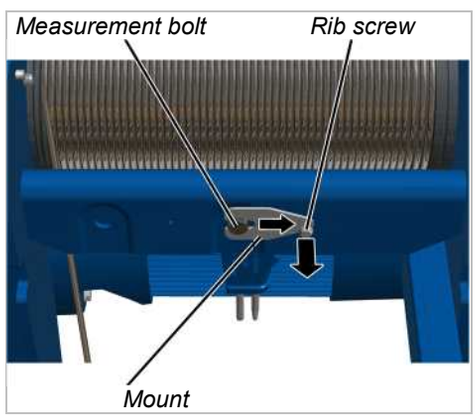
➔ Pull the rope socket out of the fixed point crosshead.

ONLY WITH WIRE ROPE HOISTS WITH LIS-SV OR ABUCONTROL AND EXCHANGE OF ROPE SOCKET WITH ROPE WEDGE

If the rope socket or the rope wedge are damaged, they must be replaced.

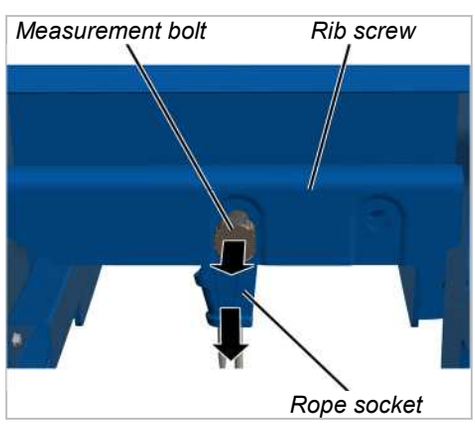
The removal of the rope socket on a fixed point crosshead for hoist drive GM 800 to GM 3000 is shown below. Removal from a fixed point crosshead on hoist drive GM 5000 to GM 7000 is very similar.

DISASSEMBLING THE ROPE SOCKET



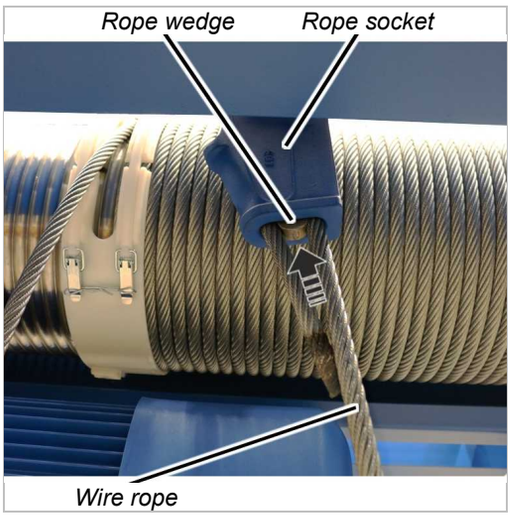
➔ Release rib screw M6x12.

➔ Slide the mount from the measurement bolt.



➔ Pull the measurement bolt out of the fixed point crosshead and the rope socket.

➔ Pull the rope socket out of the fixed point crosshead.



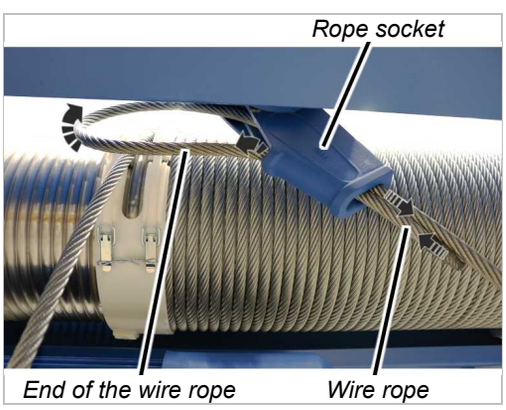
➔ Drive the rope wedge out of the rope socket and take it out.

If the rope wedge is stuck:

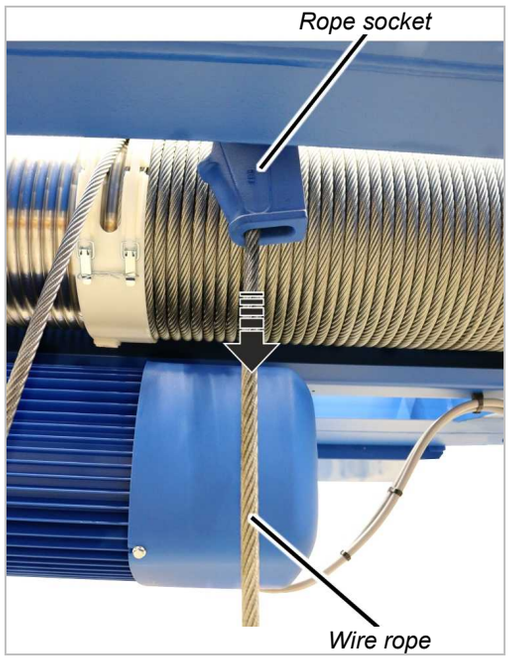
➔ Disassemble the rope socket.

➔ Turn the rope socket over and knock out the rope wedge from above.

The rope wedge can in this way be more easily knocked out of the rope socket.



➔ Push the end of the wire rope out of the rope socket.



➔ Pull the wire rope out of the rope socket.

PULLING THE WIRE ROPE OUT OF THE BOTTOM BLOCK

The rope guard in the bottom block will be damaged if the wire rope is pulled along the rope guard quickly to remove it from the bottom block. If the rope guard is to be used again, it should be disassembled beforehand.

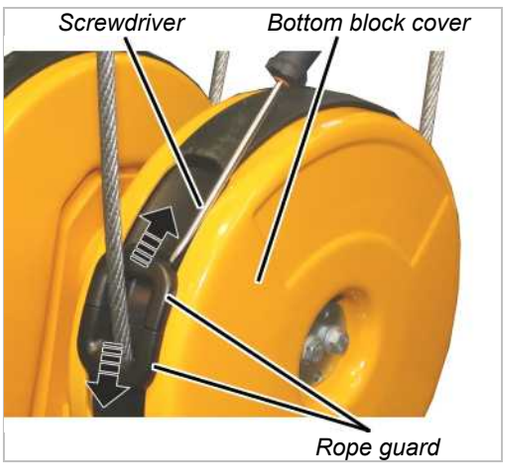
ABUS recommends replacing the rope guard together with the wire rope.

DO NOT PULL THE ROPE GUARD APART FORCEFULLY!

If the rope guard halves are pulled apart forcefully during disassembly, the catches will bend. This means the rope guard can no longer be fitted to be functional.

Carefully open the rope guard with a screwdriver.

On all openings on the bottom block:



➔ Insert the screwdriver laterally between the rope guard and bottom block cover as shown in the figure.

➔ Use the screwdriver to unhook the load hooks on the left and right of the rope guard at the same time.

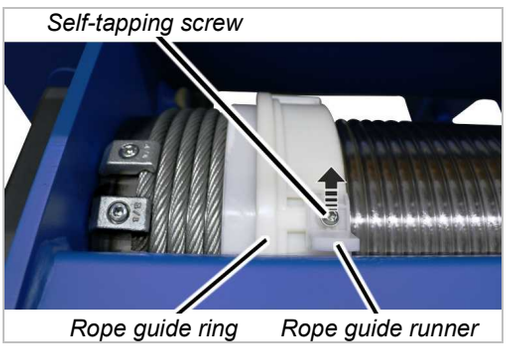
➔ Pull the rope guard (2x) apart and remove it from the bottom block.



➔ Fit the bottom block.

➔ Pull the wire rope out of the bottom block.

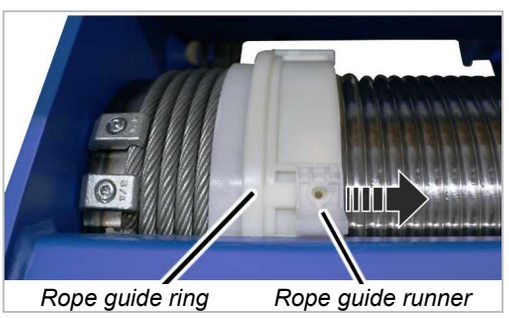
DISASSEMBLING THE CABLE GUIDE



➔ Allow the wire rope to run off the cable drum using the motor.

Several wire rope windings should remain on the cable drum.

➔ Unscrew the rope guide runner.

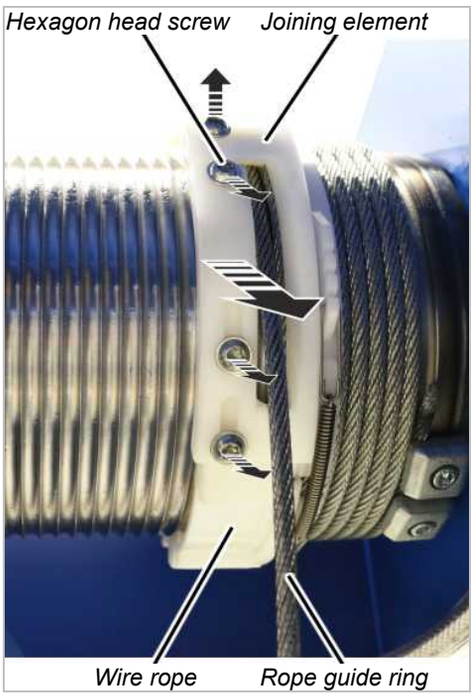


➔ Push the rope guide runner away to one side of the rope guide ring.

- The rope guide ring can now be turned freely on the cable drum.

ONLY WITH GM 800, GM 1000, GM 2000 AND GM 3000

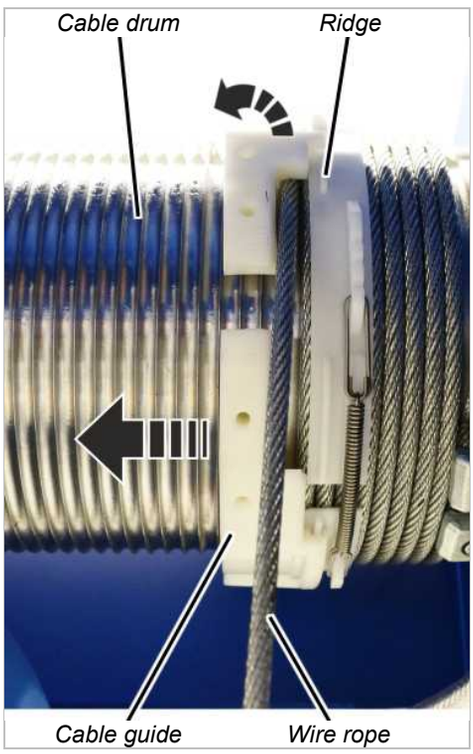
With a hoist drive of size GM 800 to GM 3000, the cable guide consists of a rope tension ring with rope guide ring and a joining element.



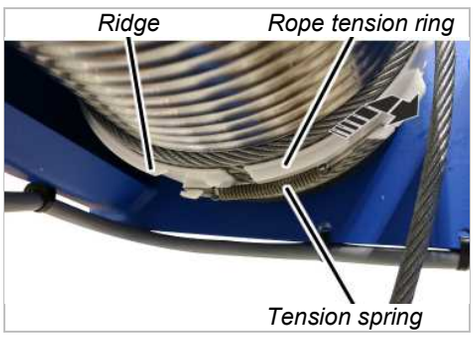
➔ Turn the rope guide ring so that the screws (4x) are easily accessible (e.g. upwards).

➔ Loosen screws (4x).

➔ Remove the joining element and unthread it from the wire rope.



➔ Force the rope guide ring apart and pull it inwards and off the cable drum.



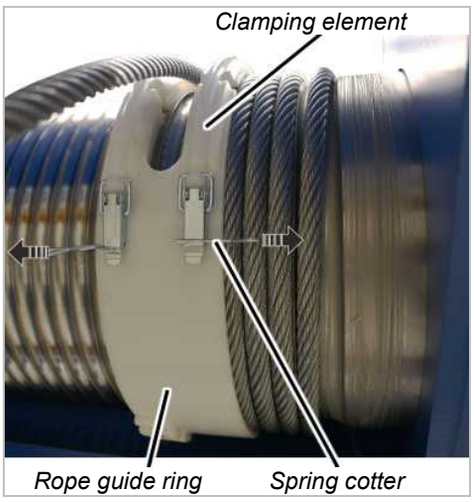
➔ Remove the tension spring from the rope tension ring.

➔ Force the rope tension ring apart and pull it inwards and off the cable drum.

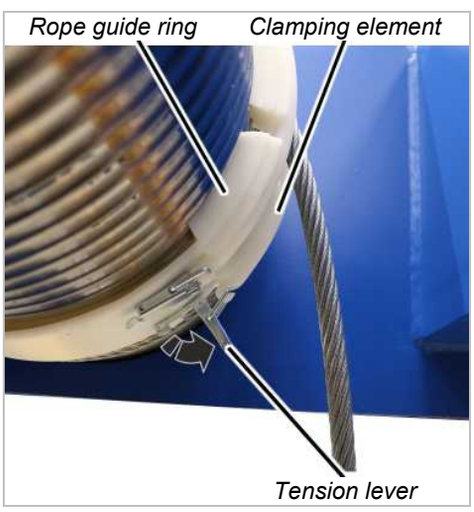
- The cable guide has now been disassembled. The wire rope is now lying loose on the cable drum.

ONLY WITH GM 5000, GM 6000 AND GM 7000

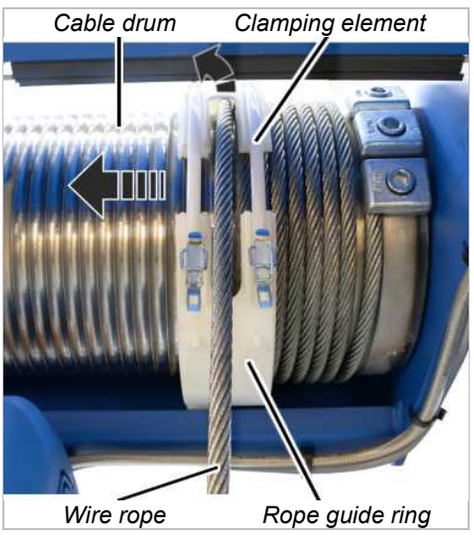
On a hoist drive of size GM 5000 to GM 7000, the cable guide consists of a rope guide ring with clamping elements.



➔ Pull the spring cotter out of the clamping elements on the rope guide ring.

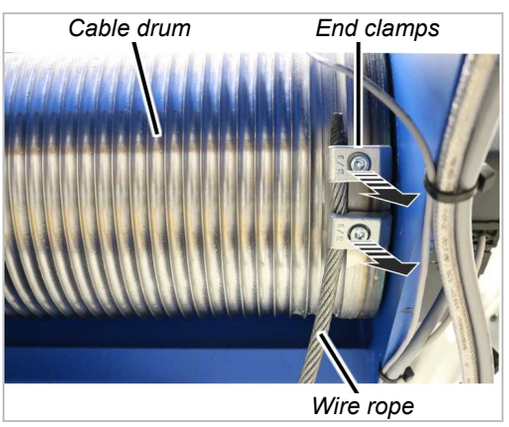


➔ Open and release the tension lever of the clamping element on the rope guide ring.



➔ Pull the rope guide ring with clamping elements from the cable drum.

REMOVING THE WIRE ROPE FROM THE CABLE DRUM



➔ Allow the wire rope to run off the cable drum using the motor.

➔ Release end clamps.

➔ Completely remove the wire rope from the cable drum.

CLEANING AND LUBRICATING THE CABLE DRUM

➔ Clean the cable drum.



➔ Lubricate the cable drum from the start to about the 8th or 10th turn.

➔ Lubricate the cable drum behind the last turn.

Lubricant: “Molykote PG-75”. . For details, see “Lubricants”, page 107.



➔ Lubricate the cable drum behind the 8th/10th turn up to the last turn.

Lubricant: “Chainlife S”. . For details, see “Lubricants”, page 107.

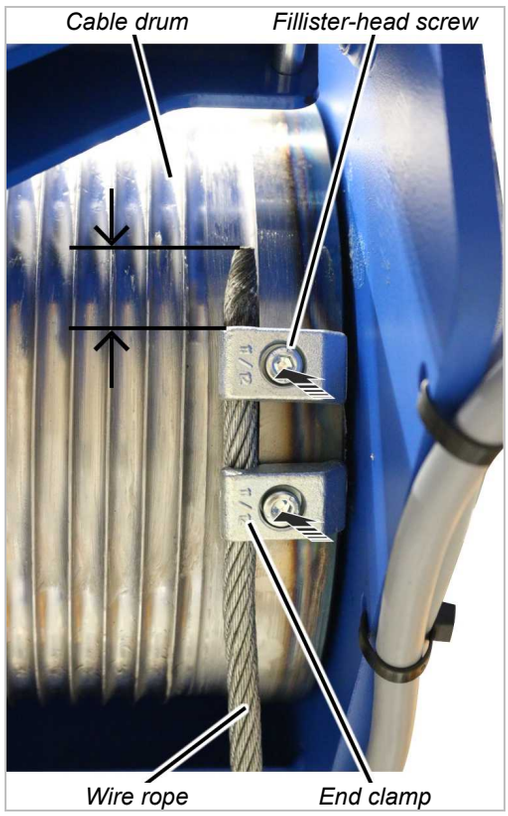
FASTENING THE WIRE ROPE TO THE CABLE DRUM

ROLL OUT THE WIRE ROPE WITHOUT TWISTING IT!

The wire rope will bend and become damaged if it is unrolled incorrectly.

Do not pull the wire rope upwards away from a horizontally-lying ring.

Unroll the ring so it is vertically upright instead, and allow the wire rope to run onto the cable drum in the same direction as it was rolled onto the ring.



➔ Unroll the wire rope without twisting it.

➔ Thoroughly clean the fillister-head screws of the end clamps.

The fillister-head screws were previously bolted with a thread lock coating or a thread lock. The residues must be completely removed before the fillister-head screw may be used again.

The thread lock coating as well as the thread lock are single-use only.

➔ Push the wire rope under the end clamps far enough that the rope protrusion is about 30 mm.

➔ Tighten the end clamps with fillister-head screws and secure with (medium strength) thread lock.

Size	Number of end clamps	Tightening torque
GM 800	2	10 Nm
GM 1000	2	25 Nm
GM 3000	2	25 Nm
GM 5000	3	210 Nm
GM 6000	3	210 Nm
GM 7000	3	425 Nm

➔ Pull the wire rope taut and use the motor to wind it onto the cable drum by 6 to 8 turns, ensuring it is taut and not twisted.

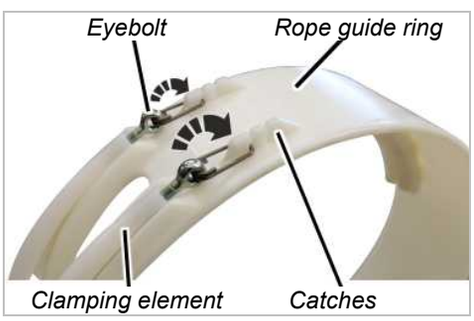
PREPARING THE CABLE GUIDE

On a cable guide for a GM 5000 to GM 7000, the clamping elements must be secured on the rope guide ring.

No pre-assembly is necessary on a cable guide for a GM 800 to GM 3000. This section can be skipped over in this case.

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 5000 AND GM 6000

On the cable guide for a wire rope hoist with hoist drive GM 5000 or GM 6000, the clamping elements are hooked into catches on the rope guide ring. The eyebolts for clamping the rope guide ring are located above on the clamping elements.



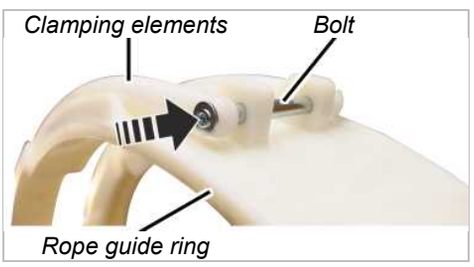
➔ Insert the clamping elements (2x) with the eyebolts in the first catch on the rope guide ring.

➔ Grease the groove of the rope guide ring from the inside.

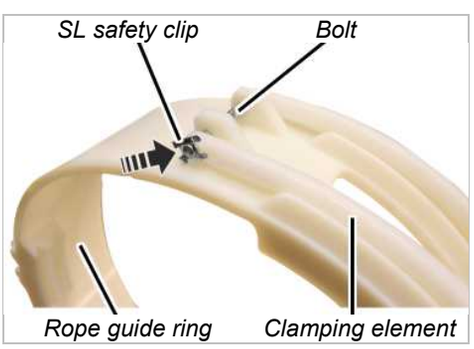
Lubricant: “Molykote PG-75”. For details, see “Lubricants”, page 107.

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 7000

On the cable guide for a wire rope hoist with hoist drive GM 7000, the clamping elements are secured on the rope guide ring using a bolt and a SL safety clip. The eyebolts for tensioning the rope guide ring are located below on the clamping elements.



➔ Push the bolt through the clamping elements and the rope guide ring.



➔ Push the SL safety clip onto the bolt.

➔ Grease the groove of the rope guide ring from the inside.

Lubricant: “Molykote PG-75”. For details, see “Lubricants”, page 107.

INSTALLING THE CABLE GUIDE

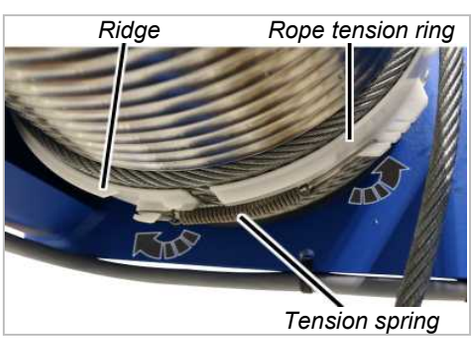
ONLY WITH GM 800, GM 1000, GM 2000 AND GM 3000

With a hoist drive of size GM 800 to GM 3000, the cable guide consists of a rope tension ring with rope guide ring and a joining element.

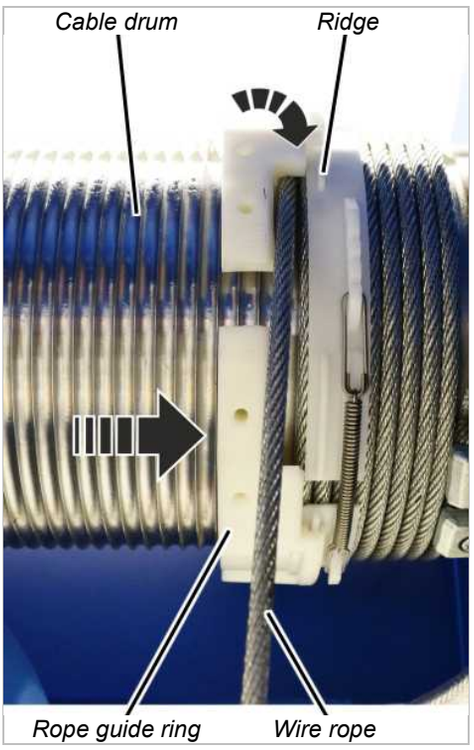


➔ Turn the rope tension ring so that the ridge is facing the hoisting gear.

➔ Force the rope tension ring apart and guide it over the wire rope on the cable drum.



➔ Hook the tension spring into one of the notches so that the rope tension ring is firmly seated on the wire rope but can still be turned freely.



➔ Grease the groove of the rope guide ring from the inside.

Lubricant: “Molykote PG-75”. For details, see “Lubricants”, page 107.

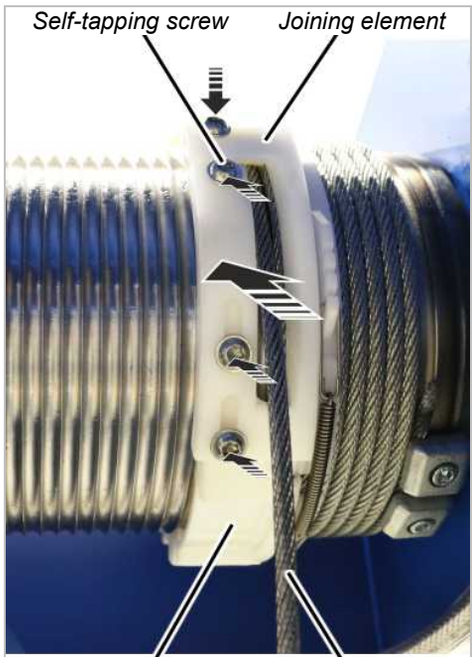
➔ Force the rope guide ring apart and push it over the cable drum.

➔ Place the rope guide ring with the groove into the ridge on the rope tension ring.

➔ Place the rope guide ring with the threaded segments into the free section of the cable drum.

➔ Check that both ends of the rope guide ring lie in the same groove of the cable drum.

The ends of the rope guide ring must be aligned.



➔ Thread the joining element onto the wire rope.

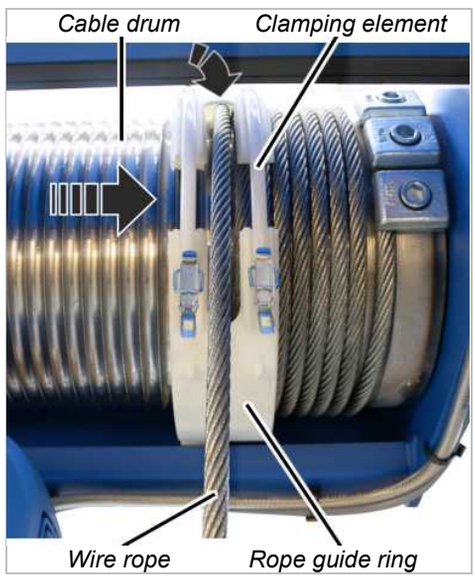
➔ Push the joining element upwards and place it onto the rope guide ring.

➔ Tighten the joining element with self-tapping screws M6x16 (4x). 5 Nm.

When doing this, first secure the upper screws which do not lie in the elongated hole.

ONLY WITH GM 5000, GM 6000 AND GM 7000

On a hoist drive of size GM 5000 to GM 7000, the cable guide consists of a rope guide ring with clamping elements.

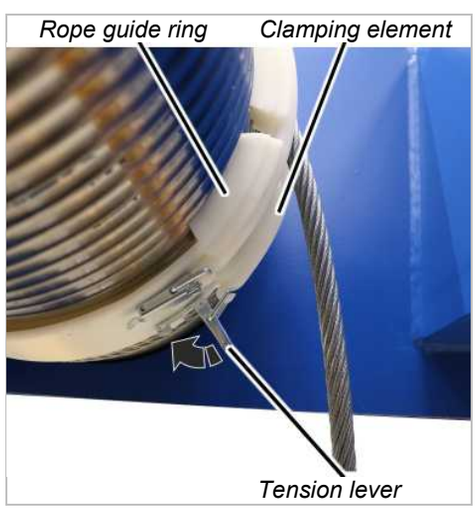


➔ Pull the rope guide ring with clamping elements onto the cable drum.

➔ Position the wire rope between the clamping elements.

➔ Check that both ends of the rope guide ring lie in the same groove of the cable drum.

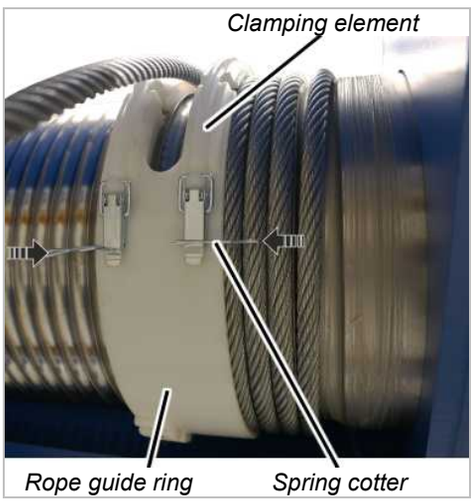
The ends of the rope guide ring must be aligned.



➔ Insert the tension lever on the clamping elements into the rope guide ring and close it.

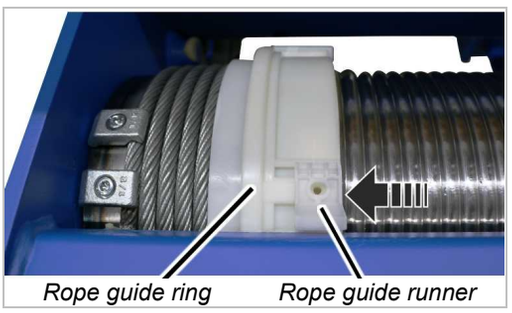
If necessary:

➔ Open the tension lever again and adjust the rope guide ring with the eyebolts so that the rope guide ring rests securely on the wire rope but can still be rotated.

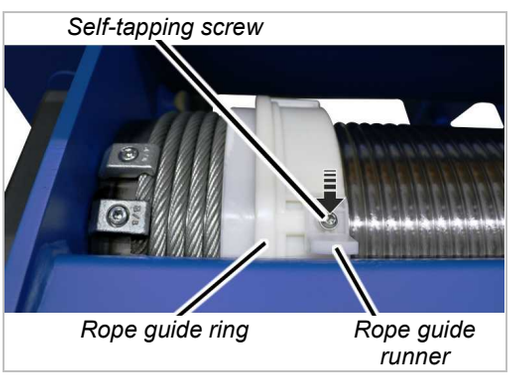


➔ Push the spring cotter into the tension lever of the clamping elements.

INSTALLING THE ROPE GUIDE RUNNER

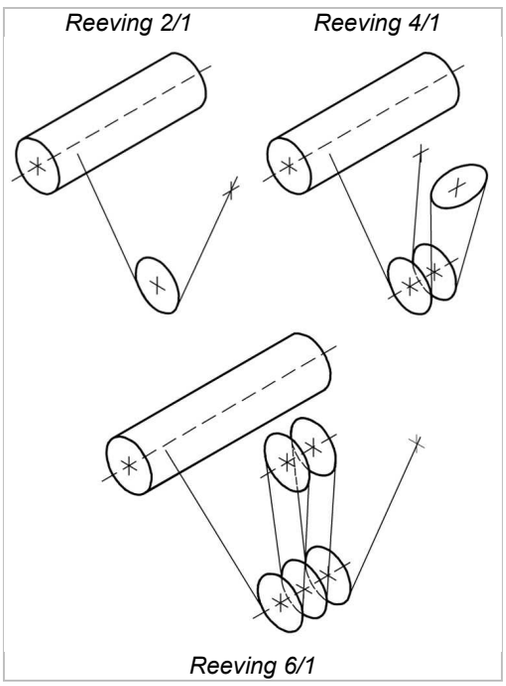


➔ Put the rope guide runner on the drum housing and push it laterally into the rope guide ring.



➔ Screw on the rope guide runner with the self-tapping screw M6x16. 5 Nm.

REEVING THE WIRE ROPE



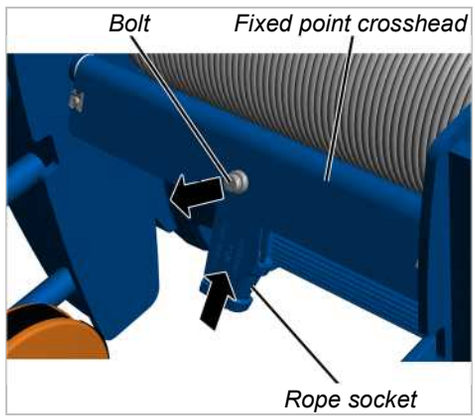
- For reeving 2/1: pull the wire rope from the cable drum and through the bottom block, without twisting it.
- For reeving 4/1: pull the wire rope from the cable drum and through the bottom block, without twisting it, and through the deflection roller and once again through the bottom block.
- For reeving 6/1: pull the wire rope from the cable drum and through the bottom block, without twisting it, and through the deflection roller and once again through the bottom block, through the deflection roller and again through the bottom block.

INSTALLING THE FIXED POINT

ONLY WITH WIRE ROPE HOISTS WITH LIS-SE AND EXCHANGE OF ROPE SOCKET WITH ROPE WEDGE

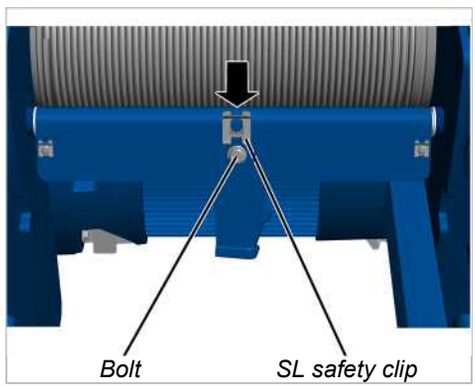
The installation of the rope socket on a fixed point crosshead for hoist drive GM 800 to GM 3000 is shown below. Installation on a fixed point crosshead on hoist drive GM 5000 to GM 7000 is very similar.

INSTALLING THE ROPE SOCKET



- Push the rope socket into the fixed point crosshead.

  Note the marking on the fixed point crosshead.
- Push the bolt through the fixed point crosshead and the rope socket.

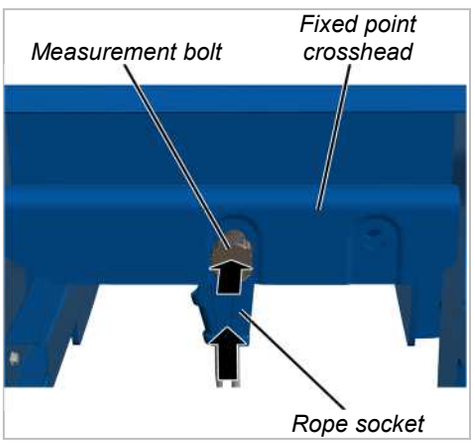


- Push the SL safety clip onto the bolt.

ONLY WITH WIRE ROPE HOISTS WITH LIS-SV OR ABUCONTROL AND EXCHANGE OF ROPE SOCKET WITH ROPE WEDGE

The installation of the rope socket on a fixed point crosshead for hoist drive GM 800 to GM 3000 is shown below. Installation on a fixed point crosshead on hoist drive GM 5000 to GM 7000 is very similar.

INSTALLING THE ROPE SOCKET

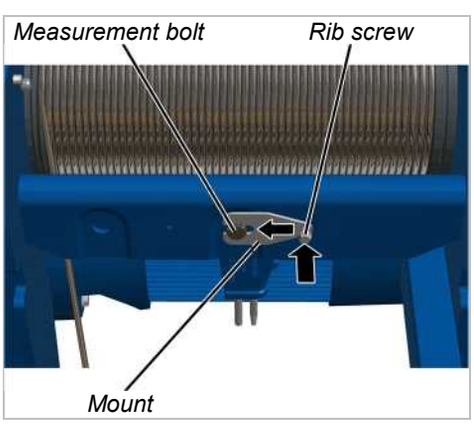


➔ Push the rope socket into the fixed point crosshead.

Note the marking on the fixed point crosshead.

➔ Push the measurement bolt through the fixed point crosshead and the rope socket.

The arrow on the measurement bolt must point downward!

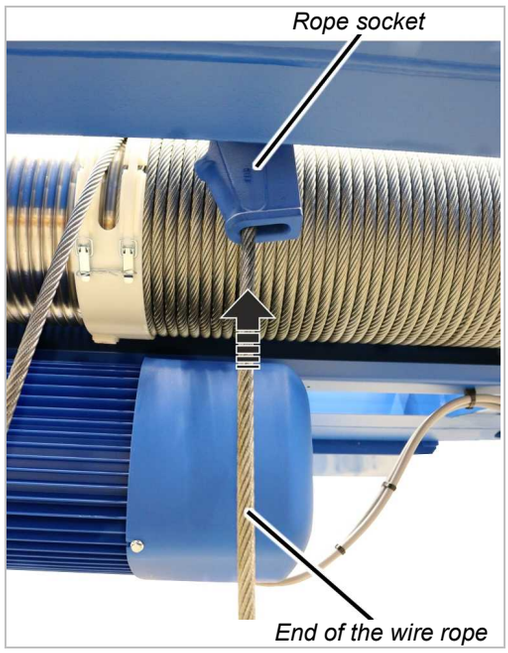


➔ Slide the mount onto the measurement bolt.

The mount must allow itself to be inserted into the groove of the measurement bolt without tension.

The measurement bolt should not have any pretension!

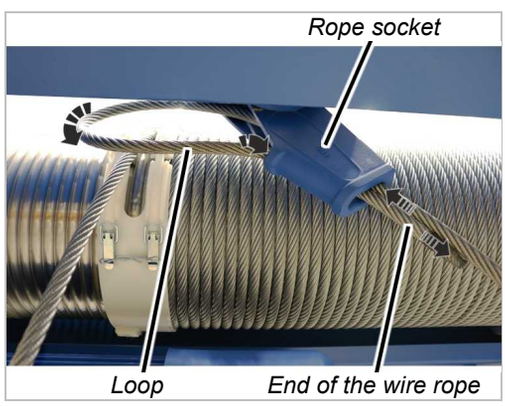
➔ Screw on the mount with the rib screw M6x12. 19 Nm.



➔ Check the lubrication of the rope socket and grease it if necessary.

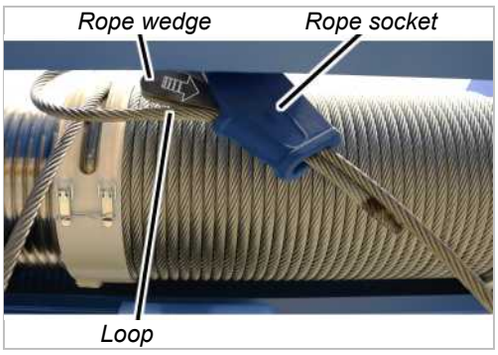
Lubricant: “High temperature paste PBC 1574”. For details, see “Lubricants”, page 107.

➔ Push the end of the wire rope into the rope socket from below.

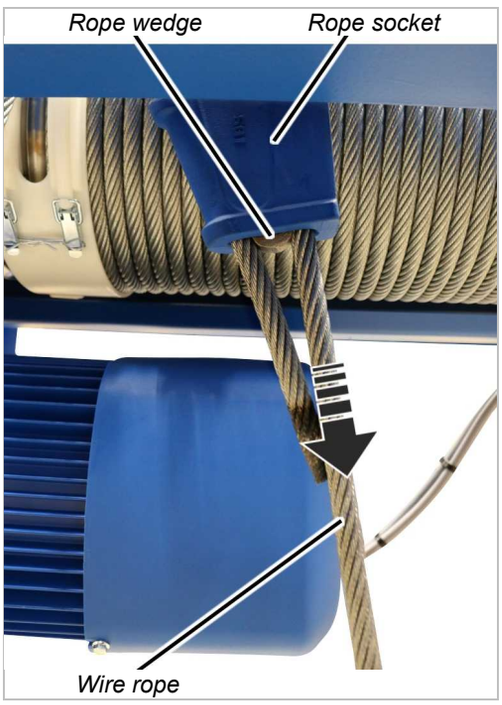


➔ Push the end of the wire rope into the rope socket from above so that a loop forms.

The end of the wire rope protrudes below about 150 mm out of the rope socket.



➔ Place the rope wedge in the loop.



➔ Give the wire rope a jolting tug.

- The rope wedge slips into the rope socket and the wire rope is tightly clamped.



➔ Push the dead end rope grip over the end of the wire rope.

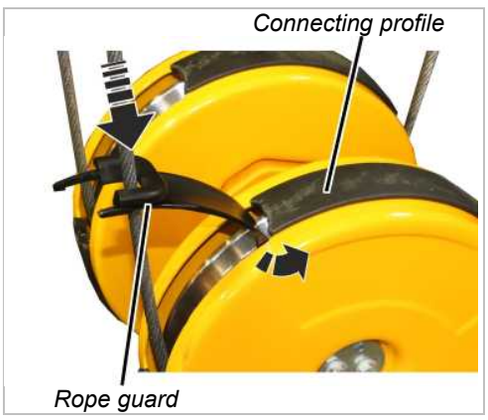
The dead end rope grip lies about 30 mm below the rope socket.

➔ Screw the dead end rope grip tight.

Size	Rib nut	Tightening torque
GM 800	M6	5 Nm
GM 1000	M8	10 Nm
GM 2000	M8	10 Nm
GM 3000	M8	10 Nm
GM 5000	M14	49 Nm
GM 6000	M14	49 Nm
GM 7000	M16	107 Nm

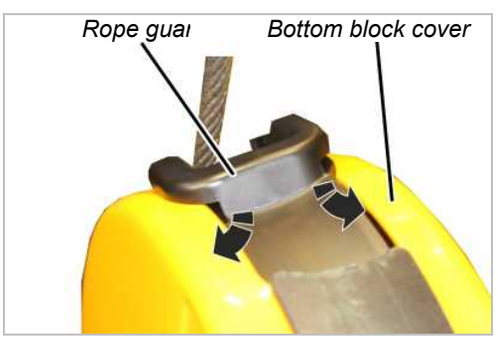
INSTALLING THE ROPE GUARD ON THE BOTTOM BLOCK

On all openings on the bottom block:



➔ Hold the rope guard on the wire rope as shown in the figure.

➔ Push the rope guard down along the wire rope with the front fork and bring the rear end under the connecting profile.



➔ First push the rope guard on the side with the short bracket into the bottom block cover.

➔ Then push the rope guard on the side with the long bracket into the bottom block cover.



➔ Push the second rope guard at the bottom into the connecting profile.

➔ Push the rope guard downwards behind the connecting profile and attach it above on the wire rope.

➔ First push the rope guard on the side with the short bracket into the bottom block cover.

➔ Then push the rope guard on the side with the long bracket into the bottom block cover.



➔ Connect the brackets of the rope guards (2x) and push them together.

- A snapping sound can be heard.

COMPLETING WIRE ROPE REPLACEMENT

➔ Use the motor to wind up the wire rope completely.

ONLY WITH ABUCONTROL

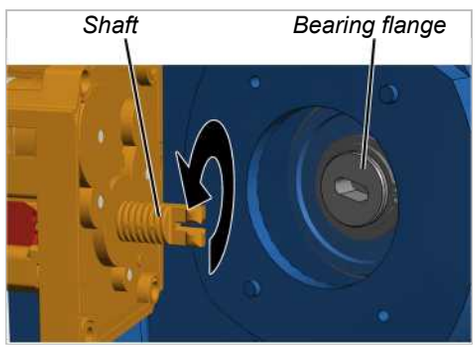
If ABUControl is installed for the wire rope hoist, refer to the product manual “ABUControl”, section “Exchanging the wire rope” to conclude the rope change.

Further steps are then described in this product manual.

ONLY WITH ABUS ELECTRICS 3 WITH CONTACTOR CONTROL OR ABULINER

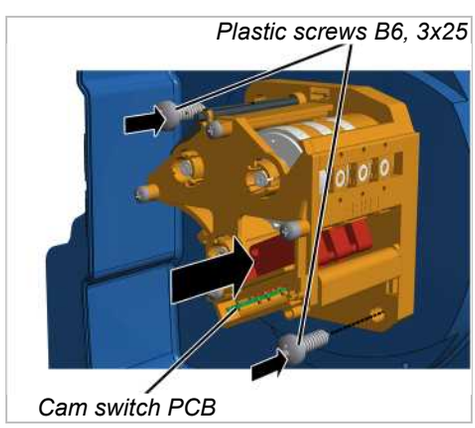
INSTALLING THE HOIST LIMIT SWITCH

➔ Move the load hooks to the position you noted previously.



➔ Open the housing cover of the hoist limit switch enclosure on the cable drum.

➔ Rotate the shaft of the gear limit switch so that it matches the elongated hole in the bearing flange of the cable drum.



➔ Insert the gear limit switch with the shaft in the elongated hole on the bearing flange and guide it onto the two pins.

➔ Screw the gear limit switch tight with the plastic screws B6, 3x25. 2.5 Nm.

CONNECTING THE HOIST LIMIT SWITCH

➔ Plug in the connector of the connection cable on the cam switch PCB.

➔ Close the housing cover.

If the switching points have moved:

Normally, the adjustment of the hoist limit switch after replacing the wire rope can in many cases be omitted, since the position of the load hook was previously measured and then travelled to again. In special cases (e.g. if the wire rope is severely damaged or has been incorrectly rolled onto the cable drum), the switching points may need to be readjusted.

➔ Reset switching points for the bottom hoist limiter, the top hoist limiter and the backup limiter.

With ABUS electrics 3 with contactor control: see product manual “ABUS gear limit switch”.

With ABUS electrics 3 with ABULiner: see product manual “Frequency converter control of the lifting speed for ABUS electrics 3”.

CHECKING THE ROPE WEDGE PROTRUSION



➔ Suspend the load from the load hooks, raise it and load the wire rope this way.

➔ Measure the rope wedge protrusion and check whether the dimensions specified in the table have been complied with.

Size	Rope wedge protrusion when the wire rope hoist is new	Maximum rope wedge protrusion	Minimum rope wedge protrusion
GM 800	18 mm	19 mm	18 mm
GM 1000	3 mm	4 mm	3 mm
GM 2000	0 mm	2 mm	0 mm
GM 3000	16 mm	18 mm	16 mm
GM 5000	19 mm	22 mm	19 mm
GM 6000	28 mm	31 mm	28 mm
GM 7000	28 mm	28 mm	26 mm

REPLACING WIRE ROPE WITH REEVING 4/2 OR 8/2

If the wire rope is damaged or has more broken wires than permitted (see “Inspecting the wire rope”, page 25), it must be replaced.

If the cable guide is damaged (see “Inspecting the cable guide”, page 33), it must be replaced.

The wire rope and cable guides are wearing parts that are subject to heavy loads. They subject each other to wear during operation. This is why the wire rope and cable guide should always be replaced together.

The wire rope is a special lifting rope with a very high breaking strength. This ensures it has a longer service life than a normal wire rope. This is why only genuine wire ropes from ABUS may be used.

These safety instructions apply specifically to changing the wire rope:

- Cordon off a generous space for the work area under the wire rope hoist.
- Ensure the floor is swept clean.

  This reduces soiling of the new wire rope.
- Scissors lift platform with adequate surface area for setting down the bottom block.

  Observe the load capacity of the lift platform!
- If an unrolling device is available to unroll the wire rope, sufficient free space for this is required.
- Position a scrap container or similar receptacle in the immediate vicinity of the lift platform and bottom block for the old wire rope.
- For a wire rope diameter larger than 11 mm, the rope change should be performed with the aid of a second person.

PREPARING FOR WIRE ROPE REPLACEMENT

➔ Run the load hook to the highest hook position.

Only with ABUS electrics 3 with ABULiner: move the load hook under the upper limit switch range.

➔ Measure, and note, the position of the load hook.

This makes adjusting the hoist limit switch after replacing the wire rope in many cases unnecessary.

ONLY WITH ABUCONTROL

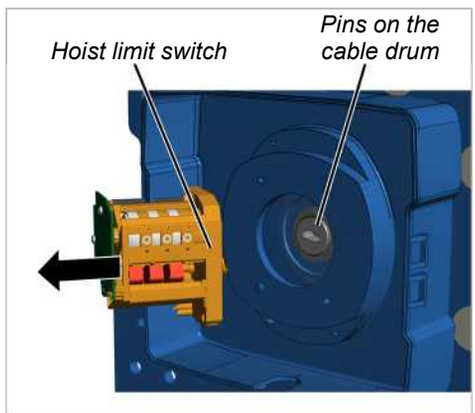
If ABUControl is installed for the wire rope hoist, refer to the product manual “ABUControl”, section “Exchanging the wire rope” to prepare for the rope change.

Further steps for the wire rope change are then described in this product manual.

ONLY FOR ABUS ELECTRICS 3 WITH CONTACTOR CONTROL OR ABULINER

DISASSEMBLING THE HOIST LIMIT SWITCH

In order for the wire rope to be unwound from the cable drum using a motor, the hoist limit switch must be removed. Otherwise the cable drum will stop at the lowest hook position.



➔ Open the hoist limit switch housing.

➔ Unscrew and remove screws (2x).

➔ Pull off the hoist limit switch.

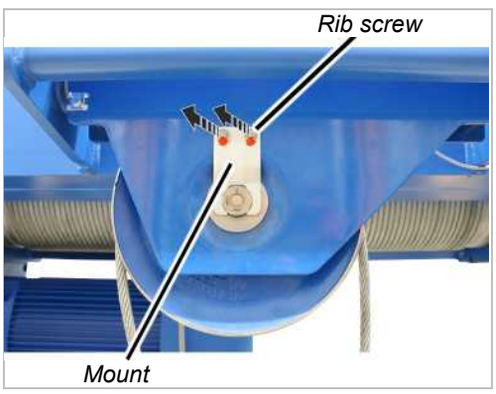
➔ Leave the shaft of the hoist limit switch in the same position, and do not turn it.

This makes adjusting the hoist limit switch after replacing the wire rope in many cases unnecessary.

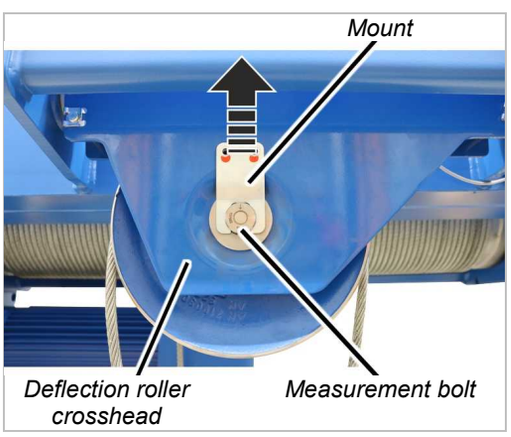
➔ Secure the hoist limit switch to ensure it does not fall down or hang from the connection cables.

DISASSEMBLING THE DEFLECTION ROLLER

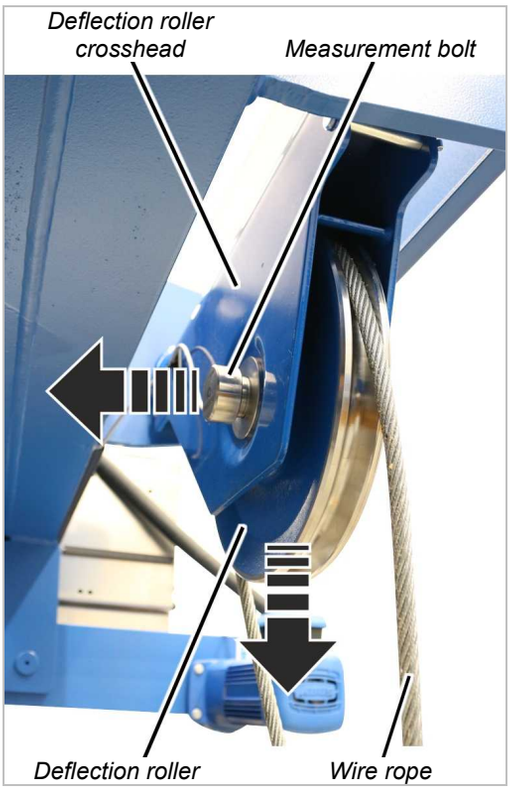
For reeving 4/2 and 8/2, both fixed points for the wire rope are on the cable drum. To replace the wire rope, the deflection roller on the trolley frame side must be removed first so the wire rope can then be wound from the cable drum.



➔ Release the rib screws on the mount.



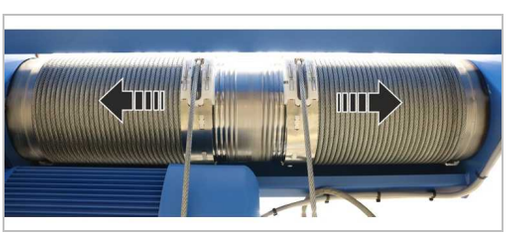
➔ Pull the mount of the measurement bolt.



➔ Pull the measurement bolt out of the deflection roller crosshead and deflection roller.

➔ Pull the deflection roller out of the deflection roller crosshead.

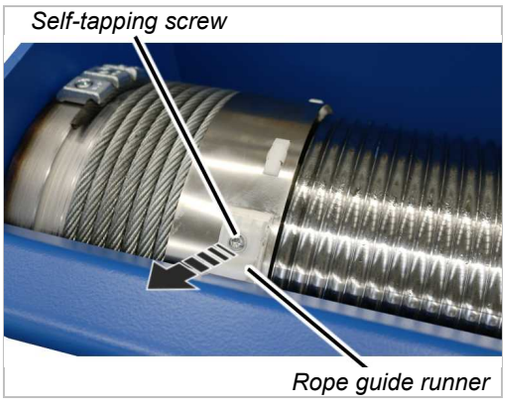
DISASSEMBLING THE CABLE GUIDE



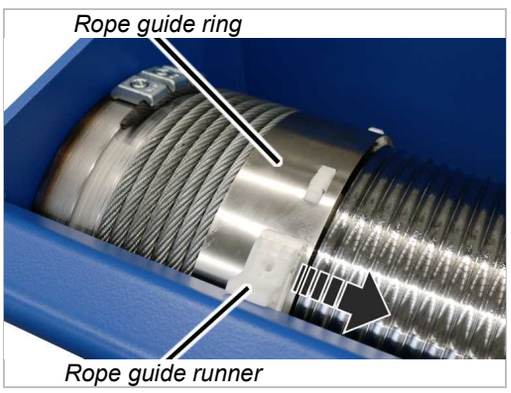
➔ Allow the wire rope to run off the cable drum using the motor.

Several wire rope windings should remain on the cable drum.

On both sides of the cable drum:



➔ Unscrew the rope guide runner.

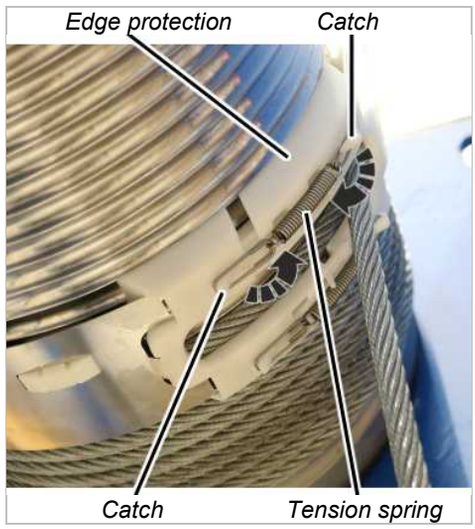


➔ Push the rope guide runner away to one side of the rope guide ring.

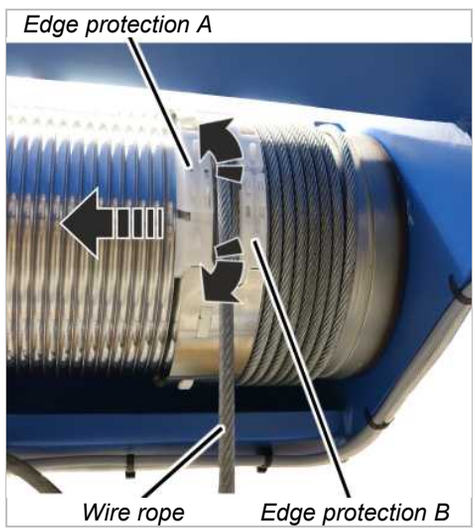
- The rope guide ring can now be turned freely on the cable drum.

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 5000

On a wire rope hoist with hoist drive GM 5000, the cable guide consists of a plate, guide elements, edge protection and tension springs.



➔ Release the tension springs from the catches.

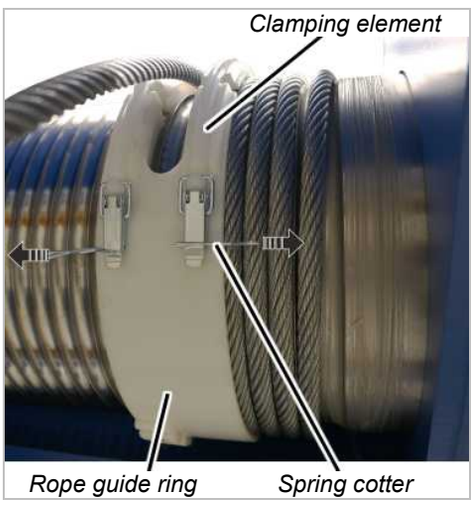


➔ Pull edge protections A and B apart.

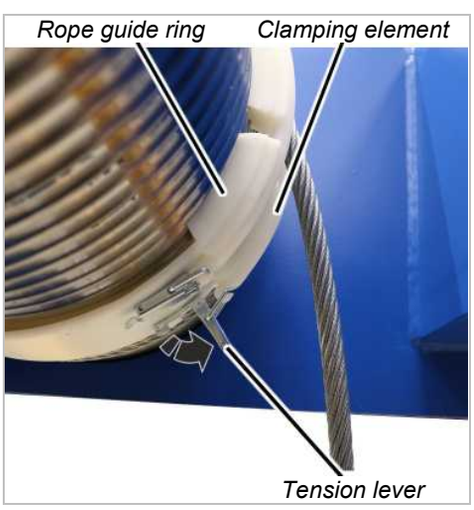
➔ Pull the cable guide off the cable drum.

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 7000

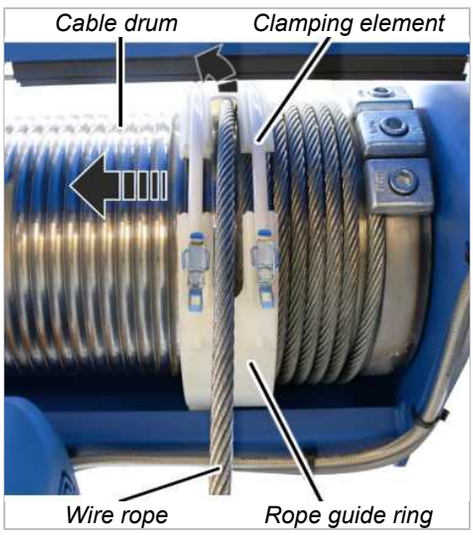
On a wire rope hoist with hoist drive GM 7000, the cable guide consists of a rope guide ring with clamping elements.



➔ Pull the spring cotter out of the clamping elements on the rope guide ring.



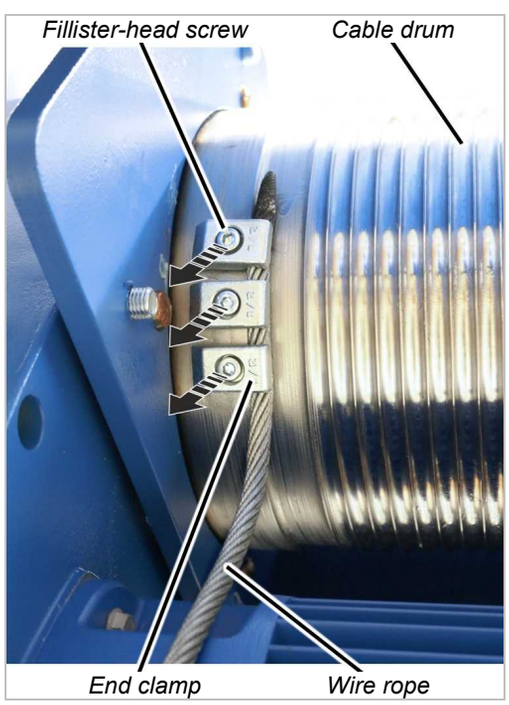
➔ Open and release the tension lever of the clamping element on the rope guide ring.



➔ Pull the rope guide ring with clamping elements from the cable drum.

REMOVING THE WIRE ROPE FROM THE CABLE DRUM

On both sides of the cable drum:



➔ Allow the wire rope to run off the cable drum using the motor.

➔ Release end clamps.

➔ Completely remove the wire rope from the cable drum.

PULLING THE WIRE ROPE OUT OF THE BOTTOM BLOCK

The rope guard in the bottom block will be damaged if the wire rope is pulled along the rope guard quickly to remove it from the bottom block. If the rope guard is to be used again, it should be disassembled beforehand.

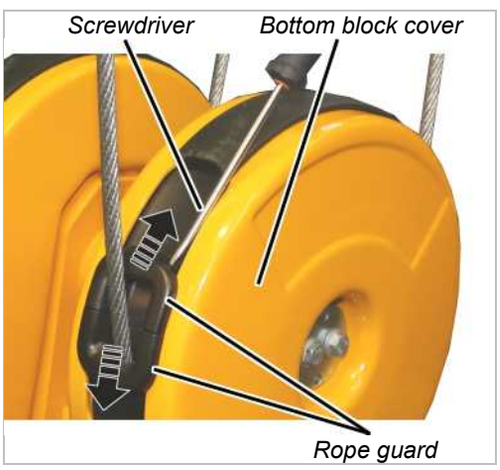
ABUS recommends replacing the rope guard together with the wire rope.

DO NOT PULL THE ROPE GUARD APART FORCEFULLY!

If the rope guard halves are pulled apart forcefully during disassembly, the catches will bend. This means the rope guard can no longer be fitted to be functional.

Carefully open the rope guard with a screwdriver.

On all openings on the bottom block:



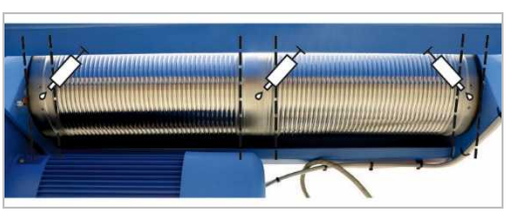
- Insert the screwdriver laterally between the rope guard and bottom block cover as shown in the figure.
- Use the screwdriver to unhook the load hooks on the left and right of the rope guard at the same time.
- Pull the rope guard (2x) apart and remove it from the bottom block.



- Fit the bottom block.
- Pull the wire rope out of the bottom block.

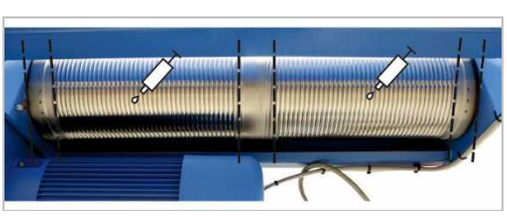
CLEANING AND LUBRICATING THE CABLE DRUM

- Clean the cable drum.



- Lubricate the cable drum from the start to about the 8th or 10th turn on both sides.
- Lubricate the cable drum in the middle behind or in front of the last turn.

  Lubricant: “Molykote PG-75”. For details, see “Lubricants”, page 107.



- Lubricate the cable drum from both sides behind the 8th or 10th turn up to the last turn in the centre.

  Lubricant: “Chainlife S”. For details, see “Lubricants”, page 107.

REEVING THE WIRE ROPE

ROLL OUT THE WIRE ROPE WITHOUT TWISTING IT!

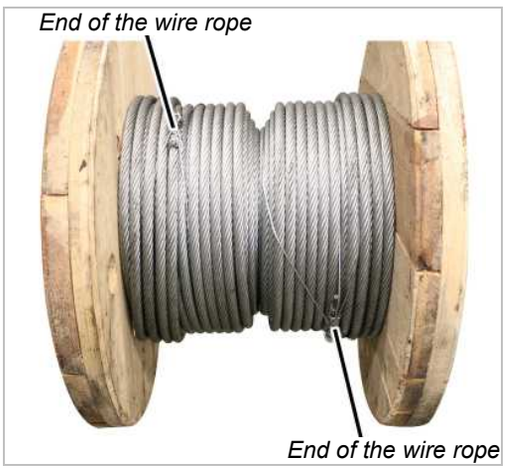
The wire rope will bend and become damaged if it is unrolled incorrectly.

Do not pull the wire rope upwards away from a horizontally-lying ring.

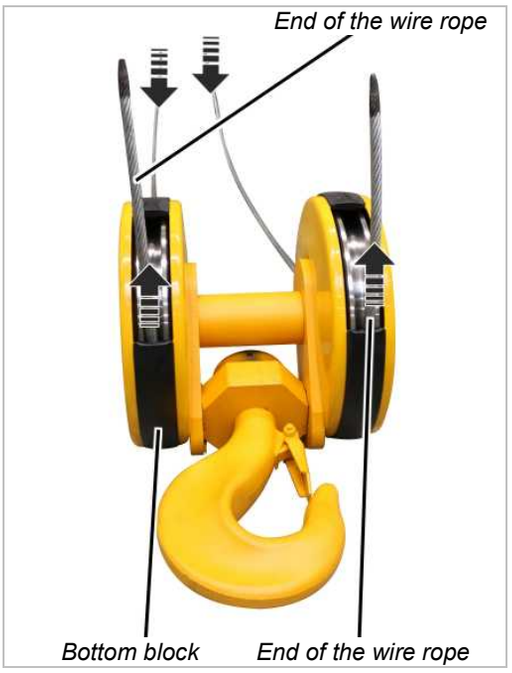
Unroll the ring so it is vertically upright instead, and allow the wire rope to run onto the cable drum in the same direction as it was rolled onto the ring.

Note

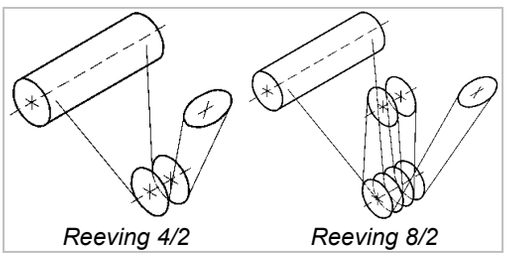
The wire rope is completely wound up. When unrolling the wire rope make sure the wire rope is unrolled evenly so that later, the wire rope marking lies centrally in the groove of the deflection roller and the wire rope is wound up onto the cable drum evenly.



➔ Unroll the wire rope evenly and without twisting it.



➔ Pull both ends of the wire rope through the bottom block without twisting the wire rope.

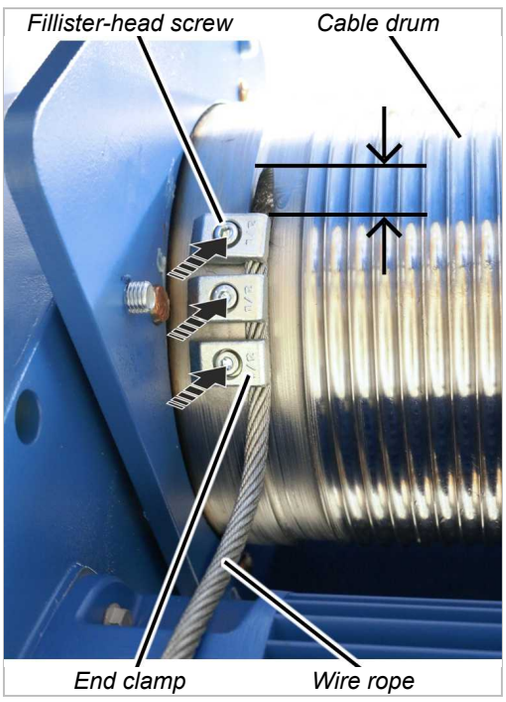


➔ For reeving 4/2: pull the wire rope through the bottom block, without twisting it.

➔ For reeving 8/2: pull the wire rope through the central reels of the bottom block, without twisting it, and through the reels of the upper block and once again through the outer reels of the bottom block.

FASTENING THE WIRE ROPE TO THE CABLE DRUM

On both sides of the cable drum:



➔ Unroll the wire rope without twisting it.

➔ Thoroughly clean the fillister-head screws of the end clamps.

The fillister-head screws were previously bolted with a thread lock coating or a thread lock. The residues must be completely removed before the fillister-head screw may be used again.

The thread lock coating as well as the thread lock are single-use only.

➔ Push the wire rope under the end clamps far enough that the rope protrusion is about 30 mm.

➔ Tighten the end clamps with fillister-head screws and secure with (medium strength) thread lock.

Size	Number of end clamps	Tightening torque
GM 5000	3	210 Nm
GM 7000	3	425 Nm

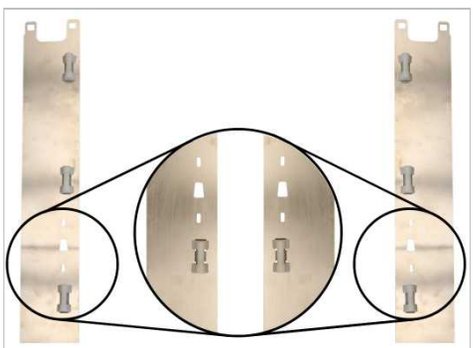
➔ Pull the wire rope taut and use the motor to wind it onto the cable drum by 6 to 8 turns, ensuring it is taut and not twisted.

PREPARING THE CABLE GUIDE

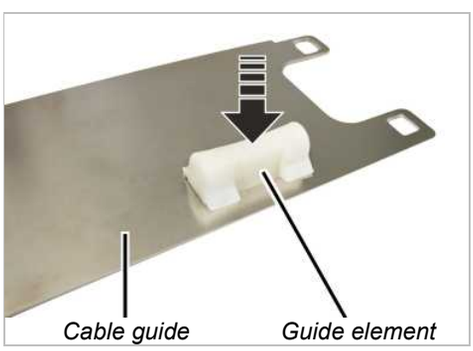
ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 5000

On a wire rope hoist with hoist drive GM 5000, the cable guide consists of a metal plate.

A cable guide for the left side and a cable guide for the right side must be fitted. The holes for the rope guide runner must be on the correct side.

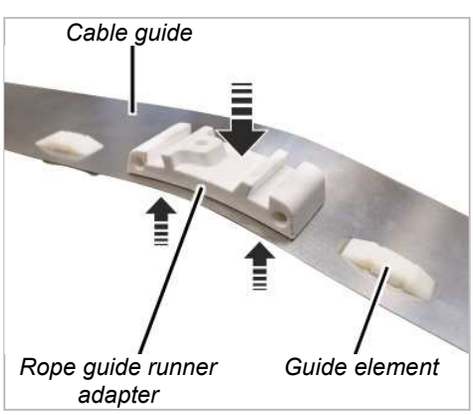


➔ Note the holes for the rope guide runner.



➔ Press guide elements into cable guide.

The hooks on the guide elements must engage and fit tightly!

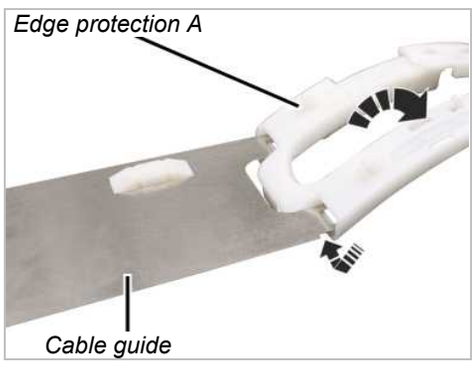


➔ Screw the rope guide runner adapter onto both cable guides from the outside. 1.5 Nm.

The thicker side of the guide elements must point inwards.

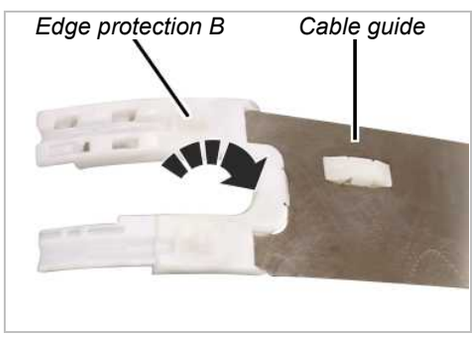
On both sides of the cable guide:

The edge protections are identified by the letters A and B.



➔ Slide edge protection A onto the cable guide and engage it.

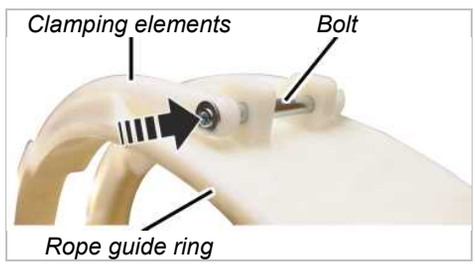
The rope guide runner adapter must be on the upper side.



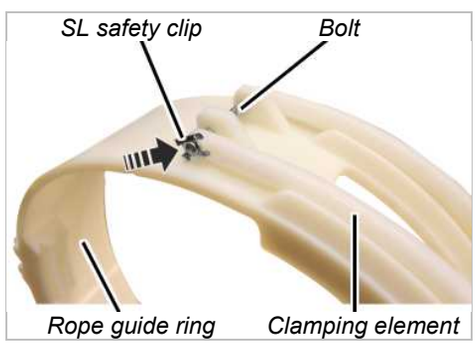
➔ Push edge protection B onto the other side of the cable guide and engage it.

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 7000

On the cable guide for a wire rope hoist with hoist drive GM 7000, the clamping elements are secured on the rope guide ring using a bolt and a SL safety clip. The eyebolts for tensioning the rope guide ring are located below on the clamping elements.



➔ Push the bolt through the clamping elements and the rope guide ring.

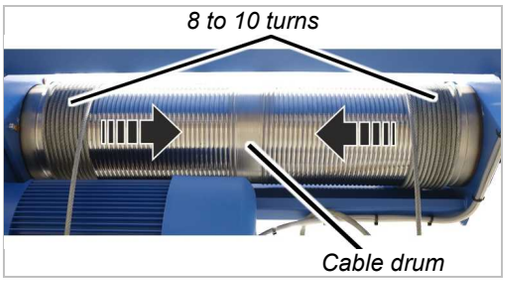


➔ Push the SL safety clip onto the bolt.

➔ Grease the groove of the rope guide ring from the inside.

Lubricant: “Molykote PG-75”. For details, see “Lubricants”, page 107.

INSTALLING THE CABLE GUIDE



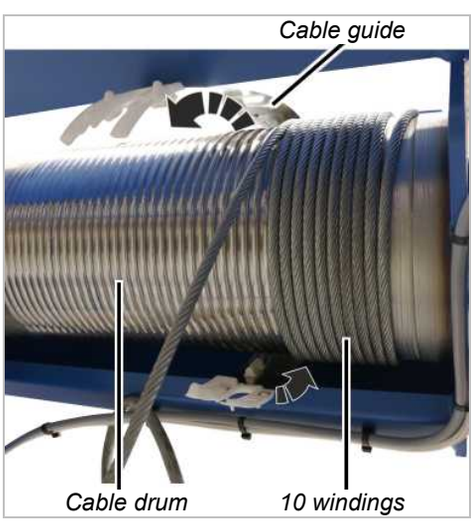
➔ Pull the wire rope taut and use the motor to wind it onto the cable drum by 8 to 10 turns, ensuring it is taut and not twisted.

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 5000

On a wire rope hoist with hoist drive GM 5000, the cable guide consists of a plate, guide elements, edge protection and tension springs.

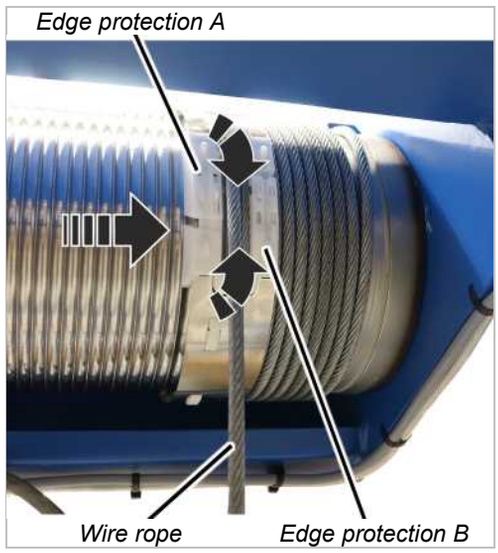
On both sides of the cable drum:

Make sure when assembling the cable guides, that there is one cable guide for the left side and one for the right side.



➔ Push the cable guide around the cable drum.

The guide elements must point inwards.

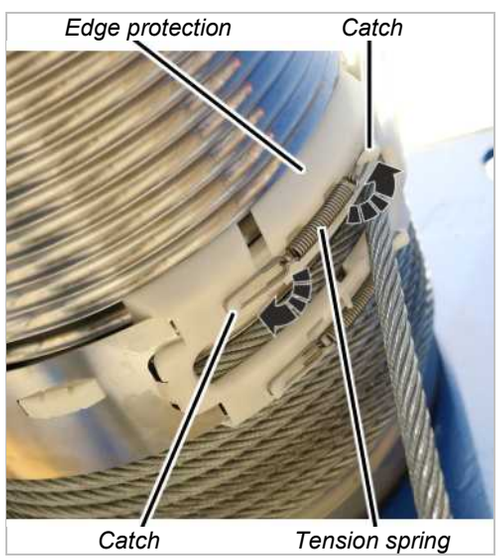


➔ Position the wire rope between edge protection A and edge protection B.

➔ Push edge protections A and B together.

Both ends of the rope guide ring must lie in the same groove of the cable drum. The ends of the rope guide ring must be aligned.

The edge protections must engage with each other.



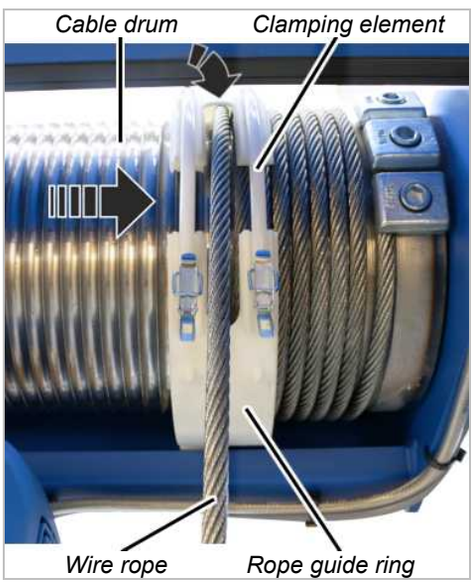
➔ Attach the tension springs in the upper and lower catches of edge protection A and B such that the rope guide ring rests firmly on the wire rope, but can still turn freely.

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 7000

On a wire rope hoist with hoist drive GM 7000, the cable guide consists of a rope guide ring with clamping elements.

On both sides of the cable drum:

Make sure when assembling the cable guides, that there is one cable guide for the left side and one for the right side.

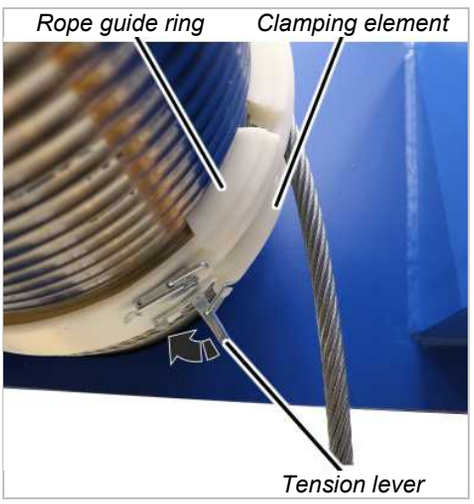


➔ Pull the rope guide ring with clamping elements onto the cable drum.

➔ Position the wire rope between the clamping elements.

➔ Check that both ends of the rope guide ring lie in the same groove of the cable drum.

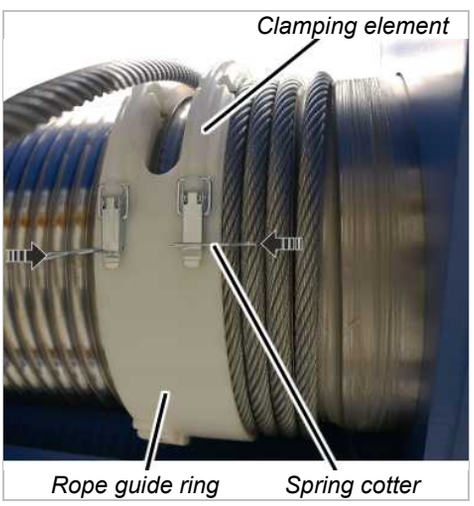
➔ The ends of the rope guide ring must be aligned.



➔ Insert the tension lever on the clamping elements into the rope guide ring and close it.

If necessary:

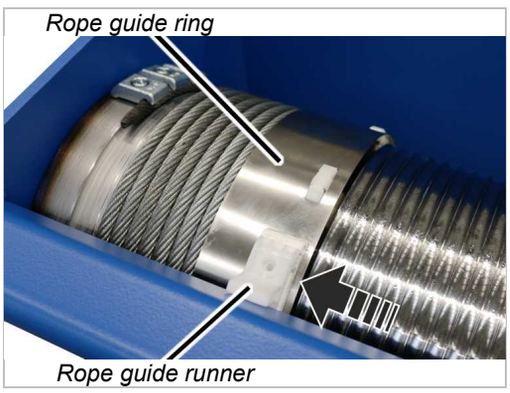
➔ Open the tension lever again and adjust the rope guide ring with the eyebolts so that the rope guide ring rests securely on the wire rope but can still be rotated.



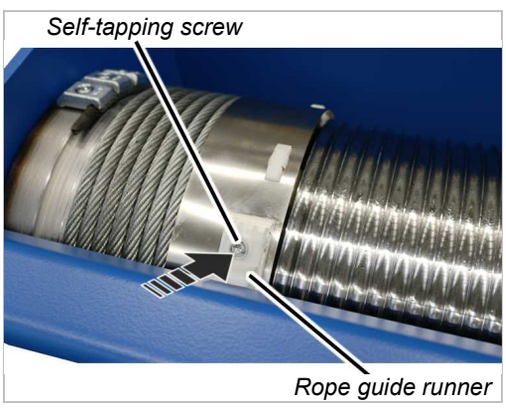
➔ Push the spring cotter into the tension lever of the clamping elements.

INSTALLING THE ROPE GUIDE RUNNER

On both sides of the cable drum:

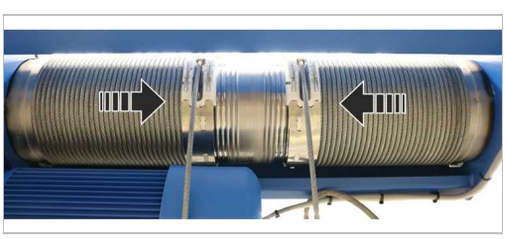


➔ Put the rope guide runner on the drum housing and push it laterally into the rope guide ring.



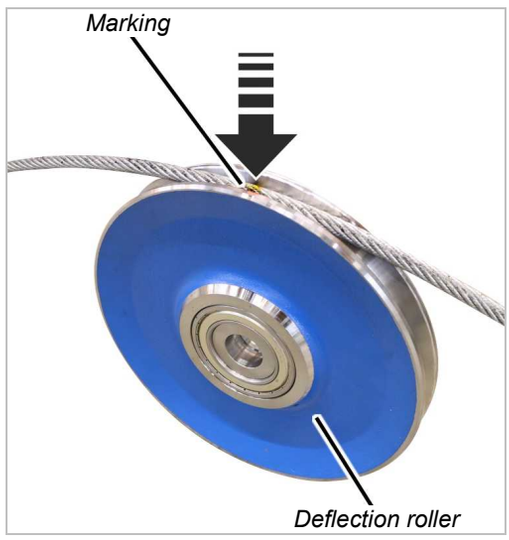
➔ Screw on the rope guide runner with the self-tapping screw M6x16. 5 Nm.

ASSEMBLING THE DEFLECTION ROLLER

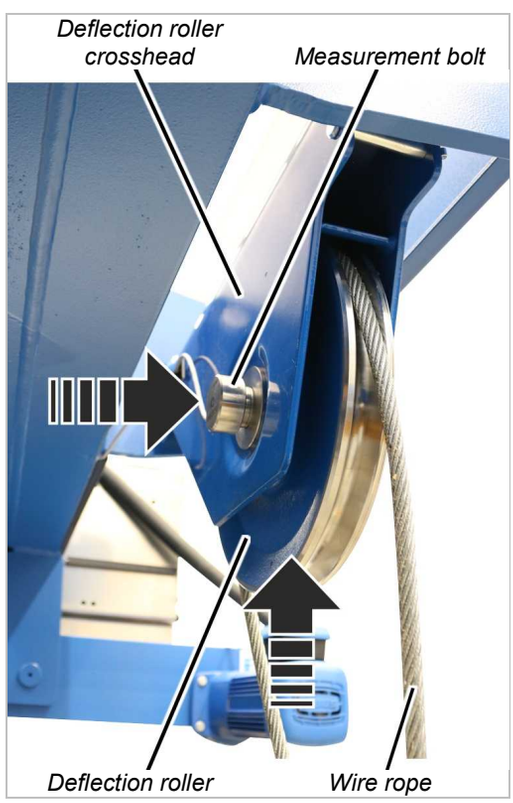


➔ Use the motor to wind up the wire rope.

The wire rope must not be completely wound up.



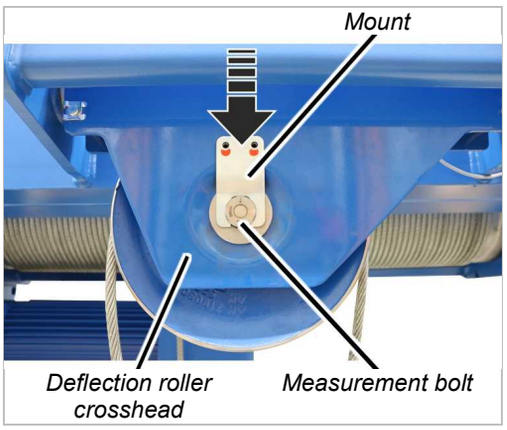
➔ Lay the wire rope into the groove of the deflection roller with the marking in the centre.



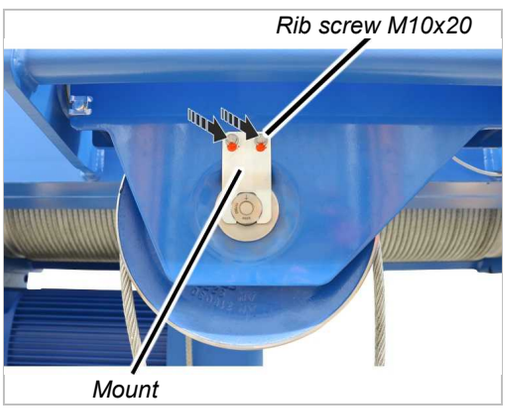
➔ Push the deflection roller with the wire rope into the deflection roller crosshead.

➔ Push the measurement bolt through the deflection roller and the deflection roller crosshead.

The arrow on the measurement bolt must point downward!



➔ Slide the mount onto the measurement bolt.

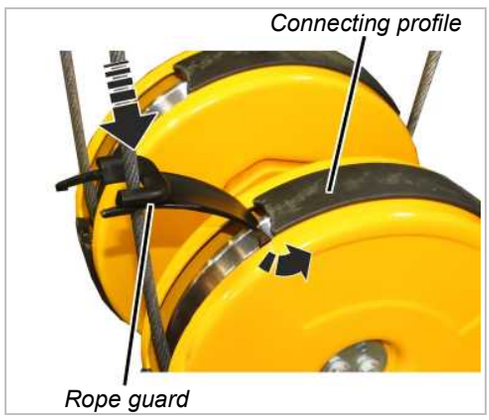


➔ Screw in the mount with the rib screws M10x20. 75 Nm.

➔ Secure the rib screws with screw lock lacquer.

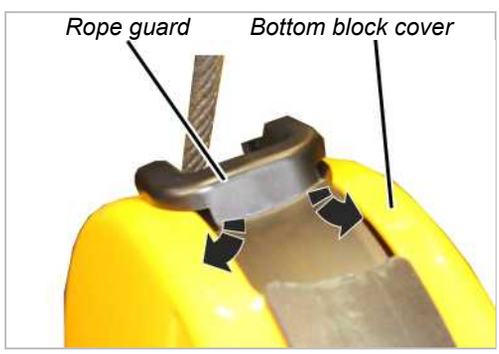
INSTALLING THE ROPE GUARD ON THE BOTTOM BLOCK

On all openings on the bottom block:



➔ Hold the rope guard on the wire rope as shown in the figure.

➔ Push the rope guard down along the wire rope with the front fork and bring the rear end under the connecting profile.



➔ First push the rope guard on the side with the short bracket into the bottom block cover.

➔ Then push the rope guard on the side with the long bracket into the bottom block cover.



➔ Push the second rope guard at the bottom into the connecting profile.

➔ Push the rope guard downwards behind the connecting profile and attach it above on the wire rope.

➔ First push the rope guard on the side with the short bracket into the bottom block cover.

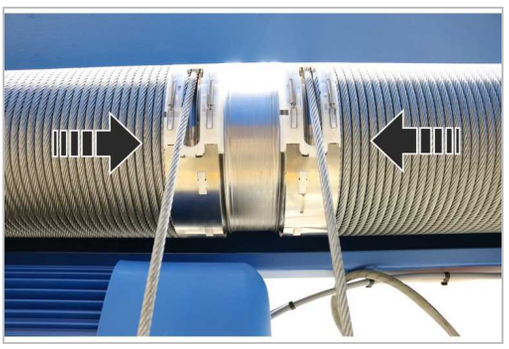
➔ Then push the rope guard on the side with the long bracket into the bottom block cover.



➔ Connect the brackets of the rope guards (2x) and push them together.

- A snapping sound can be heard.

COMPLETING WIRE ROPE REPLACEMENT



➔ Use the motor to wind up the wire rope completely.

ONLY WITH ABUCONTROL

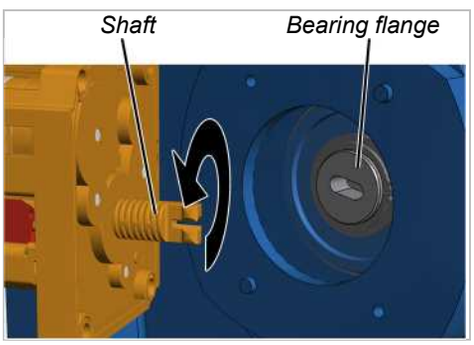
If ABUControl is installed for the wire rope hoist, refer to the product manual “ABUControl”, section “Exchanging the wire rope” to conclude the rope change.

Further steps are then described in this product manual.

ONLY WITH ABUS ELECTRICS 3 WITH CONTACTOR CONTROL OR ABULINER

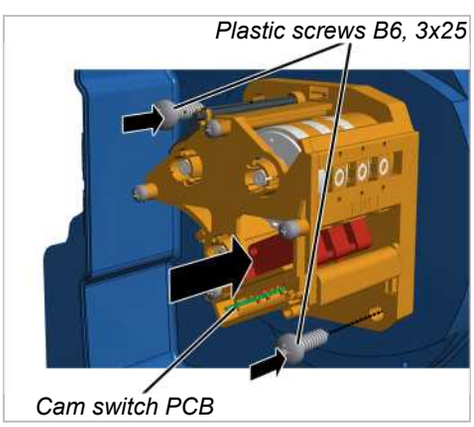
INSTALLING THE HOIST LIMIT SWITCH

➔ Move the load hooks to the position you noted previously.



➔ Open the housing cover of the hoist limit switch enclosure on the cable drum.

➔ Rotate the shaft of the gear limit switch so that it matches the elongated hole in the bearing flange of the cable drum.



➔ Insert the gear limit switch with the shaft in the elongated hole on the bearing flange and guide it onto the two pins.

➔ Screw the gear limit switch tight with the plastic screws B6, 3x25. 2.5 Nm.

CONNECTING THE HOIST LIMIT SWITCH

➔ Plug in the connector of the connection cable on the cam switch PCB.

➔ Close the housing cover.

If the switching points have moved:

Normally, the adjustment of the hoist limit switch after replacing the wire rope can in many cases be omitted, since the position of the load hook was previously measured and then travelled to again. In special cases (e.g. if the wire rope is severely damaged or has been incorrectly rolled onto the cable drum), the switching points may need to be readjusted.

➔ Reset switching points for the bottom hoist limiter, the top hoist limiter and the backup limiter.

With ABUS electrics 3 with contactor control: see product manual “ABUS gear limit switch”.

With ABUS electrics 3 with ABULiner: see product manual “Frequency converter control of the lifting speed for ABUS electrics 3”.

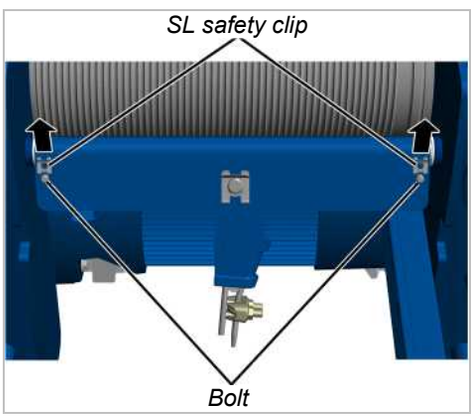
EXCHANGING THE FIXED POINT CROSSHEAD WITH ROPE SOCKET AND ROPE WEDGE

If the fixed point crosshead, the rope socket or the rope wedge are damaged, they must be replaced.

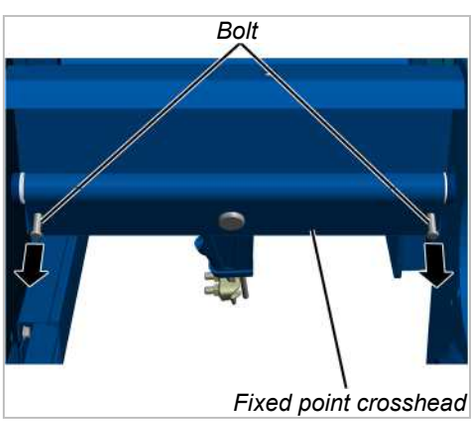
DISASSEMBLING THE FIXED POINT CROSSHEAD

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 800 TO GM 3000

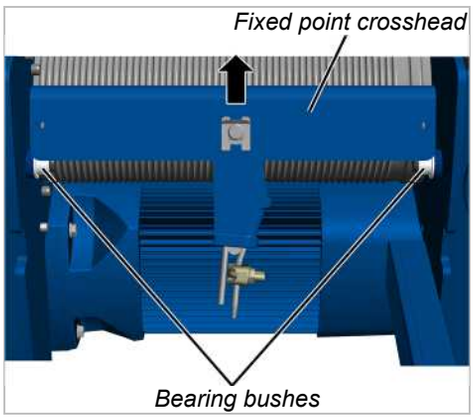
On a wire rope hoist with hoist drive GM 800 to GM 3000, the fixed point crosshead hangs on bearing bushes and is secured with bolts and SL safety clips.



➔ Pull the SL safety clips (2x) from the bolts.



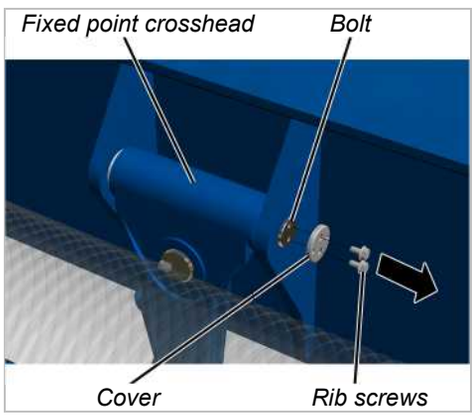
➔ Pull the bolts (2x) out of the fixed point crosshead.



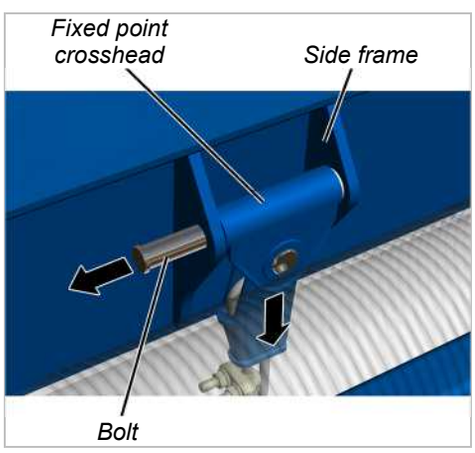
➔ Lift the fixed point crosshead from the bearing bushes.

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 5000 TO GM 7000

On a wire rope hoist with hoist drive GM 5000 to GM 7000, the fixed point crosshead hangs on bearing bushes and is secured to the side panels on the trolley frame using a laterally inserted bolt with a cover.



- Release the rib screws on the cover.
- Take the cover off the bolt.



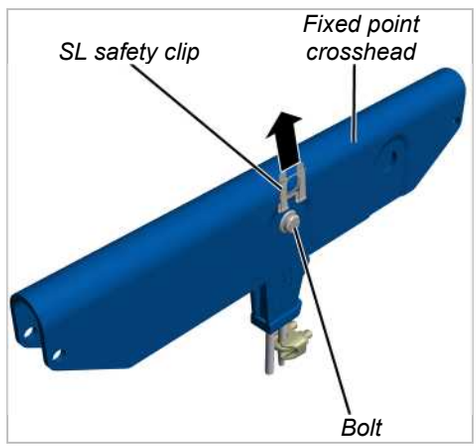
- Pull the bolt out of the fixed point crosshead and the side panels.
- Pull the fixed point crosshead out of the trolley frame.

DISASSEMBLING THE ROPE SOCKET

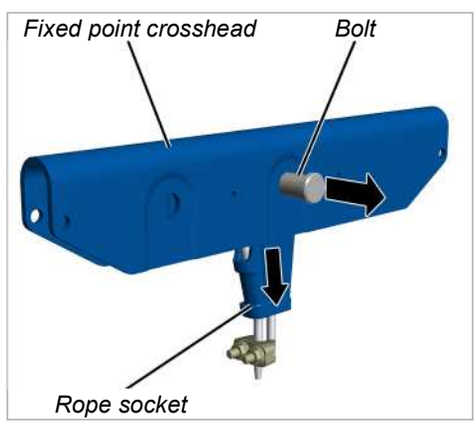
ONLY WITH WIRE ROPE HOISTS WITH HOIST DRIVE GM 800 TO GM 3000 WITH LIS-SE

This section only applies to a wire rope hoist with hoist drive GM 800 to GM 3000 with a load indicator system LIS-SE.

For a wire rope hoist with a load indicator system LIS-SE, the rope socket is fastened to the fixed point crosshead with a bolt and an SL safety clip.



- Pull SL safety clip from the bolt.

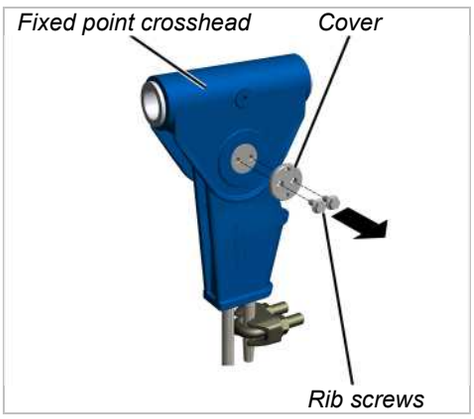


- Pull the bolt out of the fixed point crosshead and the rope socket.
- Pull the rope socket out of the fixed point crosshead.

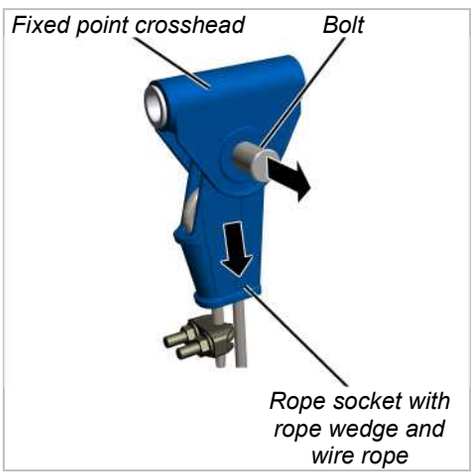
ONLY WITH WIRE ROPE HOISTS WITH HOIST DRIVE GM 5000 TO GM 7000 WITH LIS-SE

This section only applies to a wire rope hoist with hoist drive GM 5000 to GM 7000 with a load indicator system LIS-SE.

For a wire rope hoist with a load indicator system LIS-SE, the rope socket is fastened to the fixed point crosshead with a measurement bolt and a cover.



- Release the rib screws on the cover.
- Take the cover off the bolt.

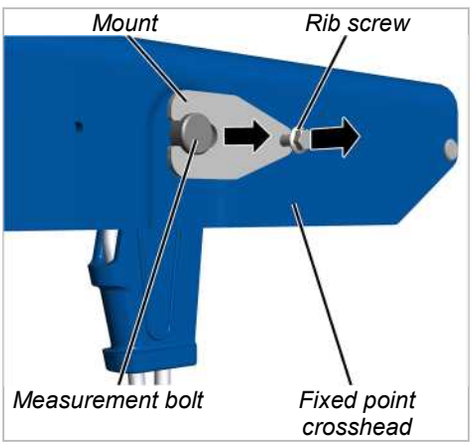


- Pull the bolt out of the fixed point crosshead and the rope socket.
- Pull the rope socket with the rope wedge and wire rope out of the fixed point crosshead.

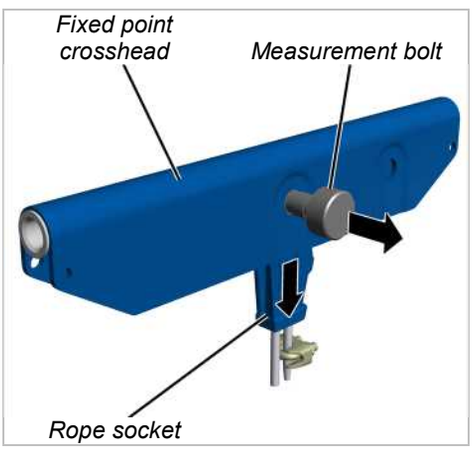
ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 800 TO GM 3000 WITH LIS-SV OR ABUCONTROL

This section only applies to a wire rope hoist with hoist drive GM 800 to GM 3000 with a load indicator system LIS-SV or ABUControl.

For a wire rope hoist with a load indicator system LIS-SV or ABUControl, the rope socket is fastened to the fixed point crosshead with a measurement bolt and a mount.



- Release rib screw M6x12.
- Slide the mount from the measurement bolt.

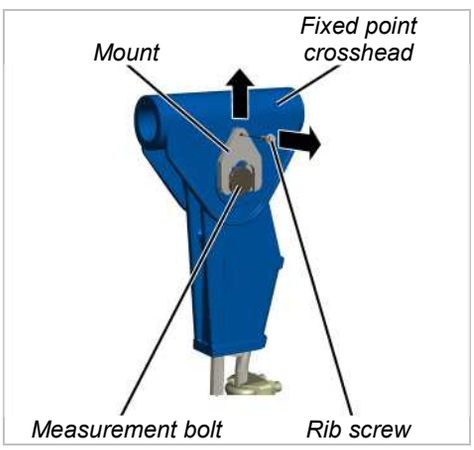


- Pull the measurement bolt out of the fixed point crosshead and the rope socket.
- Pull the rope socket out of the fixed point crosshead.

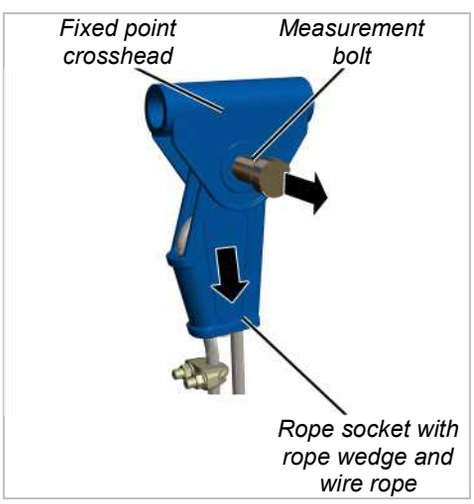
ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 5000 TO GM 7000 WITH LIS-SV OR ABUCONTROL

This section only applies to a wire rope hoist with hoist drive GM 5000 to GM 7000 with a load indicator system LIS-SV or ABUControl.

For a wire rope hoist with a load indicator system LIS-SV or ABUControl, the rope socket is fastened to the fixed point crosshead with a measurement bolt and a mount.

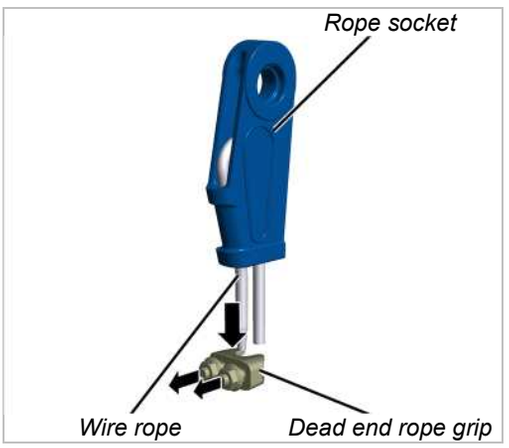


- Release the rib screw M6x20 on the mount.
- Pull the mount off the measurement bolt.

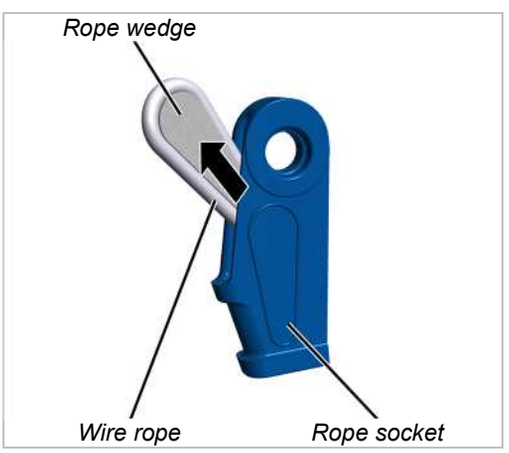


- Pull the measurement bolt out of the fixed point crosshead and the rope socket.
- Pull the rope socket with the rope wedge and wire rope out of the fixed point crosshead.

REMOVING THE ROPE WEDGE

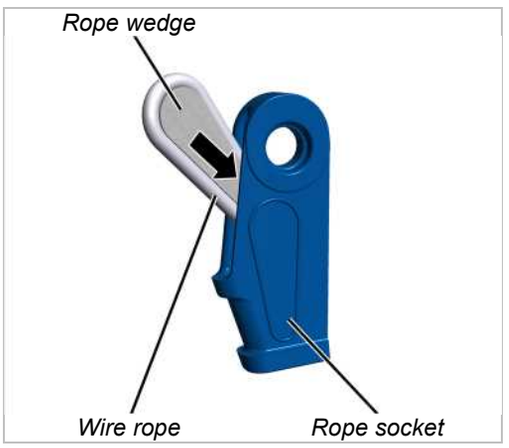


- Release the rib nuts on the dead end rope grip.
- Pull the dead end rope grip from the end of the wire rope.



- Pull the rope wedge with wire rope out of the rope socket.

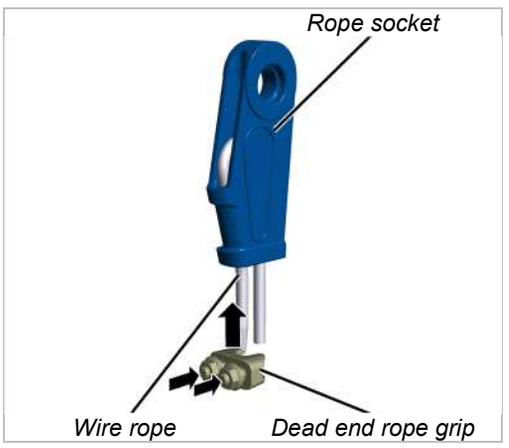
INSTALLING THE ROPE WEDGE



➔ Lay the end of the wire rope around the rope wedge.

The wire rope must lie level in the groove of the rope wedge!

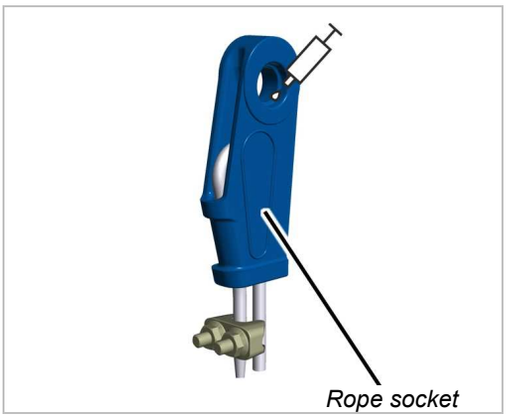
➔ Push the rope wedge with the wire rope into the rope socket.



➔ Push the dead end rope grip onto the end of the wire rope.

➔ Screw in rib nuts of the dead end rope grip.

Size	Rib nut	Tightening torque
GM 800	M6	5 Nm
GM 1000	M8	10 Nm
GM 2000	M8	10 Nm
GM 3000	M8	10 Nm
GM 5000	M14	49 Nm
GM 6000	M14	49 Nm
GM 7000	M16	107 Nm



➔ Lubricate the rope socket.

Lubricant: “High temperature paste PBC 1574”. For details, see “Lubricants”, page 107.

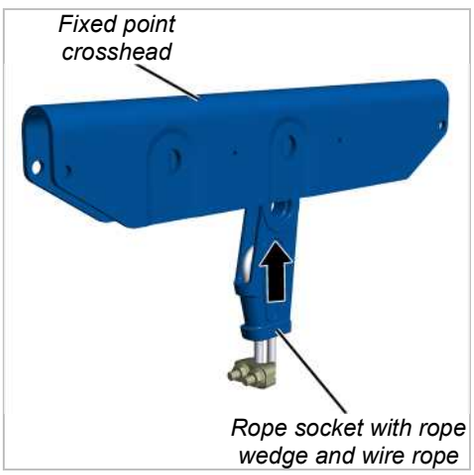
Do not lubricate the measurement bolt / collar bolt or the fixed point crosshead. No lubricant should contact the fixed point crosshead.

INSTALLING THE ROPE SOCKET

ONLY WITH WIRE ROPE HOISTS WITH HOIST DRIVE GM 800 TO GM 3000 WITH LIS-SE

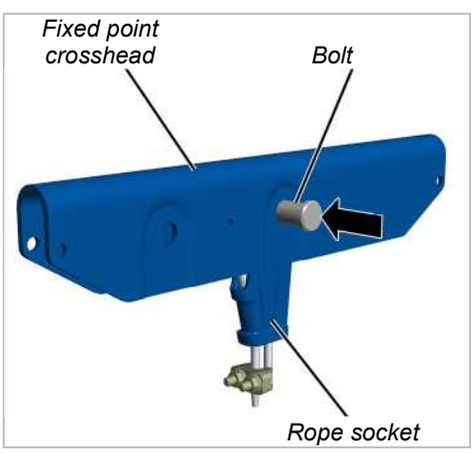
This section only applies to a wire rope hoist with hoist drive GM 800 to GM 3000 with a load indicator system LIS-SE.

For a wire rope hoist with a load indicator system LIS-SE, the rope socket is fastened to the fixed point crosshead with a bolt and an SL safety clip.

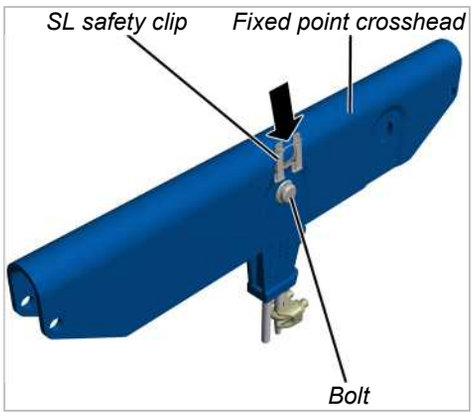


➔ Push the rope socket with rope wedge and wire rope into the fixed point crosshead.

Note the marking on the fixed point crosshead.



➔ Push the bolt through the fixed point crosshead and the rope socket.

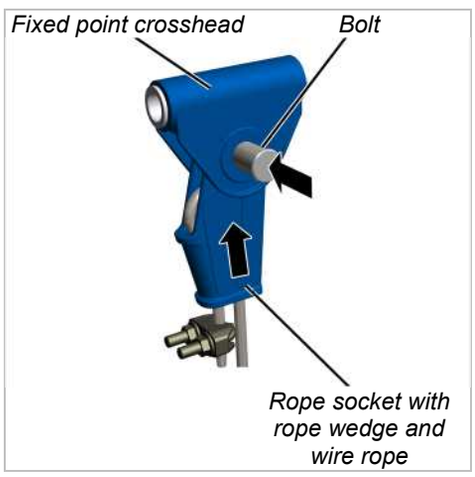


➔ Push the SL safety clip onto the bolt.

ONLY WITH WIRE ROPE HOISTS WITH HOIST DRIVE GM 5000 TO GM 7000 WITH LIS-SE

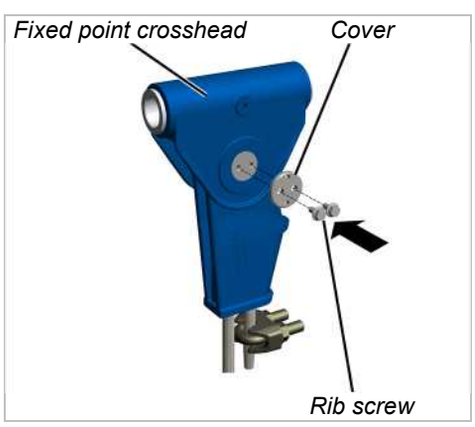
This section only applies to a wire rope hoist with hoist drive GM 5000 to GM 7000 with a load indicator system LIS-SE.

For a wire rope hoist with a load indicator system LIS-SE, the rope socket is fastened to the fixed point crosshead with a measurement bolt and a cover.



➔ Push the rope socket with rope wedge and wire rope into the fixed point crosshead.

➔ Push the bolt through the fixed point crosshead and the rope socket.



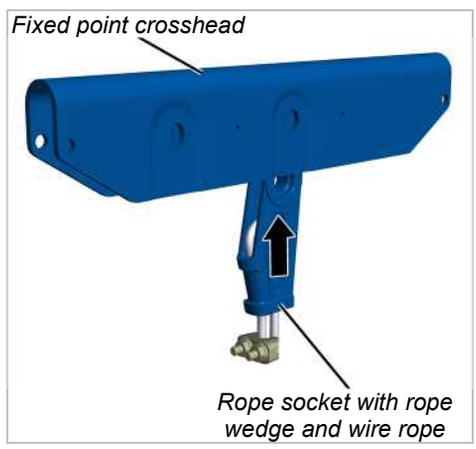
➔ Screw the cover tight with rib screws (2x) on the bolt.

Size	Rib screw	Tightening torque
GM 5000	M6x12	19 Nm
GM 6000	M6x12	19 Nm
GM 7000	M8x16	42 Nm

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 800 TO GM 3000 WITH LIS-SV OR ABUCONTROL

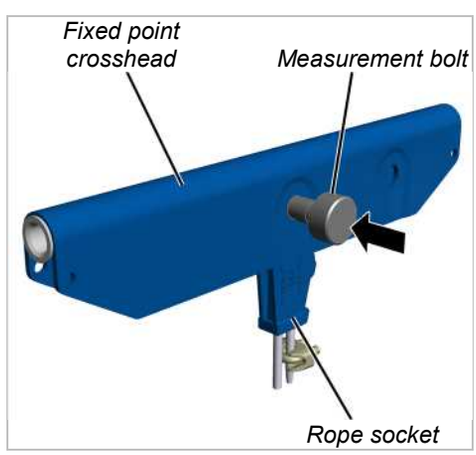
This section only applies to a wire rope hoist with hoist drive GM 800 to GM 3000 with a load indicator system LIS-SV or ABUControl.

For a wire rope hoist with a load indicator system LIS-SV or ABUControl, the rope socket is fastened to the fixed point crosshead with a measurement bolt and a mount.



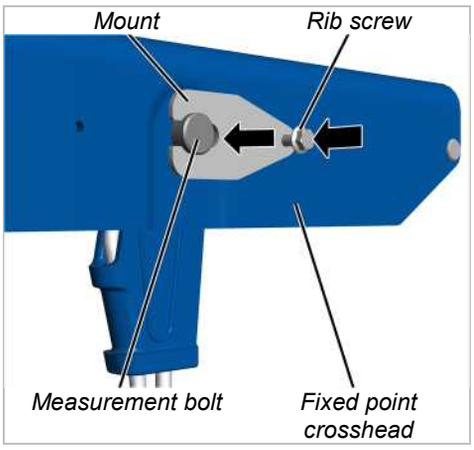
➔ Push the rope socket with rope wedge and wire rope into the fixed point crosshead.

Note the marking on the fixed point crosshead.



➔ Push the measurement bolt through the fixed point crosshead and the rope socket.

The arrow on the measurement bolt must point downward!



➔ Check the mount for deformations.

If the mount is deformed, it must be replaced.

➔ Slide the mount onto the measurement bolt.

The mount must allow itself to be inserted into the groove of the measurement bolt without tension.

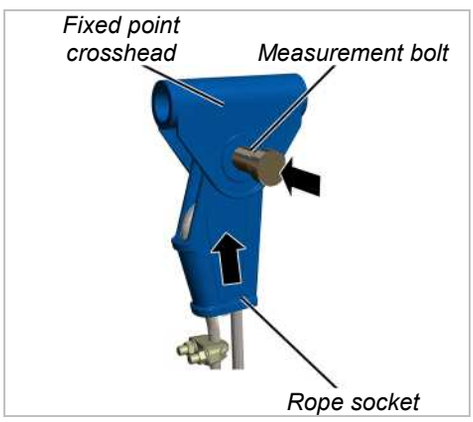
The measurement bolt should not have any pretension!

➔ Screw on the mount with the rib screw M6x12. 19 Nm.

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 5000 TO GM 7000 WITH LIS-SV OR ABUCONTROL

This section only applies to a wire rope hoist with hoist drive GM 5000 to GM 7000 with a load indicator system LIS-SV or ABUControl.

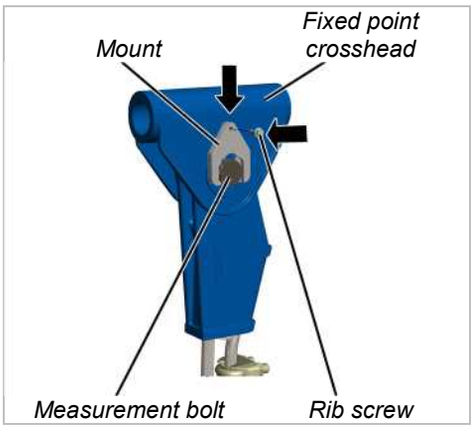
For a wire rope hoist with a load indicator system LIS-SV or ABUControl, the rope socket is fastened to the fixed point crosshead with a measurement bolt and a mount.



➔ Push the rope socket with rope wedge and wire rope into the fixed point crosshead.

➔ Push the measurement bolt through the fixed point crosshead and the rope socket.

The arrow on the measurement bolt must point downward!



➔ Check the mount for deformations.

If the mount is deformed, it must be replaced.

➔ Slide the mount onto the measurement bolt.

The mount must allow itself to be inserted into the groove of the measurement bolt without tension.

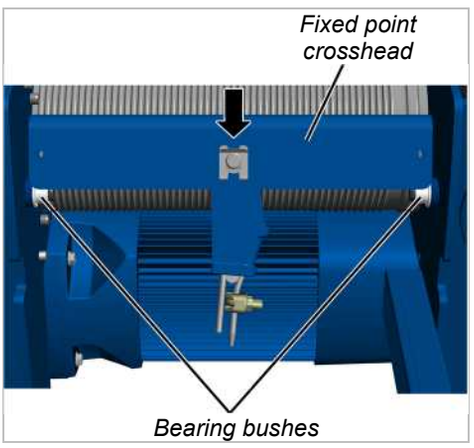
The measurement bolt should not have any pretension!

➔ Screw on the mount with the rib screw M6x20. 19 Nm.

FITTING THE FIXED POINT CROSSHEAD

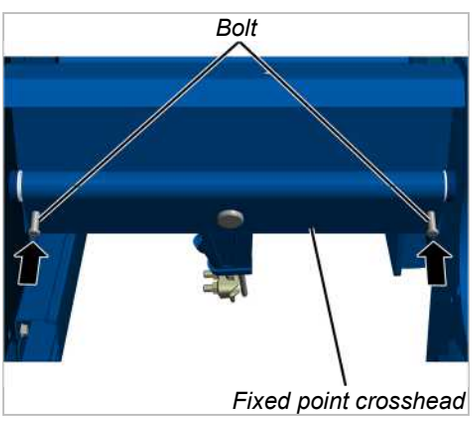
ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 800 TO GM 3000

On a wire rope hoist with hoist drive GM 800 to GM 3000, the fixed point crosshead hangs on bearing bushes and is secured with bolts and SL safety clips.

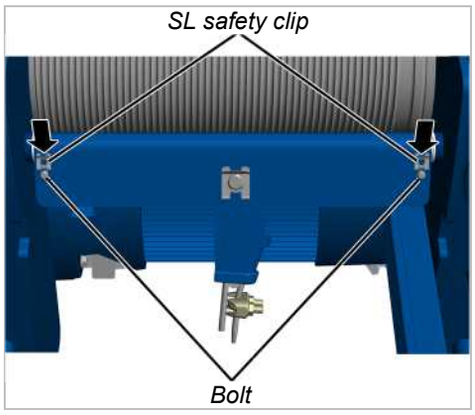


➔ Mount the fixed point crosshead on the bearing bushes.

When doing this, ensure that the marking on the fixed point crosshead points in the direction of the hoist drive.



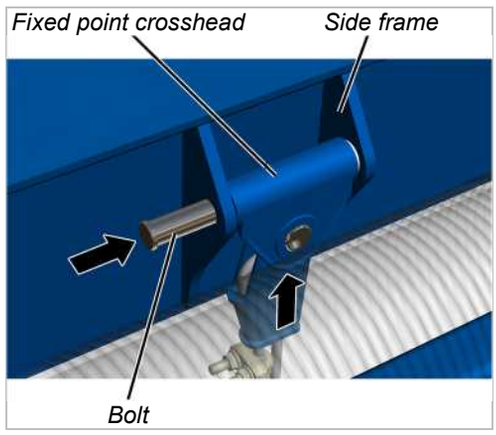
➔ Push the bolts (2x) into the fixed point crosshead.



➔ Push the SL safety clips (2x) onto the bolts.

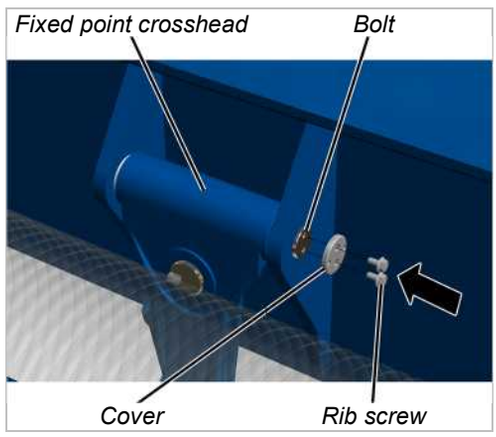
ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 5000 TO GM 7000

On a wire rope hoist with hoist drive GM 5000 to GM 7000, the fixed point crosshead hangs on bearing bushes and is secured to the side panels on the trolley frame using a laterally inserted bolt with a cover.



➔ Push the fixed point crosshead between the side panels on the trolley frame.

➔ Push the bolt through the side panels and the fixed point crosshead.



➔ Screw the cover tight with rib screws (2x) on the bolt.

Size	Rib screw	Tightening torque
GM 5000	M6x12	19 Nm
GM 6000	M6x12	19 Nm
GM 7000	M8x16	42 Nm

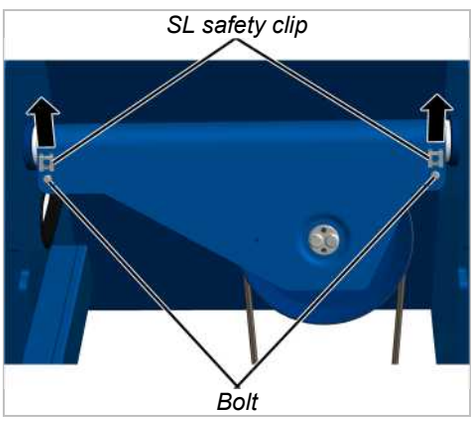
REPLACING DEFLECTION ROLLER CROSSHEAD FOR REEVING 4/1 OR 6/1

If the deflection roller crosshead is damaged, it must be replaced.

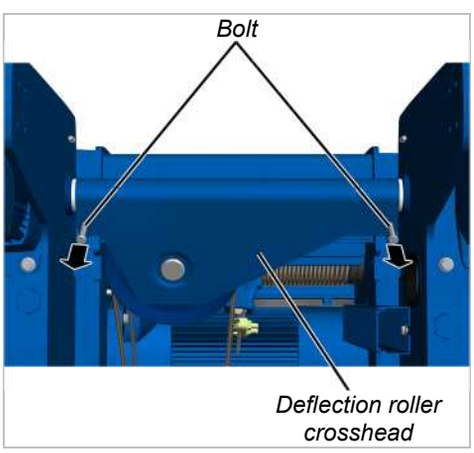
DISASSEMBLING THE DEFLECTION ROLLER CROSSHEAD

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 800 TO GM 6000 WITH REEVING 4/1

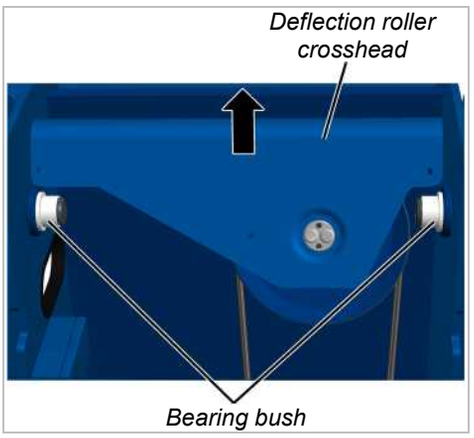
On a wire rope hoist with hoist drive GM 800 to GM 6000, the deflection roller crosshead hangs on bearing bushes and is secured with bolts and SL safety clips.



➔ Pull the SL safety clips from the bolts.



➔ Pull the bolts (2x) out of the deflection roller crosshead.

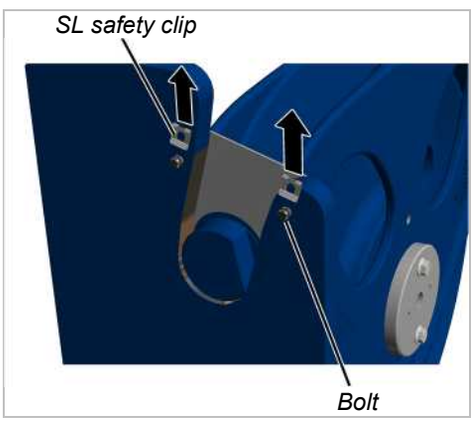


➔ Lift the deflection roller crosshead from the bearing bushes.

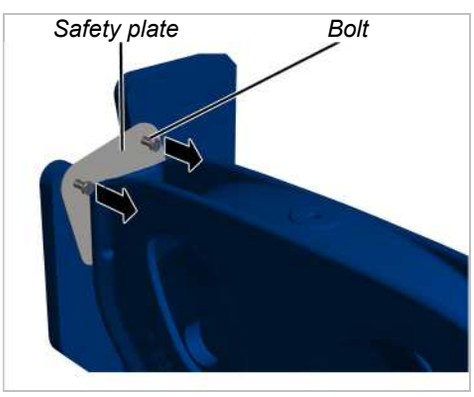
ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 7000 AND REEVING 4/1

On a wire rope hoist with hoist drive GM 7000 and a reeving 4/1, the deflection roller crosshead is installed on the outside of the trolley frame.

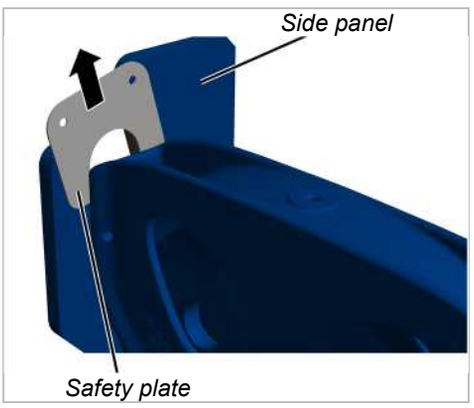
On both sides:



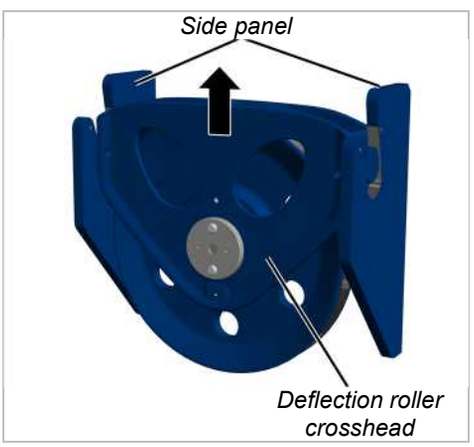
➔ Pull SL safety clip from the bolt.



➔ Pull the bolt out of the safety plate and the side panel.



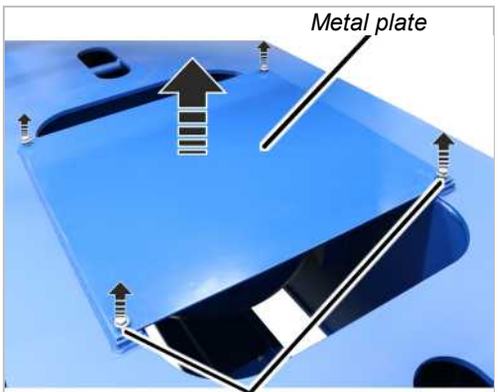
➔ Pull out the safety plate.



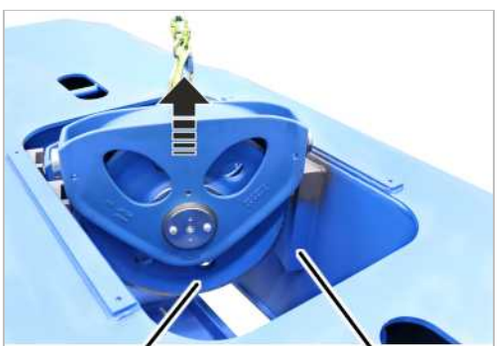
➔ Lift the deflection roller crosshead out of the side panels.

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 7000 AND REEVING 6/1

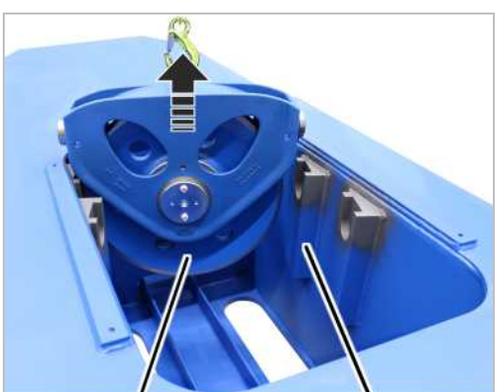
On a wire rope hoist with hoist drive GM 7000, the deflection roller crossheads are installed in the trolley frame.



- Release the rib screws M10x16 (4x) on the plate.
- Lift the plate off the trolley frame.



- Lift the deflection roller crosshead out of the trolley frame.

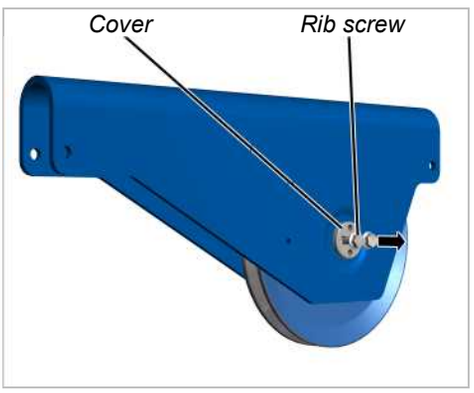


- Lift the second deflection roller crosshead out of the trolley frame.

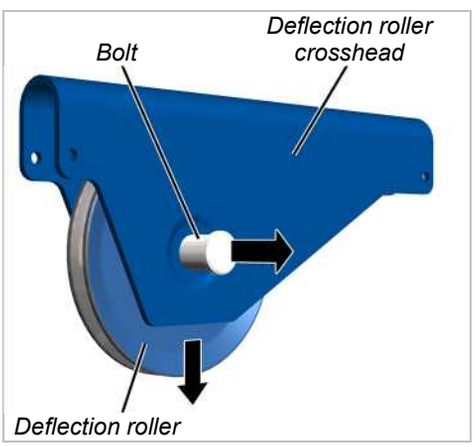
DISASSEMBLING THE DEFLECTION ROLLER

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 800 TO GM 6000 WITH REEVING 4/1

On a wire rope hoist with hoist drive GM 800 to GM 6000 and a reeving 4/1, the deflection roller is secured on the deflection roller crosshead using a bolt with a cover.



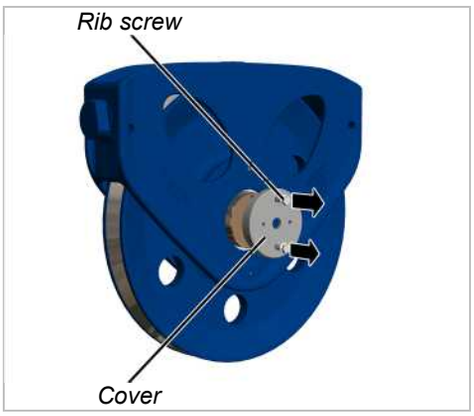
- Release the rib screws (2x) on the cover.
- Take the cover off the deflection roller crosshead.



- Pull the bolt out of the deflection roller crosshead and deflection roller.
- Pull the deflection roller out of the deflection roller crosshead.

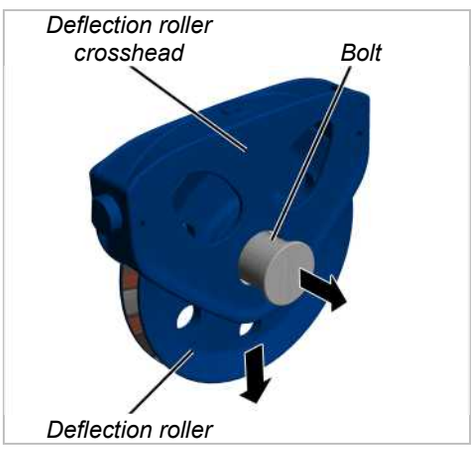
ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 7000

On a wire rope hoist with hoist drive GM 7000, the deflection roller is secured on the deflection roller using a bolt with a cover.



➔ Release the rib screws (2x) on the cover.

➔ Take the cover off the deflection roller crosshead.



➔ Pull the bolt out of the deflection roller crosshead and deflection roller.

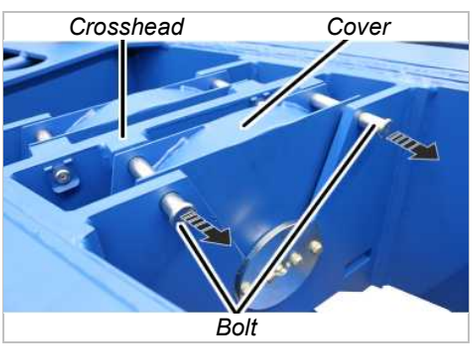
➔ Pull the deflection roller out of the deflection roller crosshead.

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 6000 AND REEVING 6/1

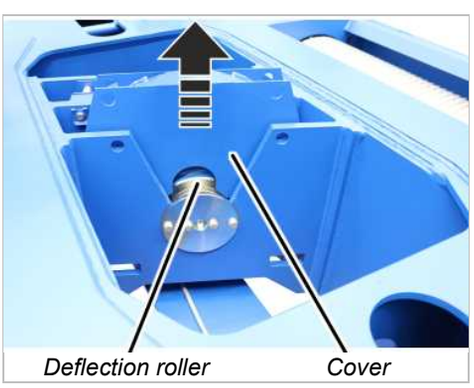
On a wire rope hoist with hoist drive GM 6000 and a reeving 6/1, the deflection roller crossheads are welded to the trolley frame. Only the deflection rollers themselves can therefore be replaced.



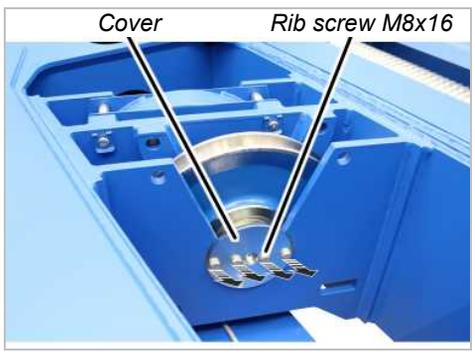
➔ Pull the SL safety clip (2x) off the bolts (2x).



➔ Pull the bolt out of the crosshead and the cover.

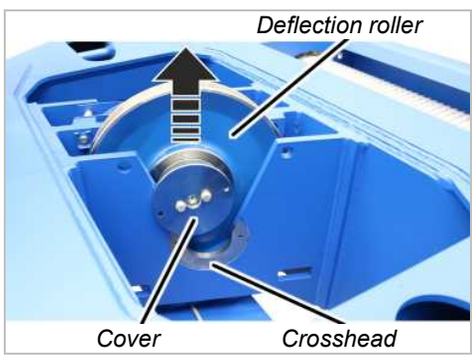


➔ Lift the cover off the deflection roller.

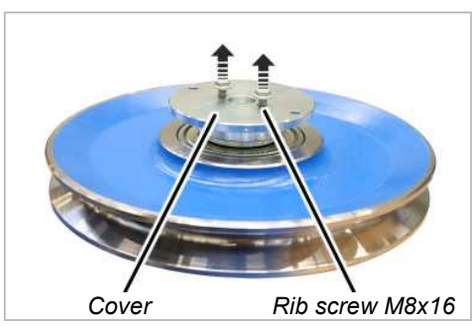


➔ Release the rib screws M8x16 (4x) on the cover.

The outer rib screws can be removed fully. The inner rib screws must stay loosely fastened on the cover because the cover is screwed to the bolt.

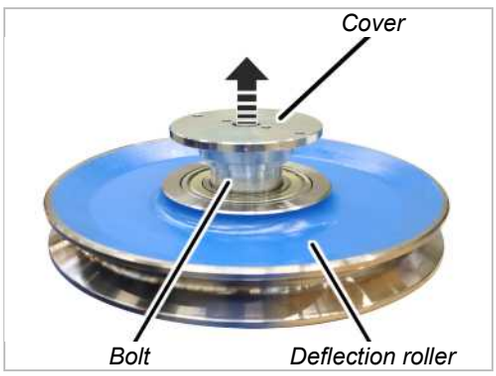


➔ Lift the deflection roller out of the crosshead.

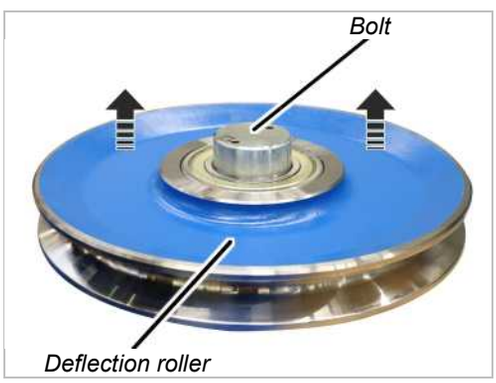


➔ Lay the deflection roller on an even surface with the bolt facing downward.

➔ Fully release the rib screws M8x16 (2x) on the cover.



➔ Lift the cover off the bolt.

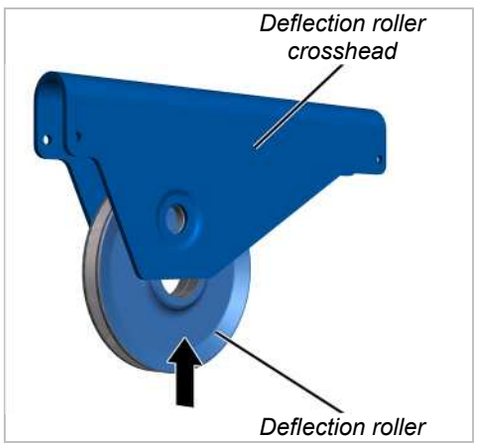


➔ Lift the deflection roller off the bolt.

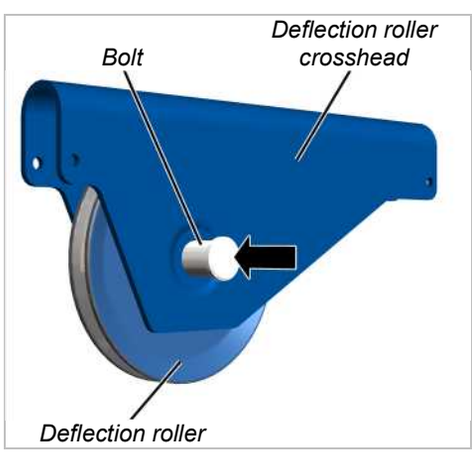
➔ Move exactly the same with the second deflection roller.

ASSEMBLING THE DEFLECTION ROLLER

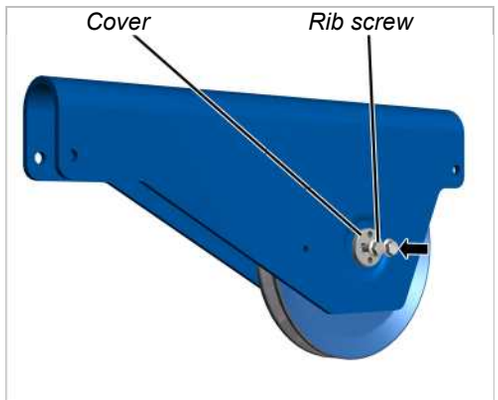
ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 800 TO GM 6000 WITH REEVING 4/1



➔ Push the deflection roller into the deflection roller crosshead.



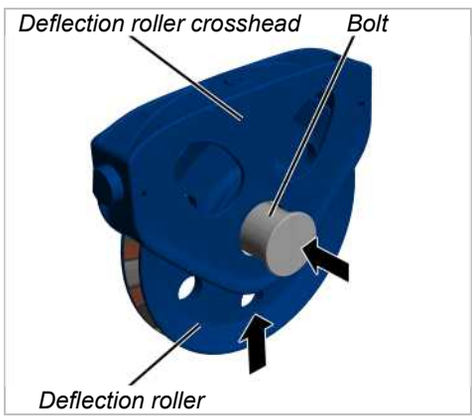
➔ Push the bolt through the deflection roller crosshead and deflection roller.



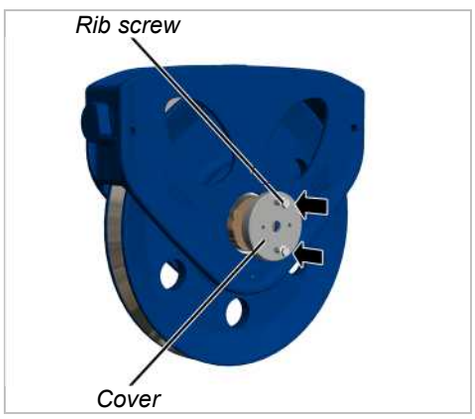
➔ Screw on the cover from the other side at the bolt with rib screws (2x).

Size	Rib screw	Tightening torque
GM 800	M6x12	19 Nm
GM 1000	M6x12	19 Nm
GM 2000	M8x16	42 Nm
GM 3000	M10x20	75 Nm
GM 5000	M10x20	75 Nm
GM 6000	M10x20	75 Nm

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 7000



➔ Push the deflection roller into the deflection roller crosshead.



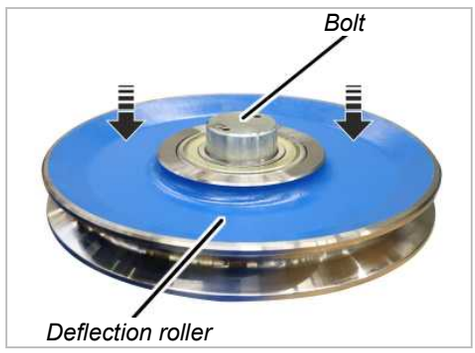
➔ Screw the cover tight with the rib screws M10x20 on the bolt. 75 Nm.

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 6000 AND REEVING 6/1

On a wire rope hoist with hoist drive GM 6000 and a reeving 6/1, the deflection roller crossheads are welded to the trolley frame. Only the deflection rollers themselves can therefore be replaced.

PREPARING THE DEFLECTION ROLLER

With both deflection rollers:

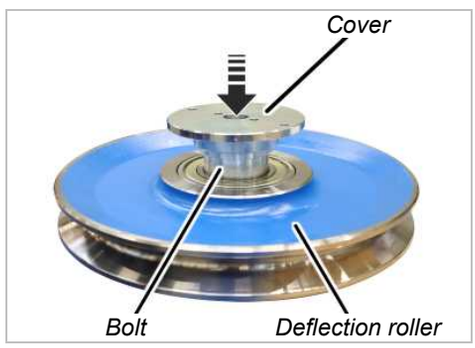


➔ Place the bolt on an even surface.

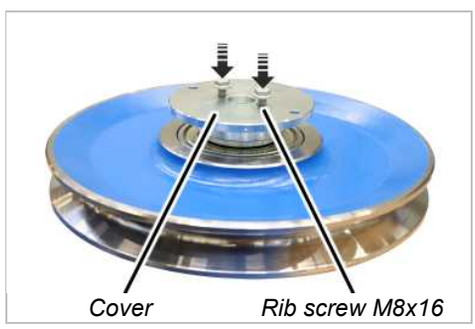
➔ Lubricate the bolt.

Lubricant: “High temperature paste PBC 1574”. For details, see “Lubricants”, page 107.

➔ Push the deflection roller onto the bolt.

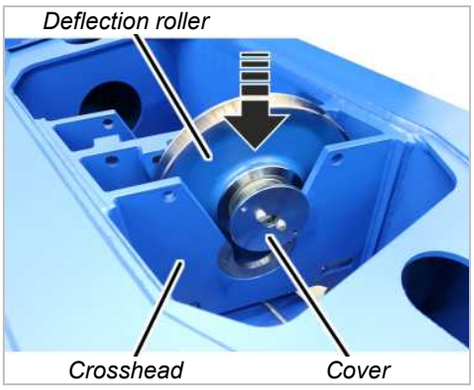


➔ Push the cover onto the bolt.



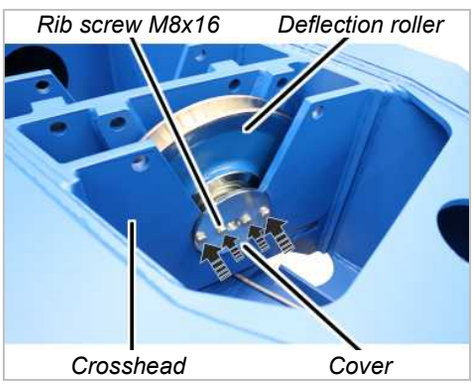
➔ Loosely screw on the cover with rib screws M8x16 (2x).

ASSEMBLING THE DEFLECTION ROLLER



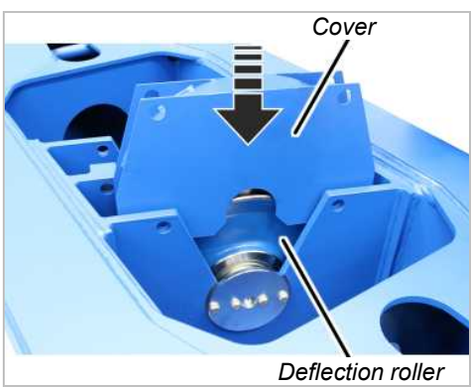
➔ Lift the deflection roller into the crosshead.

The cover must be in front of the crosshead suspension.

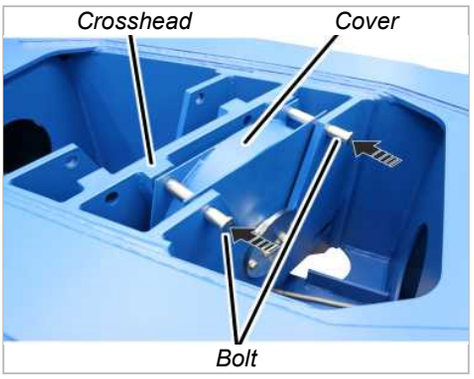


➔ Screw the deflection roller cover with rib screws M8x16 (2x) on the crosshead. 42 Nm.

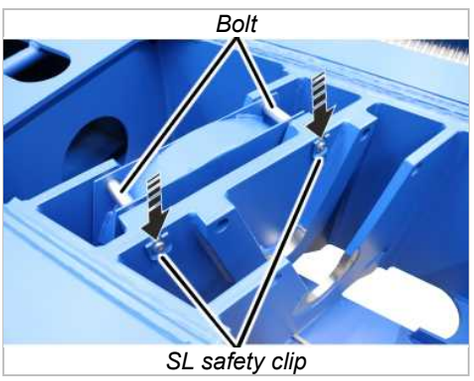
➔ Tighten the rib screws M8x16 (2x) on the cover. 42 Nm.



➔ Place the cover on the deflection roller.



➔ Push the bolts (2x) through the crosshead and the cover.



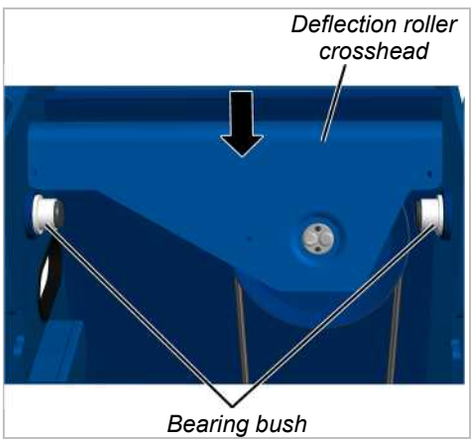
➔ Push the SL safety clips (2x) onto the bolt.

➔ Move exactly the same with the second deflection roller.

MOUNTING THE DEFLECTION ROLLER CROSSHEAD

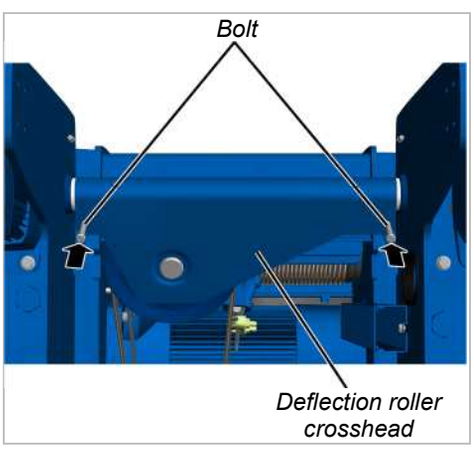
ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 800 TO GM 6000 WITH REEVING 4/1

On a wire rope hoist with hoist drive GM 800 to GM 6000, the deflection roller crosshead hangs on bearing bushes and is secured with bolts and SL safety clips.

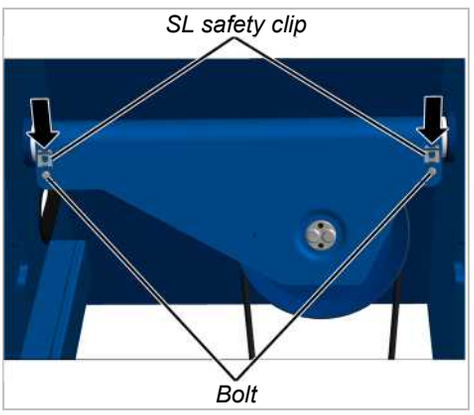


➔ Mount the deflection roller crosshead on the bearing bushes.

When doing this, ensure that the deflection roller points in the direction of the hoist drive.



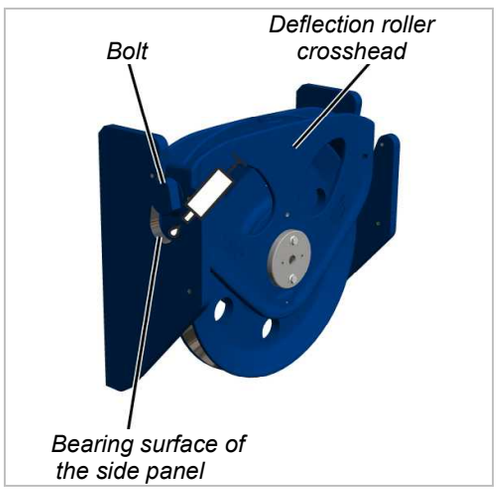
➔ Push the bolts (2x) into the deflection roller crosshead.



➔ Push the SL safety clips (2x) onto the bolts.

ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 7000 WITH REEVING 4/1

On a wire rope hoist with hoist drive GM 7000 and a reeving 4/1, the deflection roller crosshead is installed on the outside of the trolley frame.



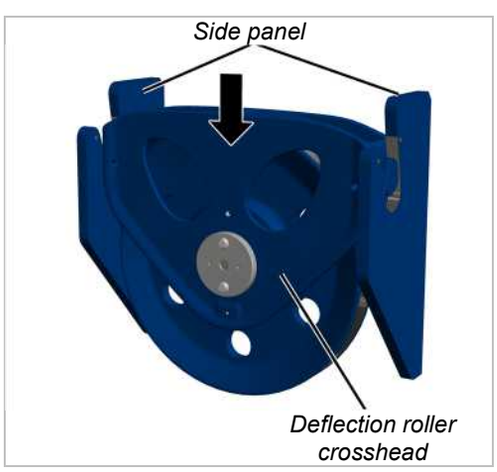
➔ Lubricate the bearing surface of the side panel.

Lubricant: “OKS 511 bonded coating”. For details, see “Lubricants”, page 107.

➔ **Let the bonded coating dry for 30 minutes!**

➔ Lubricate the side panel bearing surface again.

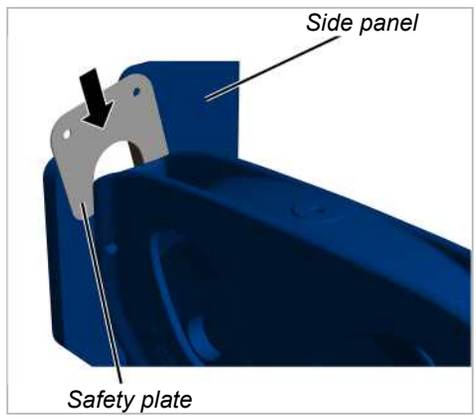
Lubricant: “High temperature paste PBC 1574”. For details, see “Lubricants”, page 107.



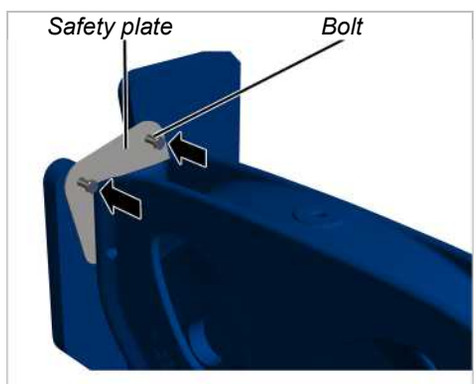
➔ Lift the deflection roller crosshead into the side panel.

You must be able to move the deflection roller crosshead back and forth by hand.

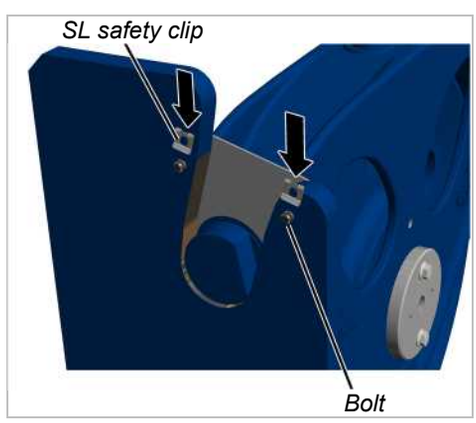
On both sides:



➔ Push the safety plate into the side panel.



➔ Push the bolt through the safety plate and the side panel.

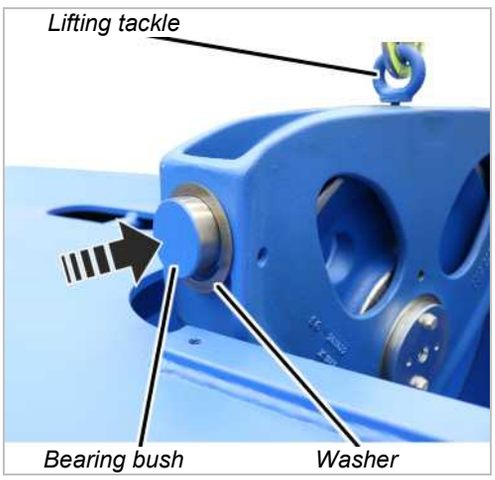


➔ Push the SL safety clip onto the bolts.

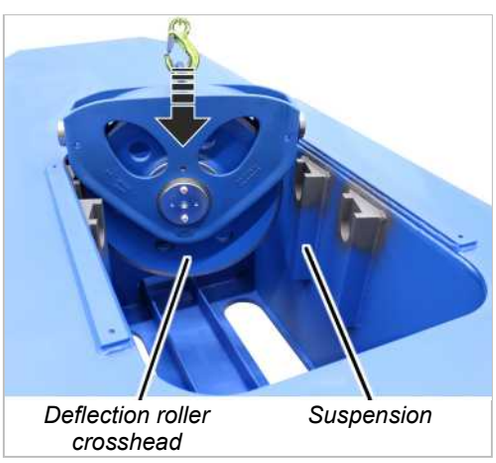
ONLY WITH WIRE ROPE HOIST WITH HOIST DRIVE GM 7000 AND REEVING 6/1

On a wire rope hoist with hoist drive GM 7000, the deflection roller crossheads are installed in the trolley frame.

Fit the first deflection roller crosshead:

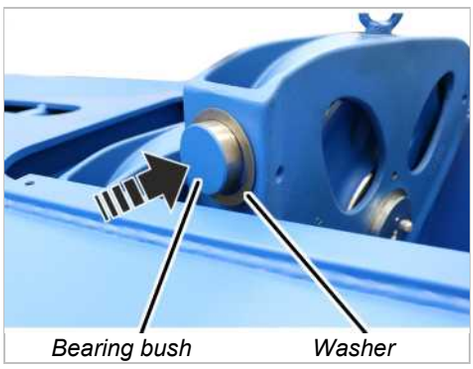


➔ Push the washer (2x) onto both bearing bushes.

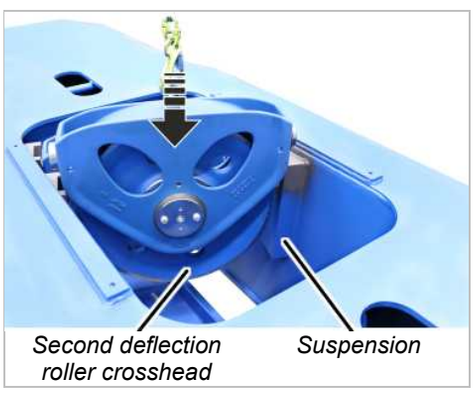


➔ Lift the deflection roller crosshead into the trolley frame.

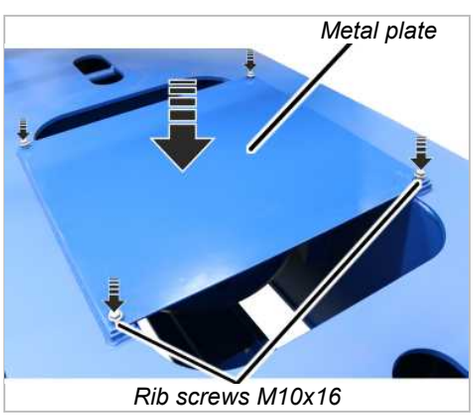
Fit the second deflection roller crosshead:



➔ Push the washer (2x) onto both bearing bushes on the deflection roller crosshead.



➔ Lift the second deflection roller crosshead into the trolley frame.



➔ Place the plate on the trolley frame over the deflection roller crossheads and screw in with rib screws M10x16 (4x). 75 Nm.

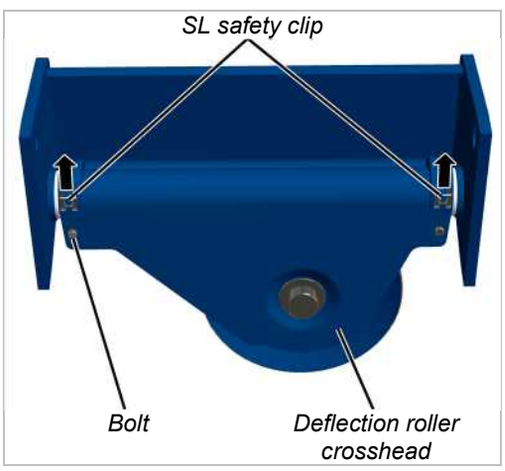
REPLACING DEFLECTION ROLLER CROSSHEAD FOR REEVING 4/2 OR 8/2

If the deflection roller crosshead is damaged, it must be replaced.

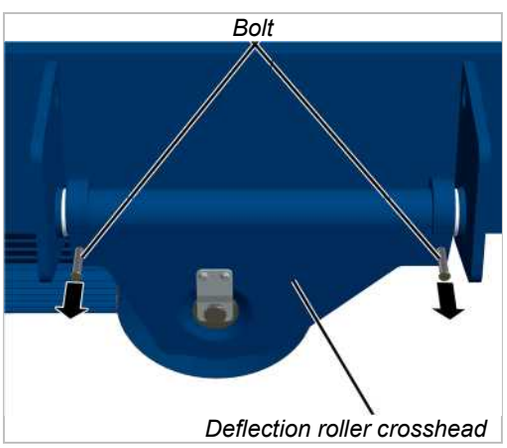
A deflection roller crosshead is installed on the trolley frame for reeving 4/2. This hangs on bearing bushes and is secured with bolts and SL safety clips

Two deflection roller crossheads are also installed in the trolley frame for reeving 8/2.

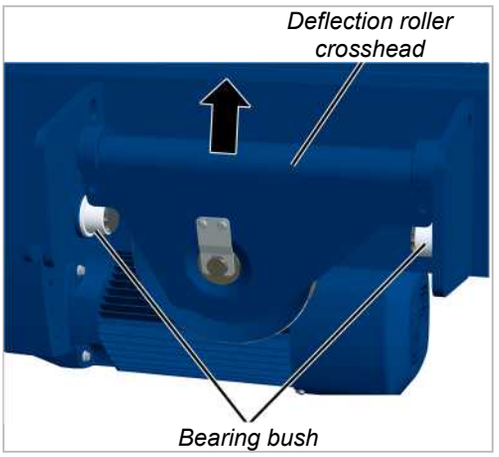
DISASSEMBLING THE DEFLECTION ROLLER CROSSHEAD



➔ Pull the SL safety clips from the bolts.



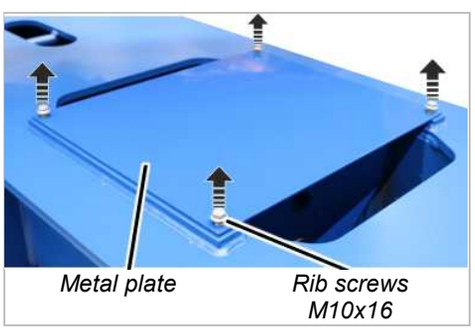
➔ Pull the bolts (2x) out of the deflection roller crosshead.



➔ Lift the deflection roller crosshead from the bearing bushes.

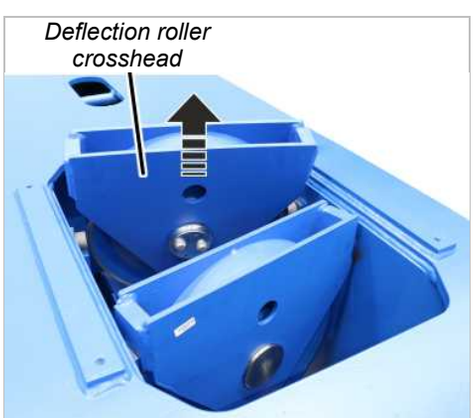
ONLY WITH WIRE ROPE HOIST WITH REEVING 8/2

On a wire rope hoist with a reeving 8/2, two deflection roller crossheads are also installed in the trolley frame.

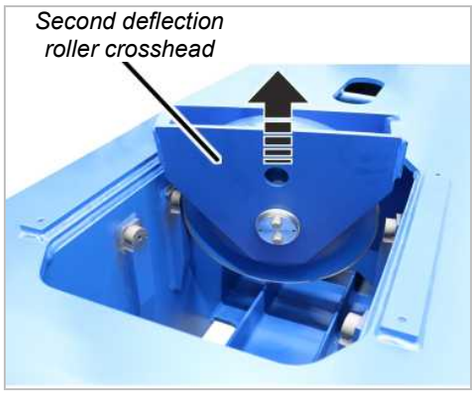


➔ Release the rib screws M10x16 (4x) on the plate.

➔ Lift the plate off the trolley frame.

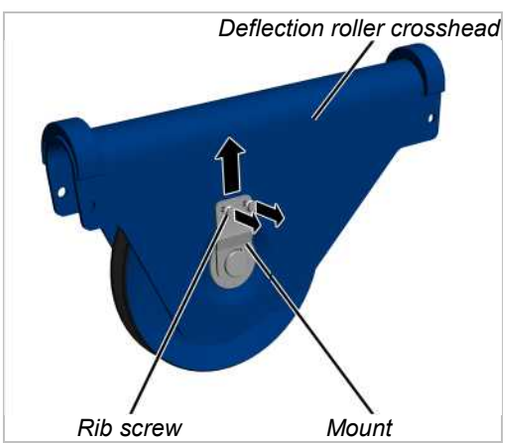


➔ Lift the deflection roller crosshead out of the trolley frame.



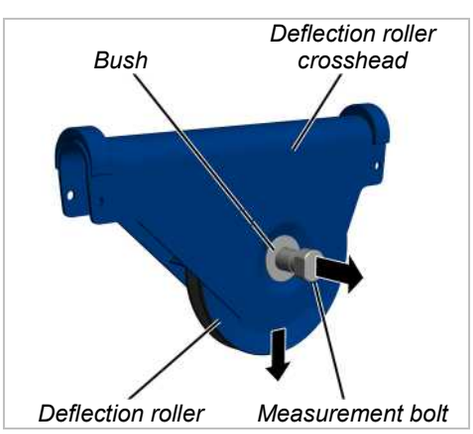
➔ Lift the second deflection roller crosshead out of the trolley frame.

DISASSEMBLING THE DEFLECTION ROLLER



➔ Release the rib screws (2x).

➔ Pull the mount of the measurement bolt.

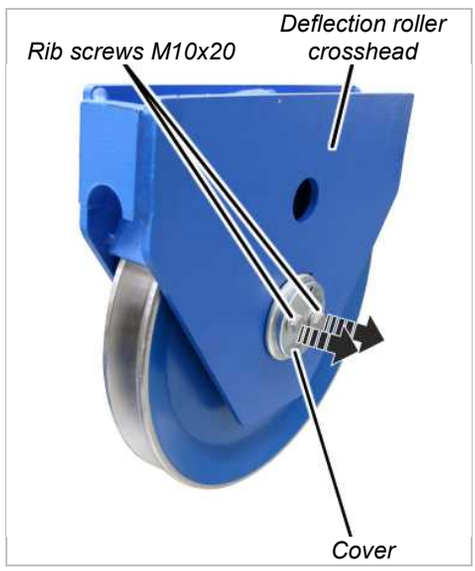


➔ Pull the measurement bolt out of the bush in the deflection roller crosshead and deflection roller.

➔ Pull the deflection roller out of the deflection roller crosshead.

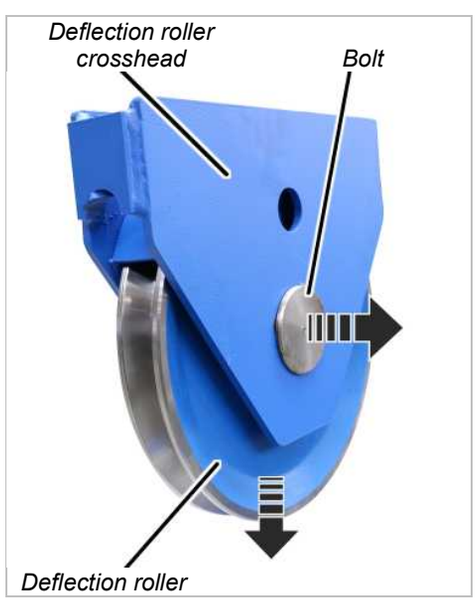
ONLY WITH WIRE ROPE HOIST WITH REEVING 8/2

On a wire rope hoist with a reeving 8/2, the deflection rollers are secured on the deflection roller crosshead using a bolt with a cover.



➔ Release the rib screws (2x) on the cover.

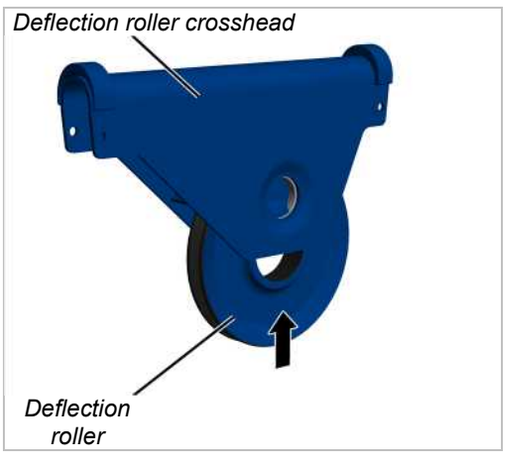
➔ Take the cover off the deflection roller crosshead.



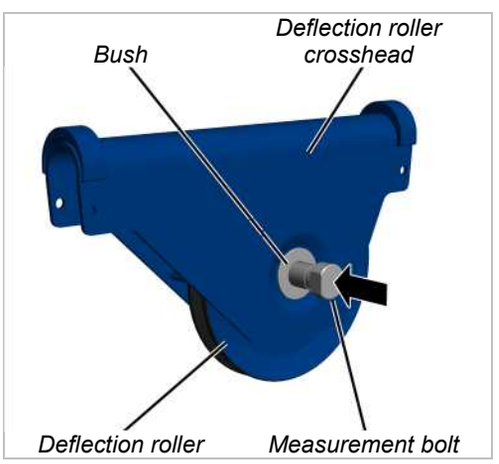
➔ Pull the bolt out of the deflection roller crosshead and deflection roller.

➔ Pull the deflection roller out of the deflection roller crosshead.

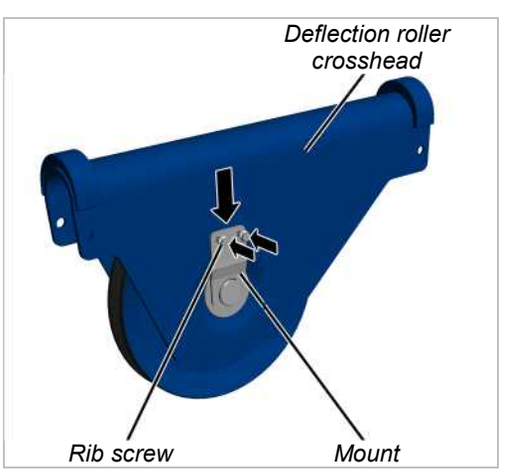
ASSEMBLING THE DEFLECTION ROLLER



➔ Push the deflection roller into the deflection roller crosshead.



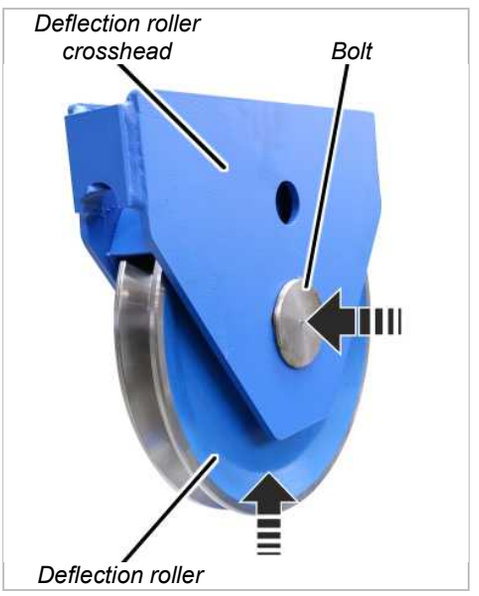
➔ Push the measurement bolt through the bush and the deflection roller in the deflection roller crosshead.



➔ Slide the mount onto the measurement bolt.

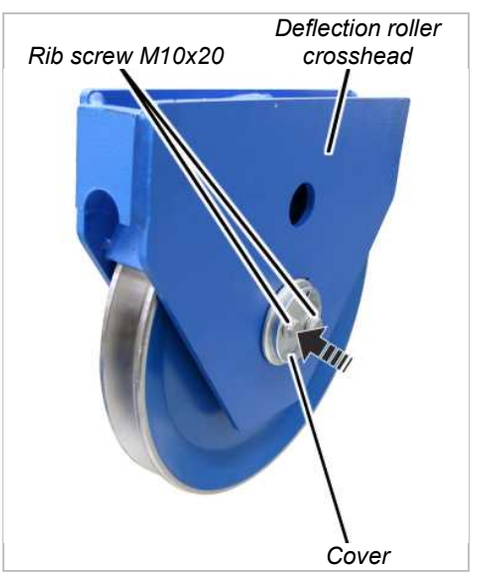
➔ Screw on the mount with the rib screws (2x) M6x12. 19 Nm.

ONLY WITH WIRE ROPE HOIST WITH REEVING 8/2



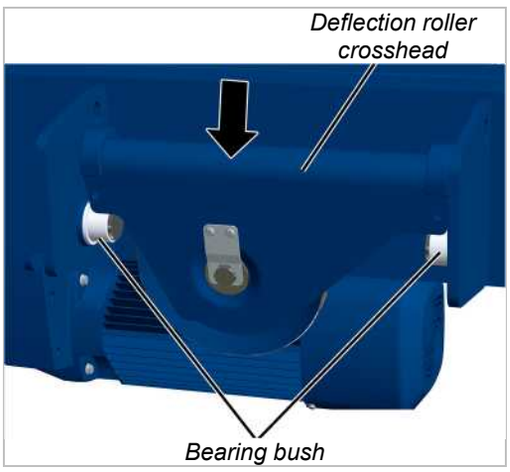
➔ Push the deflection roller into the deflection roller crosshead.

➔ Push the bolt through the deflection roller crosshead and deflection roller.



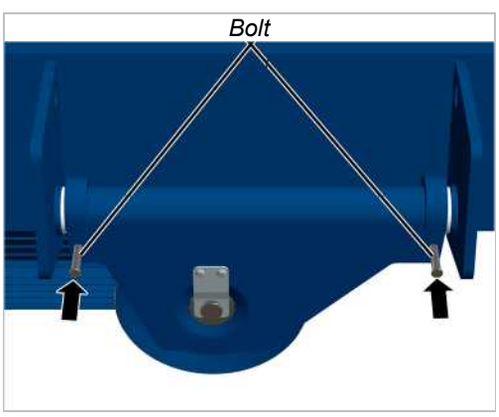
➔ Screw on the cover from the other side at the bolt with rib screws M10x20 (2x). 75 Nm.

MOUNTING THE DEFLECTION ROLLER CROSSHEAD

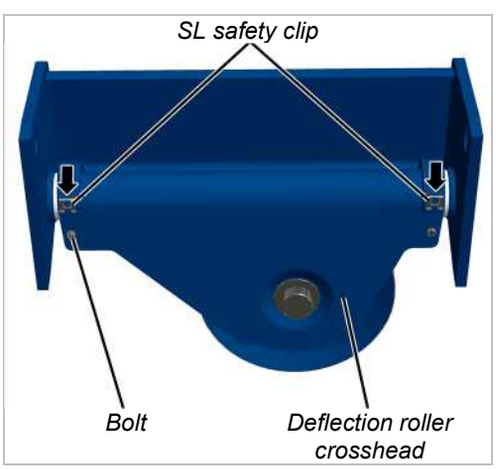


➔ Mount the deflection roller crosshead on the bearing bushes.

When doing this, ensure that the deflection roller points in the direction of the hoist drive.



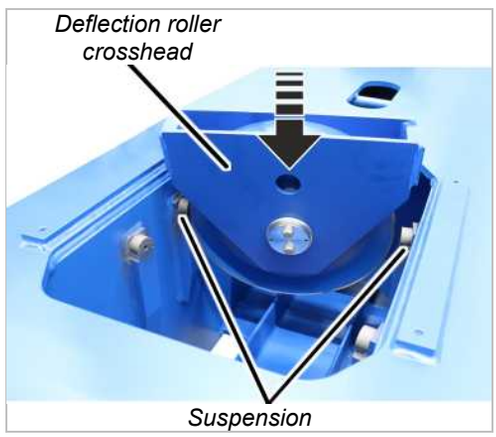
➔ Push the bolts (2x) into the deflection roller crosshead.



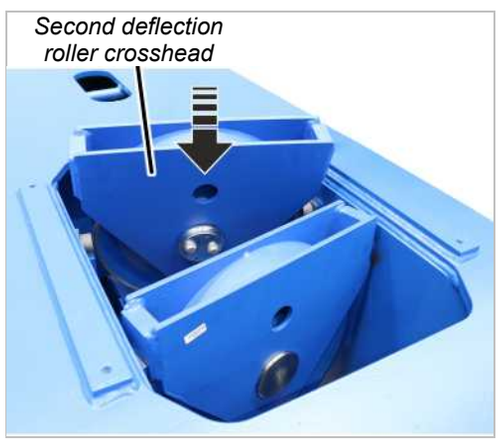
➔ Push the SL safety clips (2x) onto the bolts.

ONLY WITH WIRE ROPE HOIST WITH REEVING 8/2

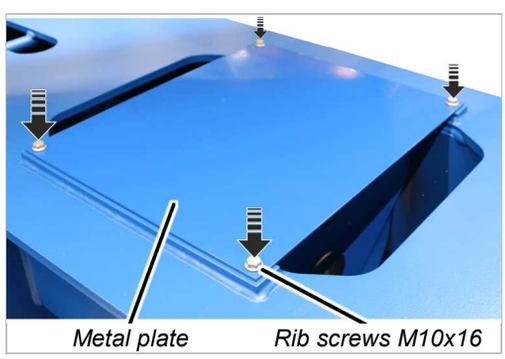
On a wire rope hoist with a reeving 8/2, two deflection rollers are installed in the trolley frame.



➔ Lift the deflection roller crosshead into the trolley frame.



➔ Lift the second deflection roller crosshead into the trolley frame.



➔ Place the plate on the trolley frame over the deflection roller crossheads and screw in with rib screws M10x16 (4x). 75 Nm.

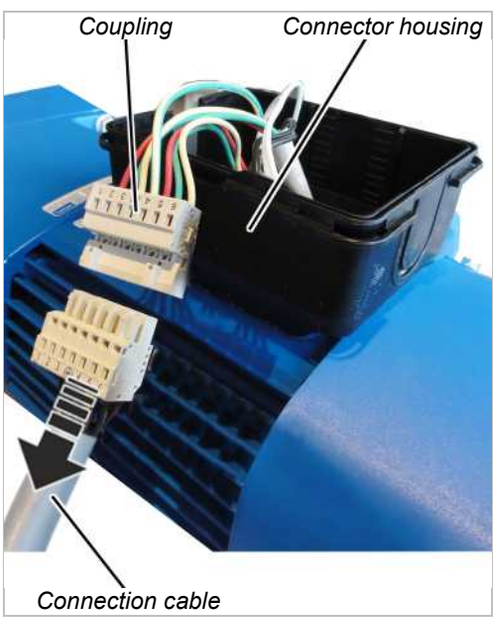
DISASSEMBLING THE POWERED WHEEL

If the wheel is damaged or worn, it must be replaced by a new wheel.

PREPARING THE WIRE ROPE HOIST

- Comply with occupational health and safety requirements.
- Relieve the load of the end carriage on the wire rope hoist near the wheel from below.

REMOVING THE DRIVE

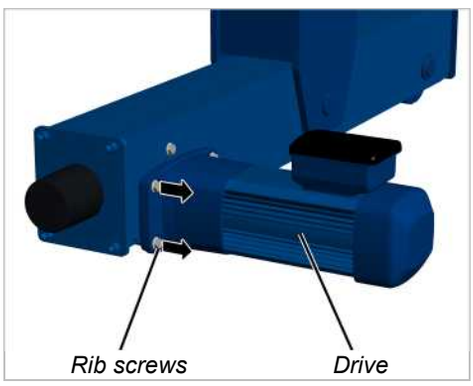


- Open the connector housing to the drive.
- Disconnect the connection cable from the coupler plug.

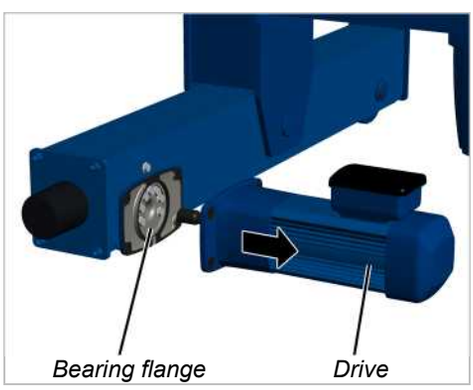
- Determine the weight of the drive. See the ABUS Drive product manual.
- Secure the drive so that the weight can be held safely and the output shaft will not be subject to load (e.g. insert in round sling).

ONLY WITH AZP DRIVES

This section only applies if an AZP drive is to be removed.



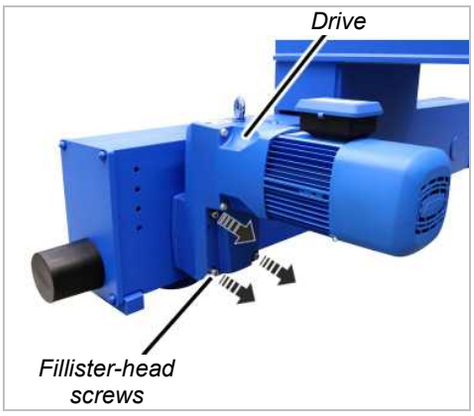
- Release the rib screws (4x).



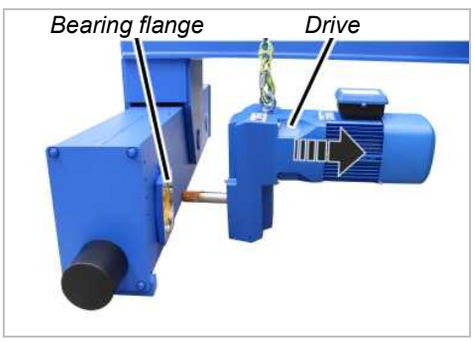
- Pull out the drive.

ONLY WITH AZF DRIVES

This section only applies if an AZF drive is to be removed.

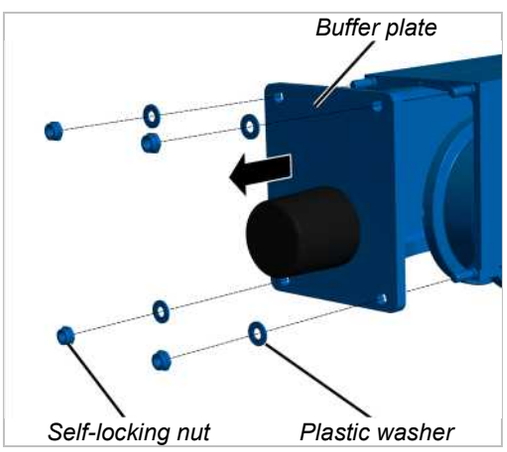


➔ Loosen the fillister-head screws (4x).



➔ Pull out the drive.

EXPOSING THE WHEEL



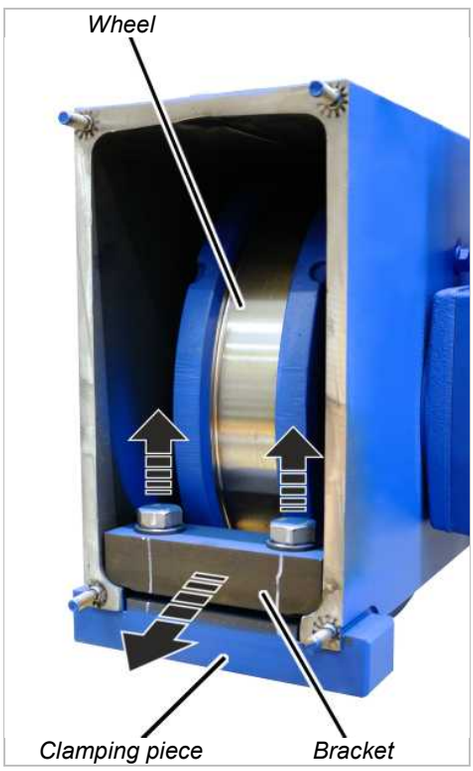
➔ Unscrew the self-locking nuts (4x).

➔ From size 160: remove the plastic washers (4x).

➔ Remove the buffer plate.

ONLY WITH END CARRIAGES OF THE SIZES 280, 350 AND 420

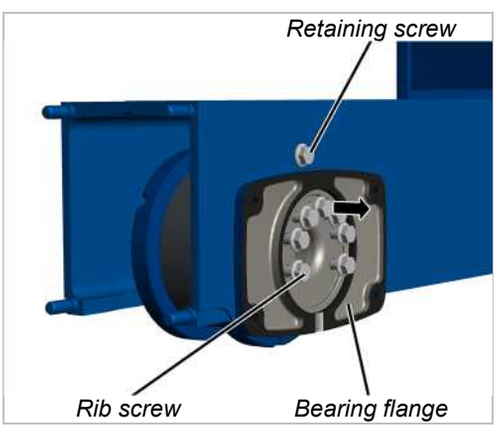
Trolleys in size 280, 350 and 420 have a clamping piece installed on the driven wheel in the end carriage. To remove the wheel, the clamping piece must be removed.



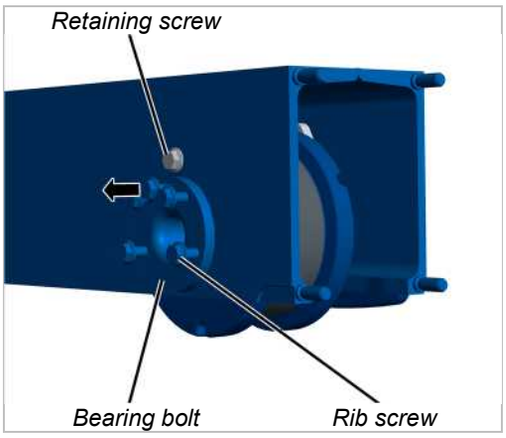
➔ Loosen screws (2x).

➔ Take out the bracket with the clamping piece.

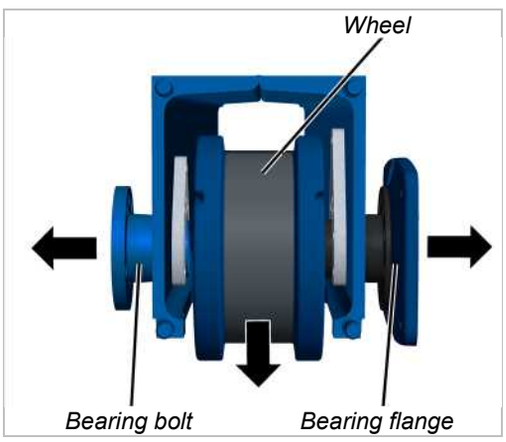
REMOVING THE WHEEL



➔ Unscrew and remove all rib screws on the bearing flange. Do not unscrew and remove the retaining screw(s).



➔ Unscrew and remove all rib screws on the bearing bolt. Do not unscrew and remove the retaining screw(s).



➔ Take out the bearing bolt, bearing flange and wheel.

Note

Only for size 420 with drive 420V: an adapter is also inserted on the bearing flange. If this cannot be easily taken off, it can remain on the bearing flange when the wheel is removed.

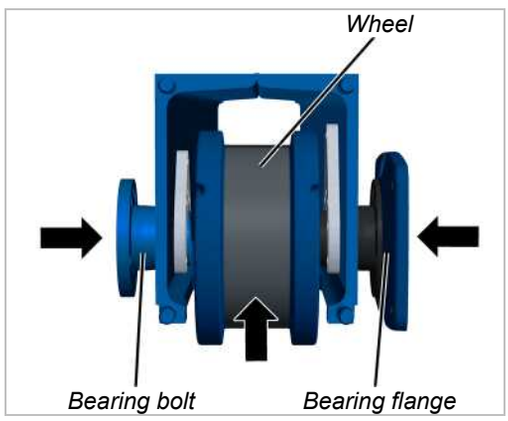
Tip:

To be able to easily remove the bearing flange from the end carriage, screw two or four screws in the corners of the bearing flange and thereby press the bearing flange off the end carriage.

INSTALLING THE POWERED WHEEL

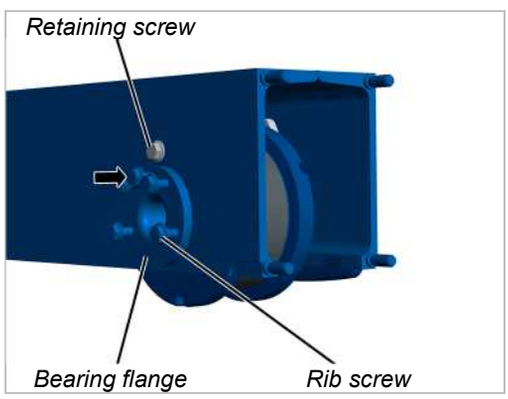
If the wheel is damaged or worn, it must be replaced by a new wheel.

FITTING THE WHEEL



➔ Hold the wheel in the end carriage. It does not matter which side faces inwards or outwards.

➔ Insert the bearing bolt and bearing flange. If necessary, lightly hammer the bearing bolt and bearing flange.



➔ Tighten the rib screws in stages.

Wheel Ø	Strength category	Number	Screw	Tightening torque
130	10.9	5x	M8x25	42 Nm
160	10.9	5x	M8x25	42 Nm
200	10.9	5x	M10x30	85 Nm
280	10.9	5x	M12x40	130 Nm
350	10.9	8x	M12x40	130 Nm
420	10.9	8x	M16x40	330 Nm

ONLY WITH SIZE 420 WITH DRIVE 420V

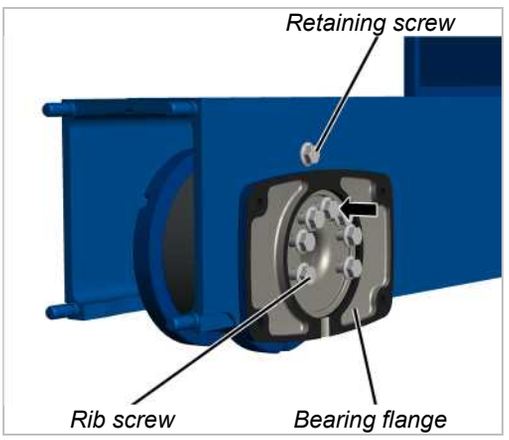
This work step only applies if an end carriage size 420 is to be installed with a drive size 420V.

MOUNTING THE ADAPTER



- If necessary: insert the pins (4x) in the bearing flange.
- Turn the adapter to match the hole pattern.
- Place the adapter on the pins on the bearing flange.
- Insert the rib screws M16x70 (10x) through the adapter and bearing flange.

FASTENING THE BEARING FLANGE



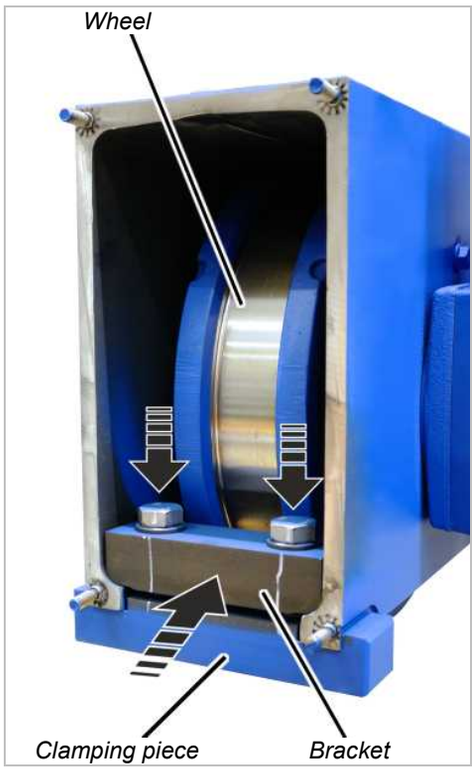
- Tighten the rib screws in stages.

Wheel Ø	Strength category	Number	Type	Tightening torque
130	10.9	7x	M8x25	42 Nm
160	10.9	7x	M8x25	42 Nm
200	10.9	7x	M10x30	85 Nm
280	10.9	7x	M12x40	130 Nm
350	10.9	10x	M12x40	130 Nm
420 with crane travel drive 420	10.9	10x	M16x40	330 Nm
420 with crane travel drive 420 V	10.9	10x	M16x70	330 Nm

ASSEMBLING THE END CARRIAGE

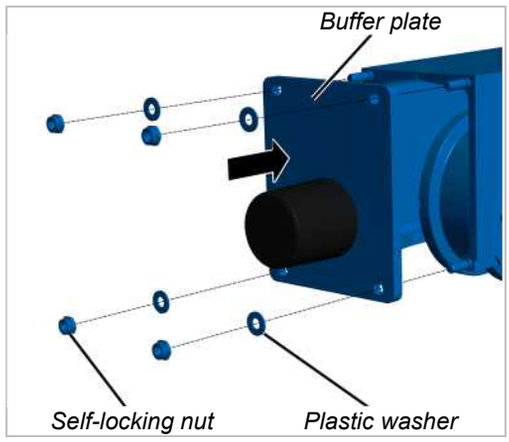
ONLY WITH END CARRIAGES OF THE SIZES 280, 350 AND 420

This work step applies only to end carriages of the sizes 280, 350 and 420.



- Insert clamping piece.
- Tighten screws (2x).

Wheel Ø	Type	Kind	Tightening torque
130	M8x45	Rib screw	42 Nm
160	M10x50	Rib screw	85 Nm
200	M10x50	Rib screw	85 Nm
280	M12x60	Hexagon head screw	130 Nm
350	M20x75 or M20x70	Hexagon head screw	300 Nm
420	M20x75 or M20x70	Hexagon head screw	300 Nm



- Slide the buffer plate onto the threaded bolt.
- From size 160: fit the plastic washers (4x) in place.
- Tightly screw the self-locking nuts M10 (4x). 6 Nm.

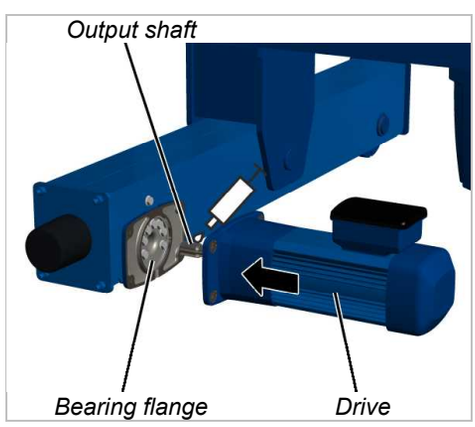
  The self-locking nuts may only be reused maximally 3 times. After this, new self-locking nuts must be used.

INSTALLING THE DRIVE

- Determine the weight of the drive. See the ABUS Drive product manual.
- Secure the drive so that the weight can be held safely and the output shaft will not be subject to load (e.g. insert in round sling).

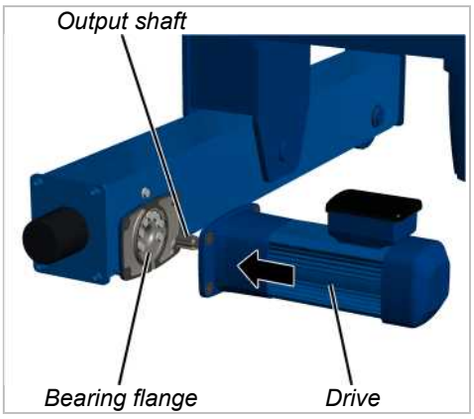
ONLY WITH AZP DRIVES

This work step only applies when an AZP drive is to be mounted on the end carriage.

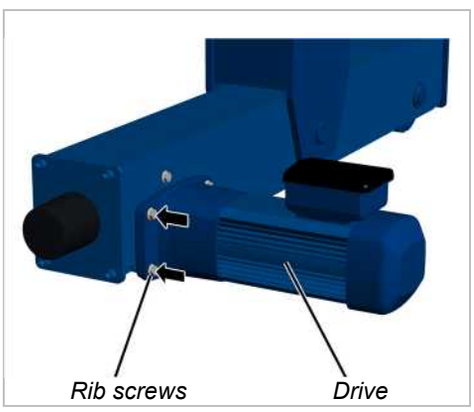


- Lubricate the gearing on the output shaft. Apply lubricant with brush to coat surface completely.

  Lubricant: “High temperature paste PBC 1574”. See the ABUS Drive product manual for details.



Push the drive into the wheel.

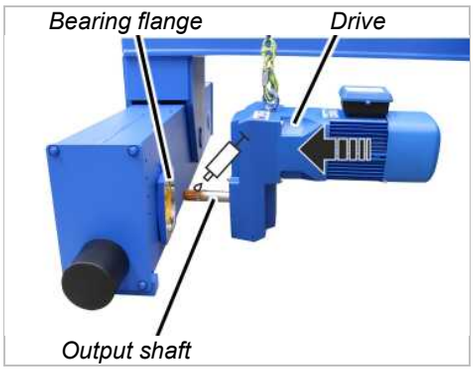


Bolt on the drive with the rib screws (4x).

Size	Type	Tightening torque
130	M8x25	35 Nm
160	M10x30	75 Nm
200	M12x30	115 Nm
280	M16x40	300 Nm

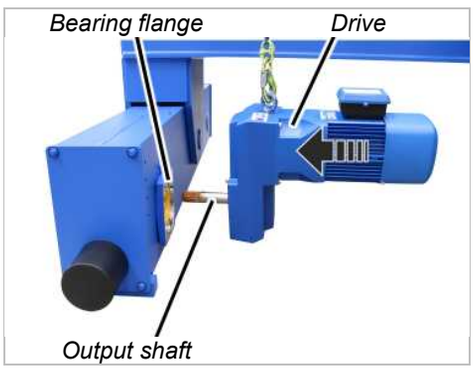
ONLY WITH AZF DRIVES

This work step only applies when an AZF drive is to be mounted on the end carriage.

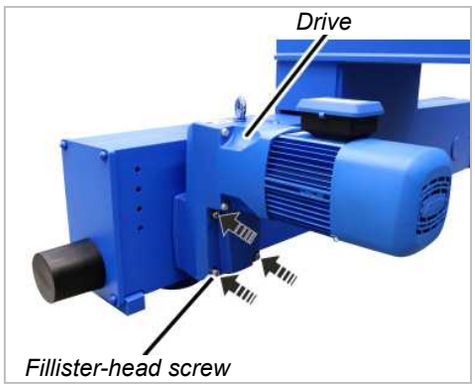


Lubricate the gearing on the output shaft. Apply lubricant with brush to coat surface completely.

Lubricant: “High temperature paste PBC 1574”. See the ABUS Drive product manual for details.



Push the drive into the wheel.

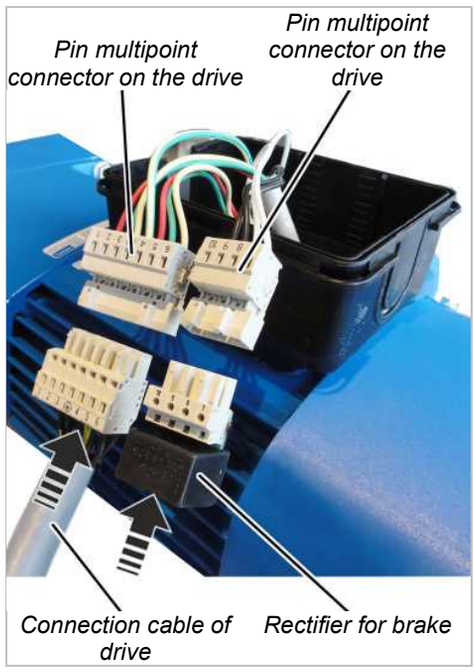


➔ Screw the drive in place using fillister-head screws and washers (4x).

Size	Type	Tightening torque
350	M12x100	85 Nm
420	M12x100	85 Nm
V420	M16x140	210 Nm
V500	M16x140	210 Nm

CONNECTING THE DRIVE

ONLY WITH CONTACTOR CONTROL

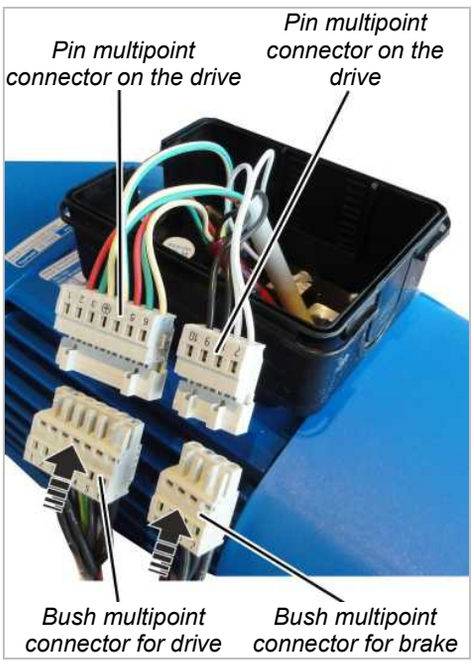


➔ Insert the connection cable for the drive in the pin multipoint connector on the drive.

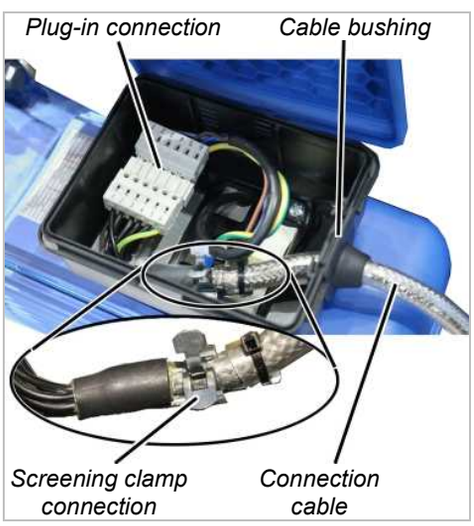
➔ Insert the rectifier for the brake in the pin multipoint connector on the drive.

➔ Lay the plug-in connections and connection cable in the connector housing.

ONLY FOR FREQUENCY CONVERTER



- Insert the bush multipoint connector for the drive on the pin multipoint connector on the drive.
- Insert the bush multipoint connector for the brake on the pin multipoint connector on the drive.



- Push the front end of the connection cable (where the sheathing around the shield has been removed) into the screening clamp connection.
- Fasten the connection cable to the screening clamp connection using cable ties.
- Lay the plug-in connections and connection cable in the connector housing.
- Close the housing cover.

CHECKING THE WIRE ROPE HOIST

- Set the wire rope hoist down again.
- Put the wire rope hoist into operation.
- Check the function of the wheel.

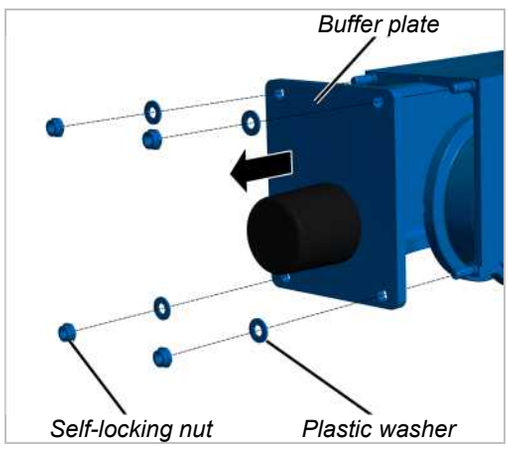
DISASSEMBLING THE NON-POWERED WHEEL

If the wheel is damaged or worn, it must be replaced by a new wheel.

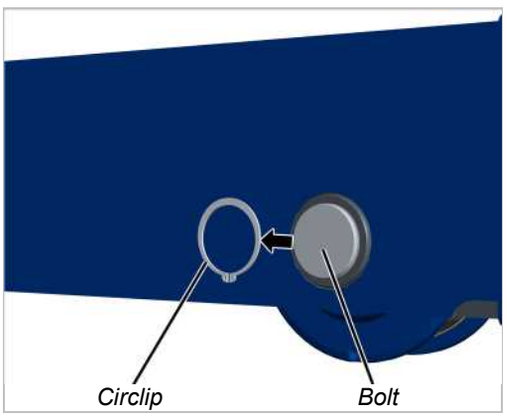
PREPARING THE WIRE ROPE HOIST

- Comply with occupational health and safety requirements.
- Relieve the load of the end carriage on the wire rope hoist near the wheel from below.

EXPOSING THE WHEEL

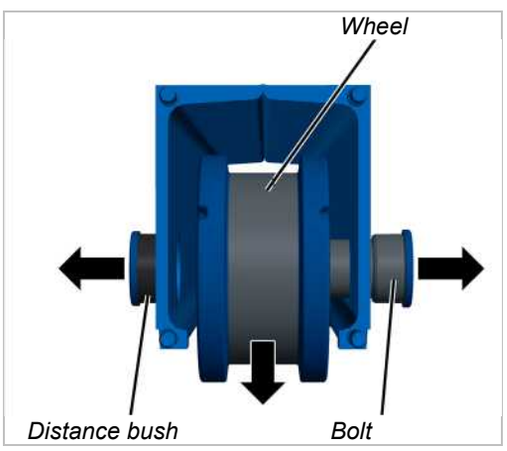


- Unscrew the self-locking nuts (4x).
- From size 160: remove the plastic washers (4x).
- Remove the buffer plate.



➔ Remove the circlip.

REMOVING THE WHEEL

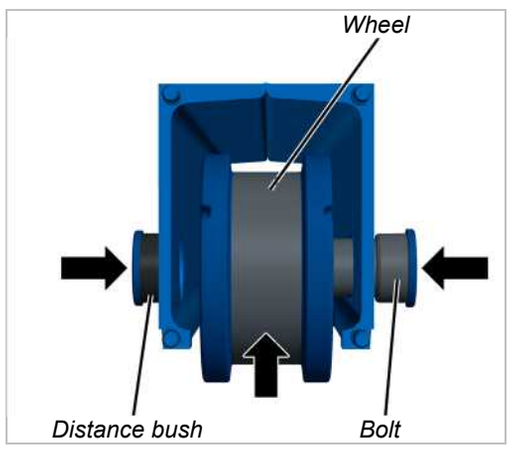


➔ Take out the bolt, distance bush and wheel.

INSTALLING THE NON-POWERED WHEEL

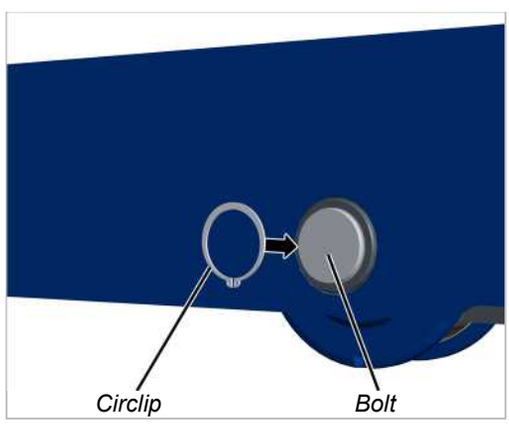
If the wheel is damaged or worn, it must be replaced by a new wheel.

FITTING THE WHEEL



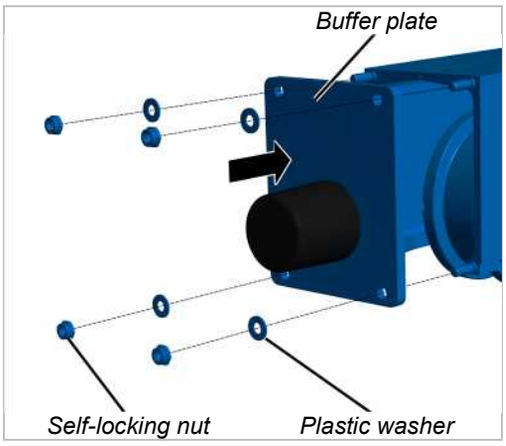
➔ Hold the wheel in the end carriage. It does not matter which side faces inwards or outwards.

➔ Insert the bolt and distance bush. If necessary, lightly hammer the bolt and distance bush.



➔ Attach the circlip and neatly lock into place.

➔ Turn the wheel. It must move easily.



➔ Slide the buffer plate onto the threaded bolt.

➔ From size 160: fit the plastic washers (4x) in place.

➔ Tightly screw the self-locking nuts M10 (4x). 6 Nm.

The self-locking nuts may only be reused maximally 3 times. After this, new self-locking nuts must be used.

CHECKING THE WIRE ROPE HOIST

➔ Set the wire rope hoist down again.

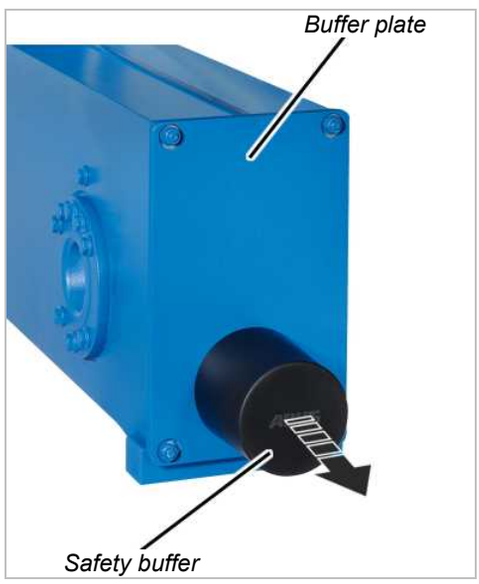
➔ Put the wire rope hoist into operation.

➔ Check the function of the wheel.

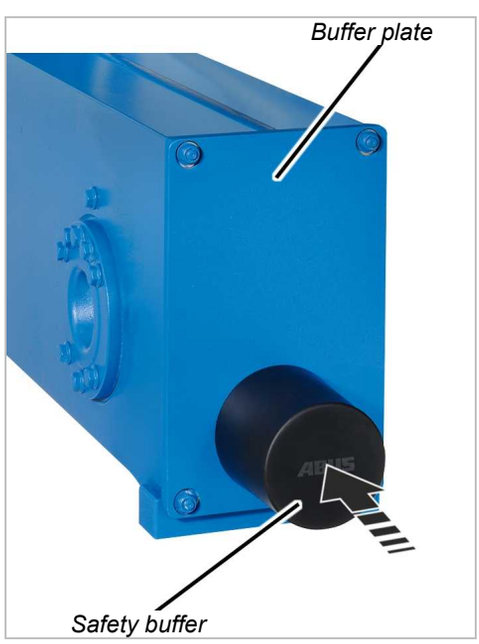
REPLACING THE SAFETY BUFFER

If the safety buffer is damaged, it must be replaced.

On both sides:



➔ Unscrew the safety buffer.



➔ Screw safety buffers into the buffer plates. Approx. 30 Nm.

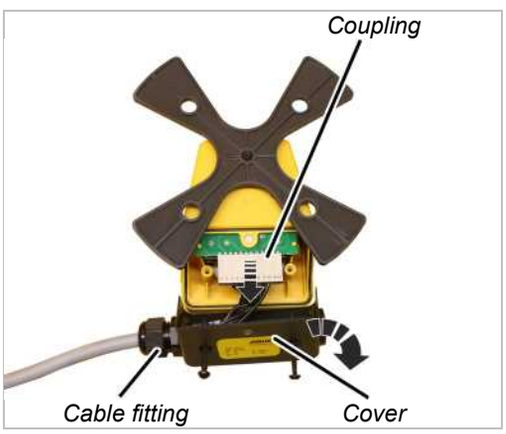
The safety buffer must be in contact with the buffer plate and there must be no gap between them, otherwise the impact will be absorbed by the screw and not by the safety buffer.

REPLACING THE CROSS-TYPE LIMIT SWITCH

If the cross-type limit switch is damaged, it must be replaced. The plug-in connection allows you to carry out the replacement without any wiring work.

DANGER DUE TO MALFUNCTION!
If the cross-type limit switch is screwed on too tightly, the parts on the inside can become jammed and no longer function properly.

The tightening torque of 3 Nm must be strictly observed.



➔ Unscrew and remove the screws (2x) on the cover.

➔ Remove the cover.

➔ Disconnect the coupling of the connecting cable from the circuit board.

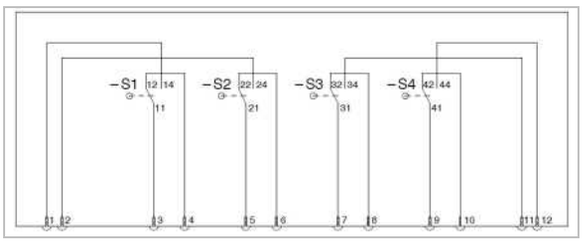
The cable fitting on the cover does not need to be unscrewed. The existing cover can be fitted with the attached connection cable to a new cross-type limit switch.

➔ Insert the coupling onto the new cross-type limit switch.

➔ Close the cover.

➔ Tighten screws (2x). 3 Nm.

WIRING DIAGRAM OF THE CROSS-TYPE LIMIT SWITCH



– The contact numbering shown on the wiring diagram starts on the left ascending from 1.

– The designation of the microswitches starts on the left of the installation position with S1.

Position arrow	Switch position	S1 closed	S1 open	S2 closed	S2 open	S3 closed	S3 open	S4 closed	S4 open
0	0°	3.4		5.6		7.8		9.10	
1	90°	3.4		5.6			7.11	9.10	
2	180°	3.4		5.6			7.11		9.12
3	270°		3.1	5.6		7.8			9.12
4	360°		3.1		5.2	7.8		9.10	
5	450°	3.4			5.2	7.8		9.10	
0	540°	3.4		5.6		7.8		9.10	

LUBRICANTS

Note:

Synthetic lubricants must not be mixed with mineral-based lubricants!

CABLE DRUM IN THE EDGE AREAS



On-site lubrication with “Molykote PG-75”, ABUS item number 5168.

Lubrication ex works with “Molykote PG-75”, ABUS item number 411.

Lubricate the wire rope during:

- Wire rope change
- Cable guide change
- Regular inspection
- General overhaul

For details, see “Cleaning and lubricating the cable drum”, page 46.

WIRE ROPE AND CABLE DRUM IN THE CENTRE



On-site lubrication with “Chainlife S”, ABUS item number 2717

Lubrication ex works with “Chainlife S”, ABUS item number 2718.

Alternatives:

- Castrol “Viscogen KL 23”
- Klüber “Grafloscon CA 901 Ultra Spray”
- Optimol “KL 23”
- Shell “Malleus GL 95”

Lubricate the wire rope during:

- Wire rope change
- Regular inspection
- General overhaul

For details, see “Cleaning and lubricating the cable drum”, page 61.

ROPE GUIDE RUNNER ON THE CABLE GUIDE



On-site lubrication with “Molykote PG-75”, ABUS item number 5168.

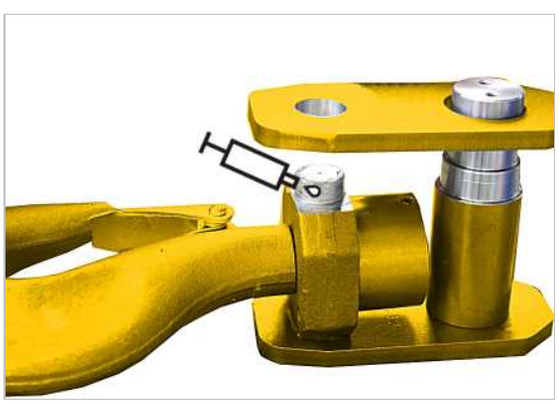
Lubrication ex works with “Molykote PG-75”, ABUS item number 411.

Lubricate the rope guide runner during:

- Wire rope change
- Cable guide change
- Regular inspection
- General overhaul

For details, see “Replacing wire rope with reeving 2/1, 4/1 or 6/1”, page 39.

LOAD HOOK CROSSHEAD ON THE BOTTOM BLOCK



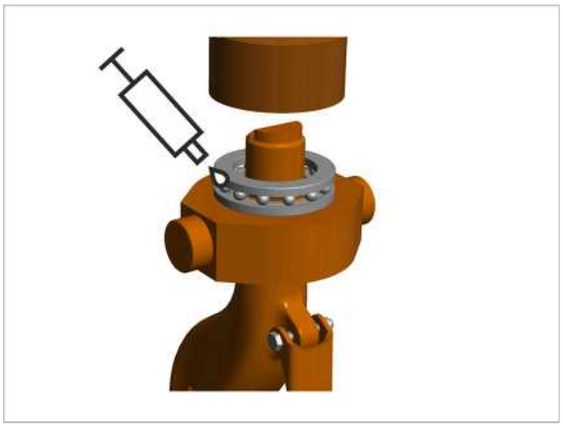
On-site lubrication with “High temperature paste PBC 1574”, ABUS item number 6758.

Lubrication ex works with “High-Lub LT1 EP”, ABUS item number 317880.

Lubricate the load hook crosshead during:

- Assembly and disassembly
- General overhaul

DEEP GROOVE BALL THRUST BEARING ON THE LOAD HOOK



On-site lubrication with “Molykote PG-75”, ABUS item number 5168.

Lubrication ex works with “High-Lub LT1 EP”, ABUS item number 317880.

Lubricate the deep groove ball thrust bearing during:

- Assembly and disassembly
- General overhaul

ONLY WITH FIXED POINT CROSSHEAD

ROPE SOCKET



On-site lubrication with “High temperature paste PBC 1574”, ABUS item number 6758.

Lubrication ex works with “High-Lub LT1 EP”, ABUS item number 317880.

Lubricate the rope socket during:

- Wire rope change
- Regular inspection
- General overhaul

For details, see “Exchanging the fixed point crosshead with rope socket and rope wedge”, page 71.

BOLT IN DEFLECTION ROLLER



On-site lubrication with “High temperature paste PBC 1574”, ABUS item number 6758.

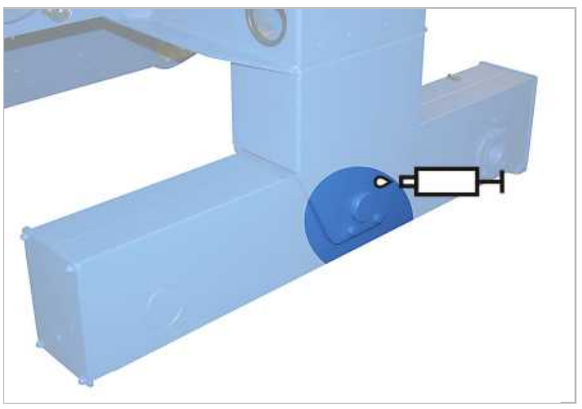
Lubrication ex works with “High-Lub LT1 EP”, ABUS item number 317880.

Lubricate bolt for:

- Assembly or disassembly of the deflection roller
- General overhaul

For details, see “Replacing deflection roller crosshead for reeving 4/1 or 6/1” on page 81 or “Replacing deflection roller crosshead for reeving 4/2 or 8/2” on page 92.

HINGE PIN

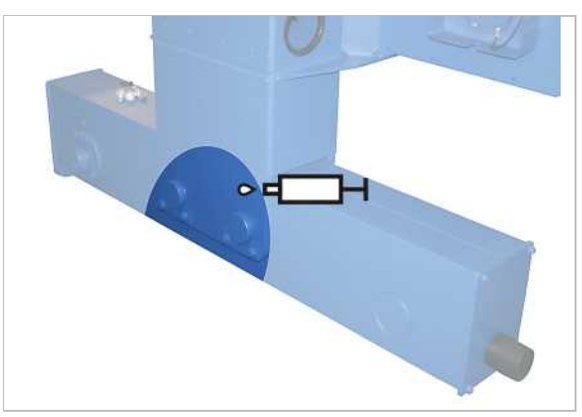


On-site lubrication with “High temperature paste PBC 1574”, ABUS item number 6758.

Lubrication ex works with “High-Lub LT1 EP”, ABUS item number 317880.

Lubricate the hinge pin for:

- Assembling or disassembling the end carriage



On-site lubrication with “High temperature paste PBC 1574”, ABUS item number 6758.

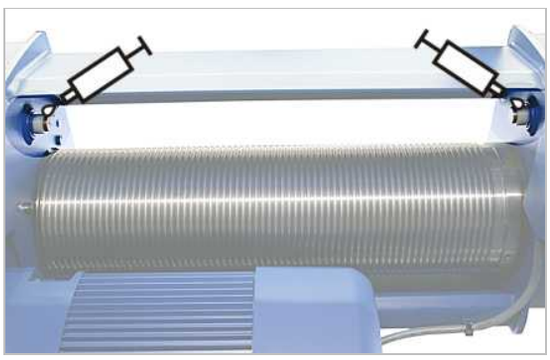
Lubrication ex works with “High-Lub LT1 EP”, ABUS item number 317880.

Lubricate the hinge pin for:

- Assembling or disassembling the end carriage

ONLY WITH WIRE ROPE HOIST GM 800, GM 1000, GM 2000, GM 3000, GM 5000 AND GM 6000

BEARING BUSH (ON COLLAR BOLT) ON THE DEFLECTION ROLLER CROSSHEAD AND ON THE FIXED POINT CROSSHEAD



On-site lubrication with “High temperature paste PBC 1574”, ABUS item number 6758.

Lubrication ex works with “High-Lub LT1 EP”, ABUS item number 317880.

Lubricate bearing bush (on collar bolt) during:

- Assembly and disassembly of the deflection roller crosshead or fixed point crosshead
- General overhaul

For details, see “Exchanging the fixed point crosshead with rope socket and rope wedge” on page 71 or “Replacing deflection roller crosshead for reeving 4/1 or 6/1” on page 81.

ONLY WITH WIRE ROPE HOIST WITH GM 7000

DEFLECTION ROLLER CROSSHEAD



On-site lubrication with “High temperature paste PBC 1574”, ABUS item number 6758.

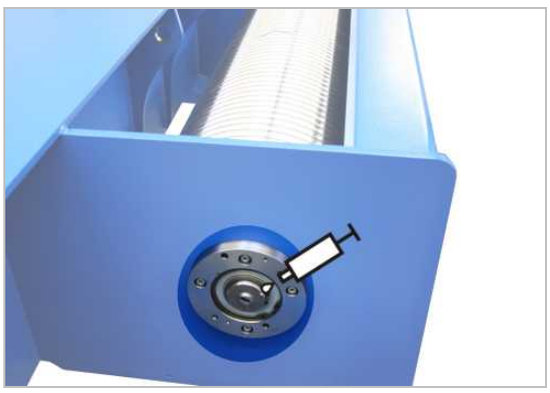
Lubrication ex works with “OKS 511 bonded coating”, item number 18930 and “High-Lub LT1 EP”, ABUS item number 317880.

Lubricate the contact surfaces of the deflection roller crosshead for:

- Assembling and disassembling the deflection roller crosshead
- General overhaul.

For details, see “Replacing deflection roller crosshead for reeving 4/1 or 6/1”, page 81.

SELF-ALIGNING ROLLER BEARING ON THE CABLE DRUM



On-site lubrication with “High-Lub LT1 EP”, ABUS item number 317880.

Lubrication ex works with “High-Lub LT1 EP”, ABUS item number 317880.

Lubricate the self-aligning roller bearing during:

- Assembling and disassembling the self-aligning roller bearing
- General overhaul

OVERVIEW OF TIGHTENING TORQUES

END CLAMP



Size	Fillister-head screw	Number	Tightening torque
GM 800	M6x16	2	10 Nm
GM 1000	M8x20	2	25 Nm
GM 2000	M8x20	2	25 Nm
GM 3000	M8x20	2	25 Nm
GM 5000	M16x35	3	210 Nm
GM 6000	M16x35	3	210 Nm
GM 7000	M20x40	3	425 Nm

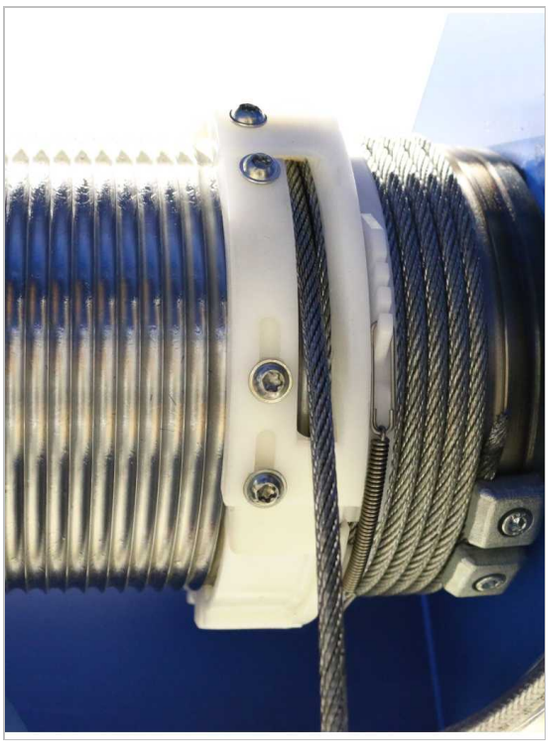
ROPE GUIDE RUNNER



Size	Screw	Tightening torque
GM 800	Self-tapping screw M6x16	5 Nm
GM 1000	Self-tapping screw M6x16	5 Nm
GM 2000	Self-tapping screw M6x16	5 Nm
GM 3000	Self-tapping screw M6x16	5 Nm
GM 5000	Self-tapping screw M6x16	5 Nm
GM 6000	Self-tapping screw M6x16	5 Nm
GM 7000	Self-tapping screw M6x16	5 Nm

ONLY WITH GM 800, GM 1000, GM 2000, AND GM 3000

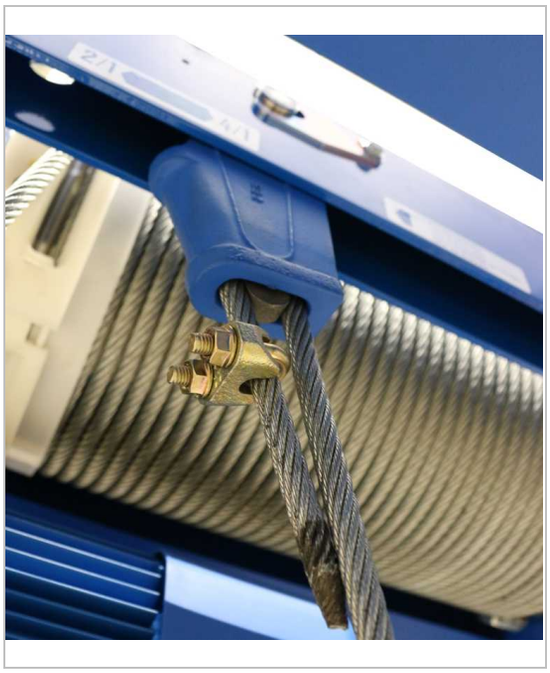
JOINING ELEMENT



Size	Screw	Number	Tightening torque
GM 800	Self-tapping screw M6x16	4x	5 Nm
GM 1000	Self-tapping screw M6x16	4x	5 Nm
GM 2000	Self-tapping screw M6x16	4x	5 Nm
GM 3000	Self-tapping screw M6x16	4x	5 Nm

ONLY WITH WIRE ROPE HOIST WITH REEVING 2/1, 4/1 OR 6/1

DEAD END ROPE GRIP



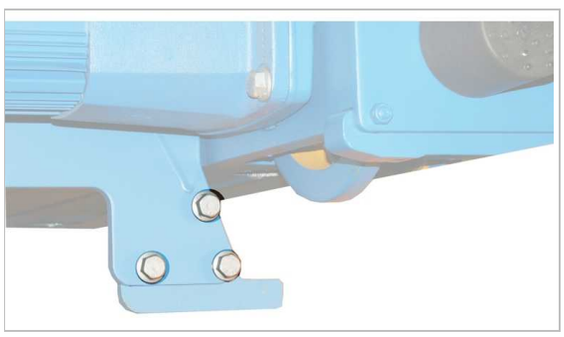
Size	Rib nut	Tightening torque
GM 800	M6	5 Nm
GM 1000	M8	10 Nm
GM 2000	M8	10 Nm
GM 3000	M8	10 Nm
GM 5000	M14	49 Nm
GM 6000	M14	49 Nm
GM 7000	M16	107 Nm

LIFT-OFF PROTECTION DEVICE ZLK WITH TROLLEY D, DA



- Rib nut M8
- 2x per lift-off prevention device
- 15 Nm

LIFT-OFF PREVENTION DEVICE TROLLEY DQA



Type and size	Number	Tightening torque
Rib screw M12x30	3x	115 Nm
Rib screw M16x35	3x	210 Nm

TROLLEY TOWING ARM TROLLEY DQA



- Hexagon head screw M8 with hexagonal nut
- 6x
- 25 Nm

ABUS SERVICE

ONLY IN GERMANY

➔ If available, have the product number, serial number and customer number handy.

➔ Call the ABUS Service Centre:

– Phone: +49-2261-37-237

➔ If calling outside the usual business hours, leave a message on the answering machine.

● ABUS Service will return your call promptly.

➔ If necessary, send a description of the problem by fax or e-mail:

– Fax: +49-2261-37-265

– E-mail: service@abus-kransysteme.de

ONLY OUTSIDE GERMANY

➔ Call the local ABUS branch or crane service partner.

Your local ABUS branch or crane service partner will provide details of contact data, contacts and availability.

ABUS Kransysteme GmbH
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51647 Gummersbach Germany
Tel. 0049 – 2261 – 37-0
Fax. 0049 – 2261 – 37-247
info@abus-kransysteme.de

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